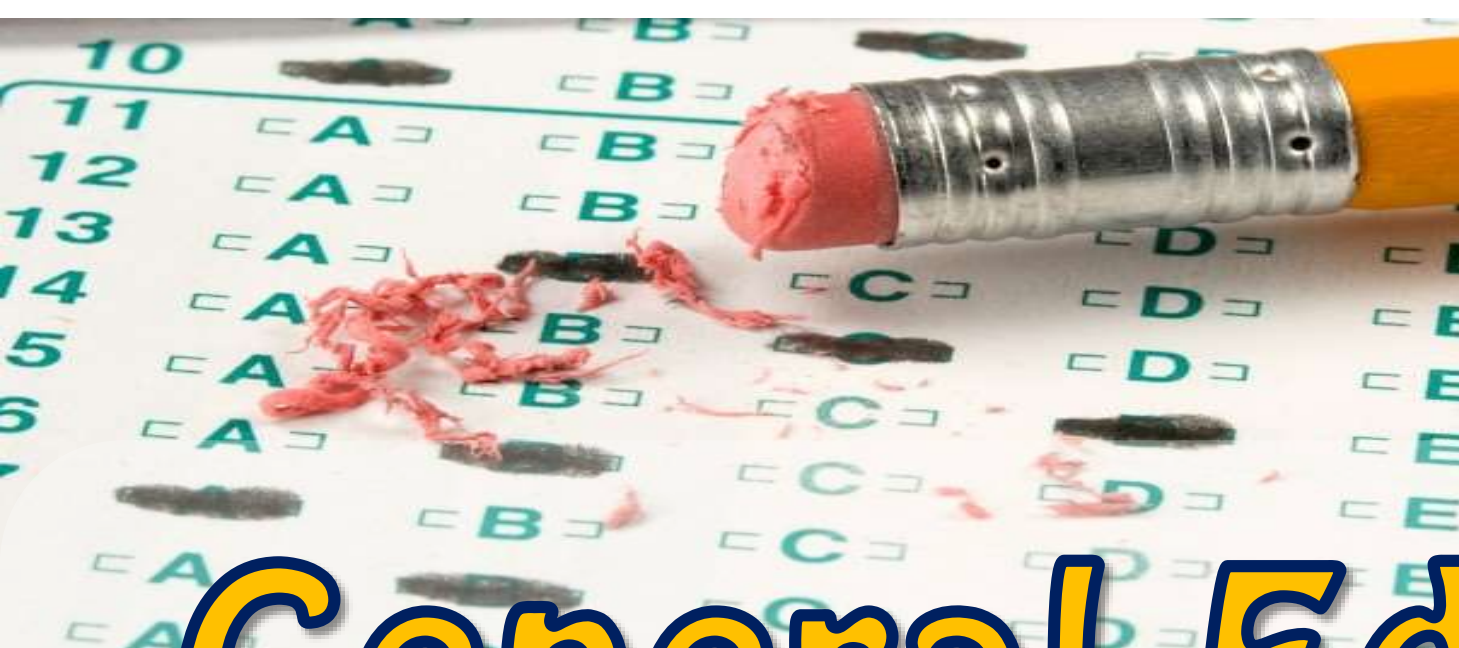




General Education Rationalization





SCORE AIM

136 – 150 – *Topnotcher*

113 – 135 – *VS Passer*

76 – 112 – *Delicate Scorer*

0 – 75 – *Try again!*

1. I can't use _____ calculator
may I use _____ , Jesson?

- A. my, yours
- B. my, his
- C. his, yours
- D. yours, mine



The Possessive Case

- The personal pronouns in the possessive case—my, mine, your, yours, his, her, hers, its, our, ours, their and theirs—are used to show ownership or possession
1. The possessive pronouns—mine, yours, his, hers, its, ours, and theirs—can be used in the same way that the personal pronouns in the nominative and objective cases are used.
 - **Mine** has a flat tire.
 - This key is **yours**.
 2. The possessive pronouns—my, your, his, her, its, our, and there are used to modify nouns and pronouns.
 - **My** bicycle has a flat tire. (**My** modifies bicycle.)
 - This is **your** key. (**Your** modifies key.)
 3. A noun or pronoun preceding a gerund generally should be in the possessive case
 - They approved of the **students'** [or **their**] organizing a cleanup campaign.

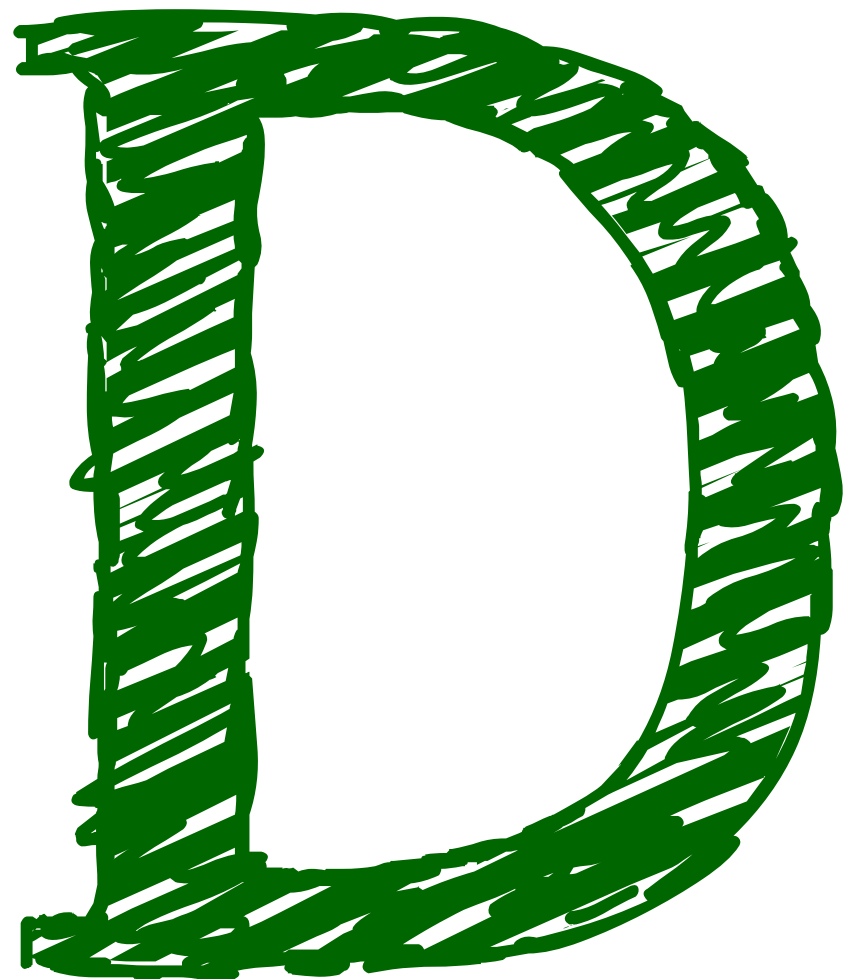
2. Of the following remarks from workers, which deserve a higher wage owing to effective speech that increases work efficiency?

A. I have strong hands, willingness and honesty in work

B. I have strong hands and I am a willing worker

C. I like people and listen to respond to their opinions and needs

D. I have work well and plan my work to accomplish more



3. Which among the functions of listening do most students do in the classroom?

A. for inspiration

B. for information and ideas

C. for “escape”

D. for enjoyment

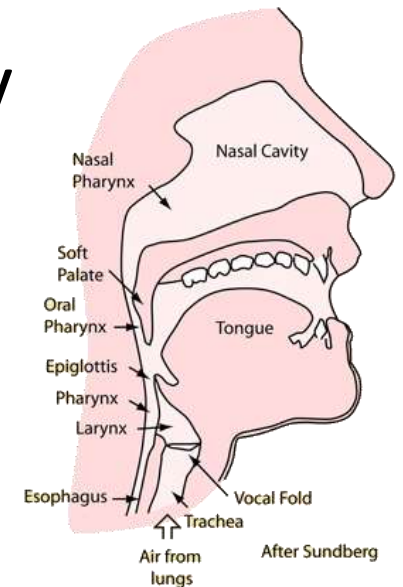
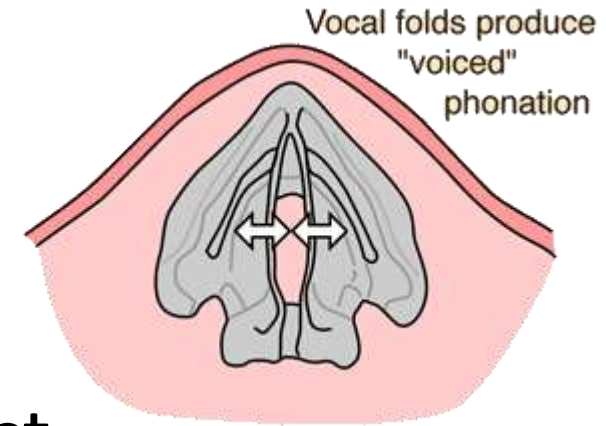
B

4. In the speech process, which is concerned with amplifying and enriching the voice using human amplifiers such as the nose, windpipe and chest?

- A. resonation
- B. inflection
- C. phonation
- D. articulation

The process of speech production by the human voice may be divided into phonation, resonance, and articulation.

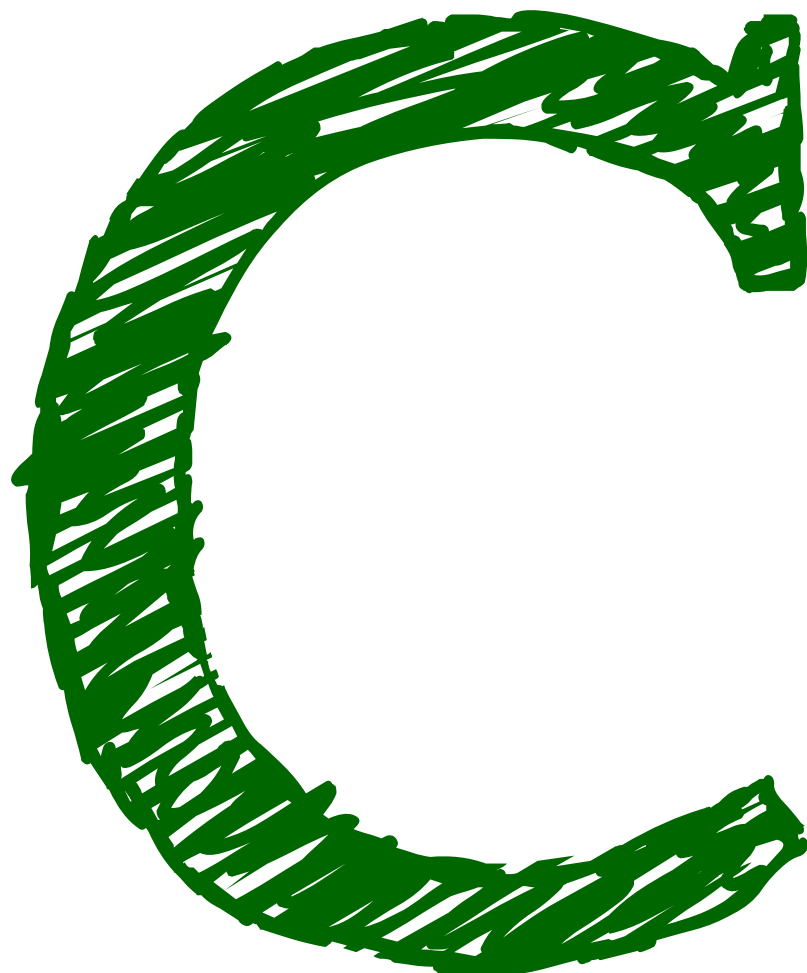
- ***Phonation*** is the process by which energy from the lungs in the form of air pressure is converted into audible vibrations.
- ***Resonation*** is the process by which the basic product of phonation is enhanced in timbre and/or intensity by the resonators, including amplification, enrichment, enlargement, etc.
- ***Articulation*** is seen as the changes in the resonances of the vocal tract, and the agents of such changes





5. “I have always been financially independent by continuing to teach even when I got married. At times though when the cares of life weighed heavily on me, I would think of my mother and her example of making the most of what she had. She did her household chores methodically, tutored us in our lesson and gave modest contributions to worthy causes.” What adjective best describes the mother?

- A. suffering
- B. sacrificing
- C. dutiful
- D. overworked



DUTIFUL

– conscientiously or
obediently fulfilling one's
duty

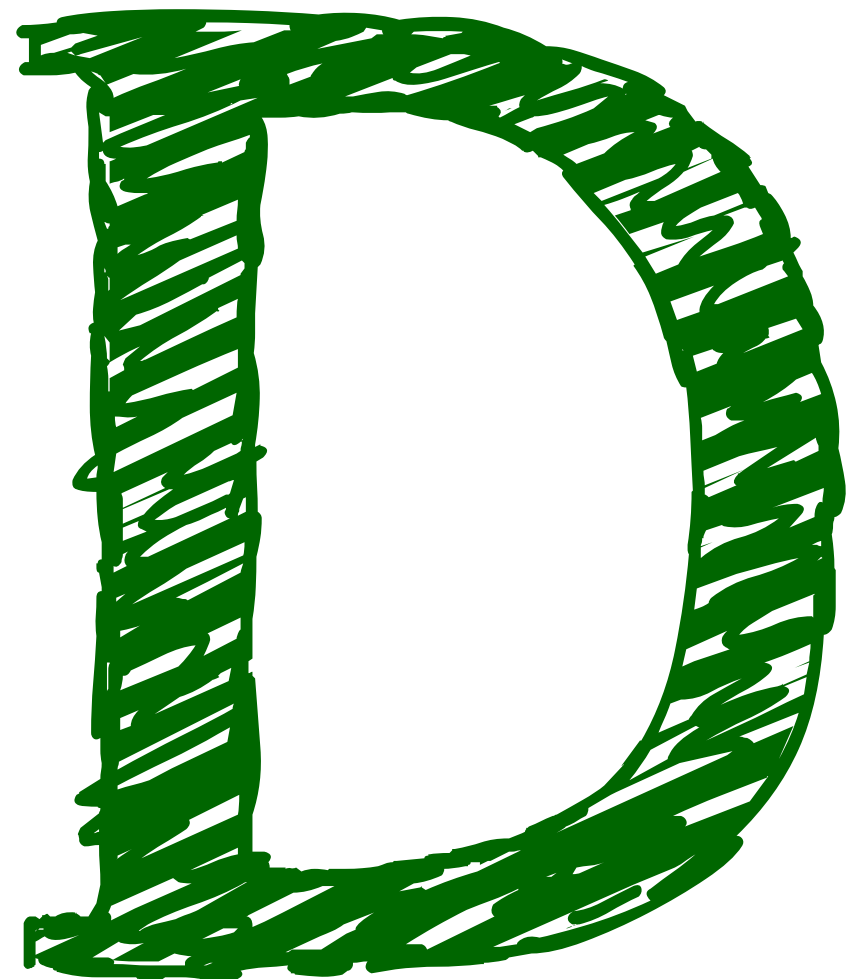
6. Because it will adversely affect the population's food supplies, the most DISADVANTAGEOUS source of energy is_____

A. wind

B. sun

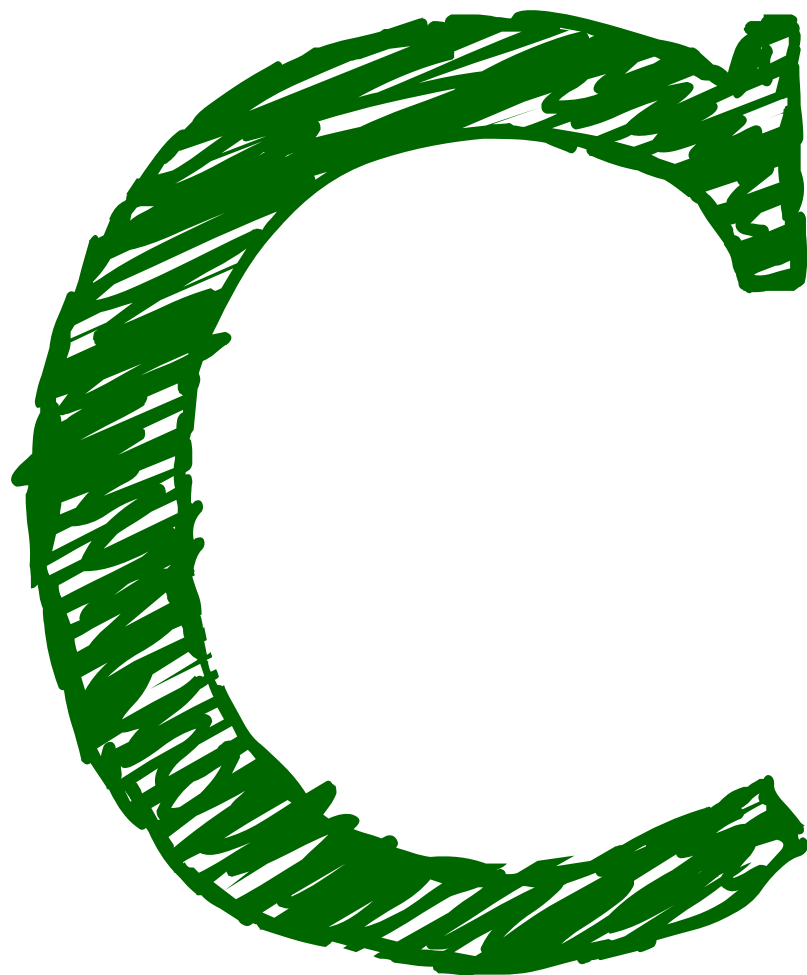
C. geo-thermal

D. plants



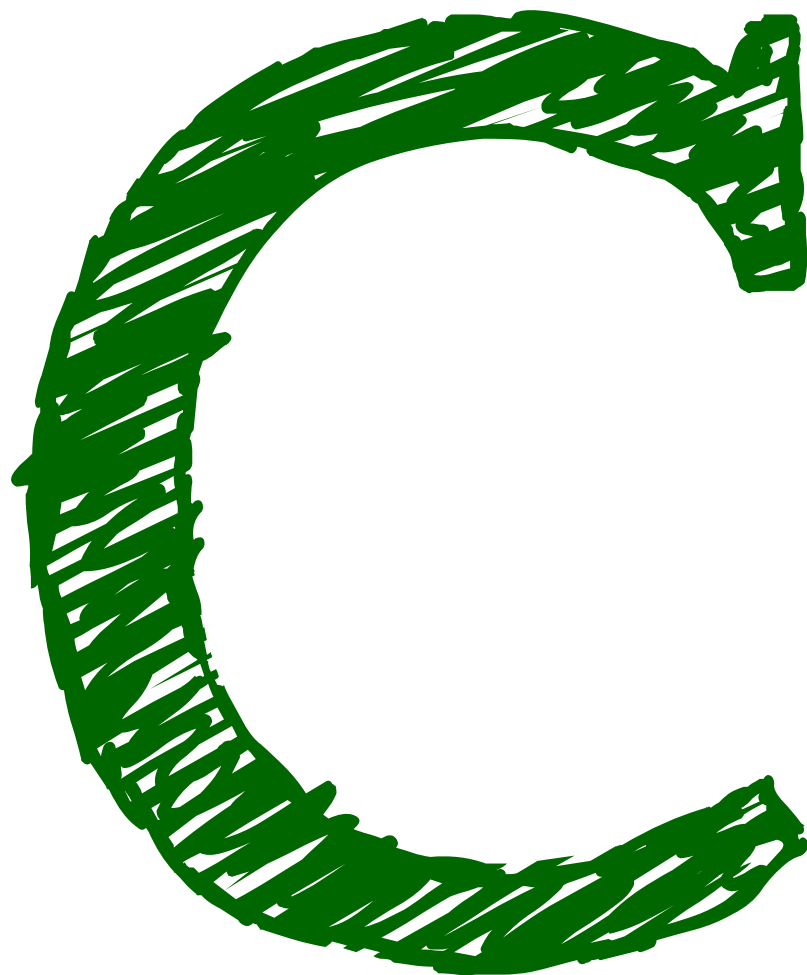
7. Imprisoned for 27 years, Nelson Mandela realized his dream of a free and non-racial south Africa by forgiving his oppressors and negotiating in good faith for the peaceful transition to democracy. Nelson Mandela demonstrated his trait of_____.

- A. friendliness
- B. determination
- C. forgiving
- D. loyalty



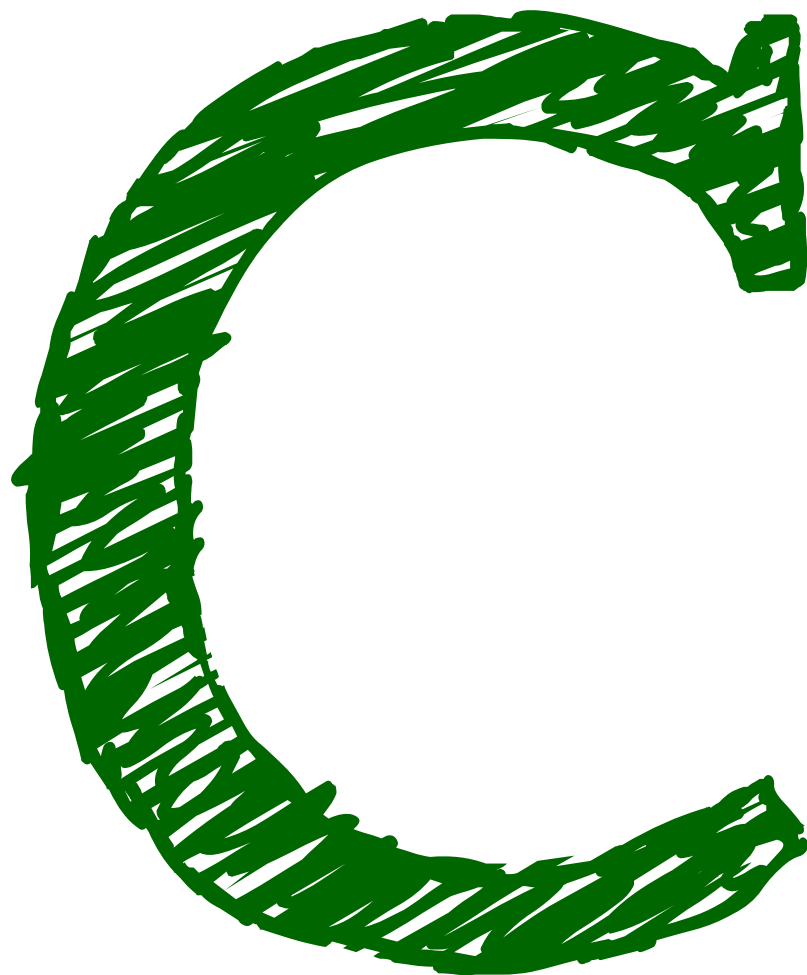
8. What is the figure of speech in the sentence: “the sun was a ball of fire over the mountain”?

- A. Personification
- B. Simile
- C. Metaphor
- D. Hyperbole



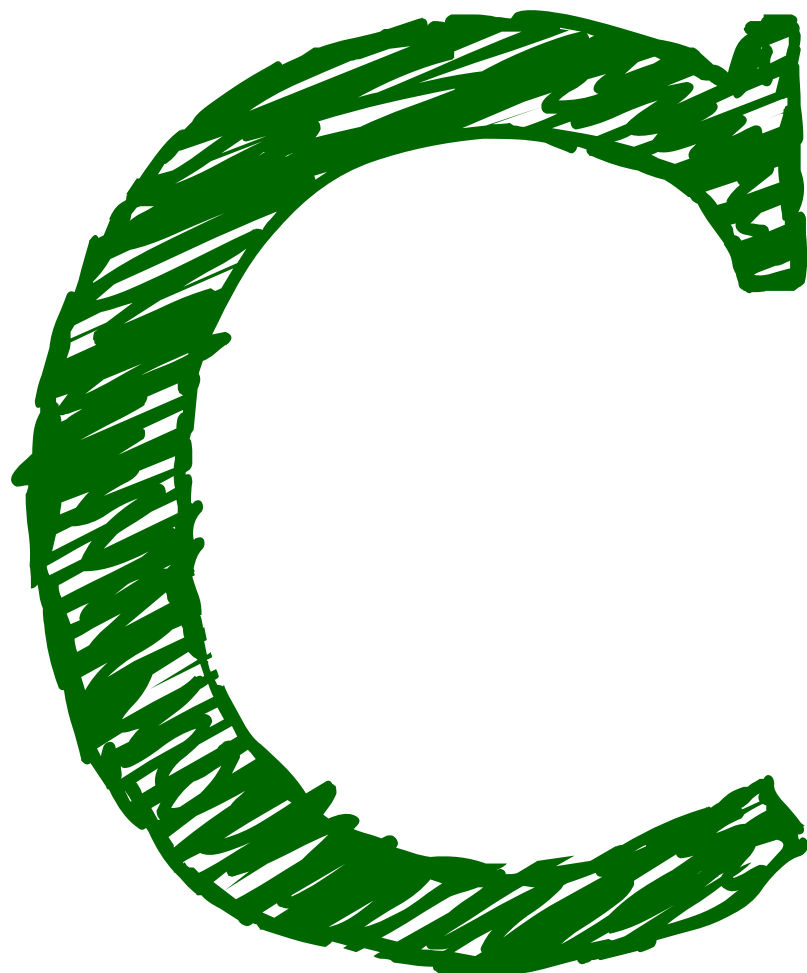
9. Let us campaign for the total ban
_____ of harmful insecticides.

- A. with the use
- B. on using
- C. on the use
- D. with using



10. “The writer’s APOCALYPTIC accounts suggest the end of the world”. The capitalized word means _____.

- A. attractive
- B. careless
- C. prophetic
- D. encouraging



11. “Your statements are IMPERTINENT to the case”. The capitalized word means_____.

- A. important
- B. irrelevant
- C. violent
- D. malicious

B

12. If I _____ known you before, we could have arranged a joint project.

A. have

B. had

C. could

D. will

IF I CAN UNDERSTAND **CONDITIONALS**, I WILL BE HAPPY!

@www.theenglishstudent.com

Likely To Happen Today Or In The Future



If

CONDITION RESULT
I save enough money, I will buy that mansion.
└──────────┘ └──────────┘
PRESENT SIMPLE WILL + VERB

1st
Conditional

Unlikely To Happen Today Or In the Future



If

CONDITION RESULT
I saved a lot of money, I would buy that mansion.
└──────────┘ └──────────┘
SIMPLE PAST WOULD + VERB

2nd
Conditional

Already Passed. Will Never Happen!



If

CONDITION RESULT
I had won the lottery, I would have bought that mansion.
└──────────┘ └──────────┘
HAD+PAST PERFECT WOULD HAVE + PAST PARTICIPLE

3rd
Conditional

B

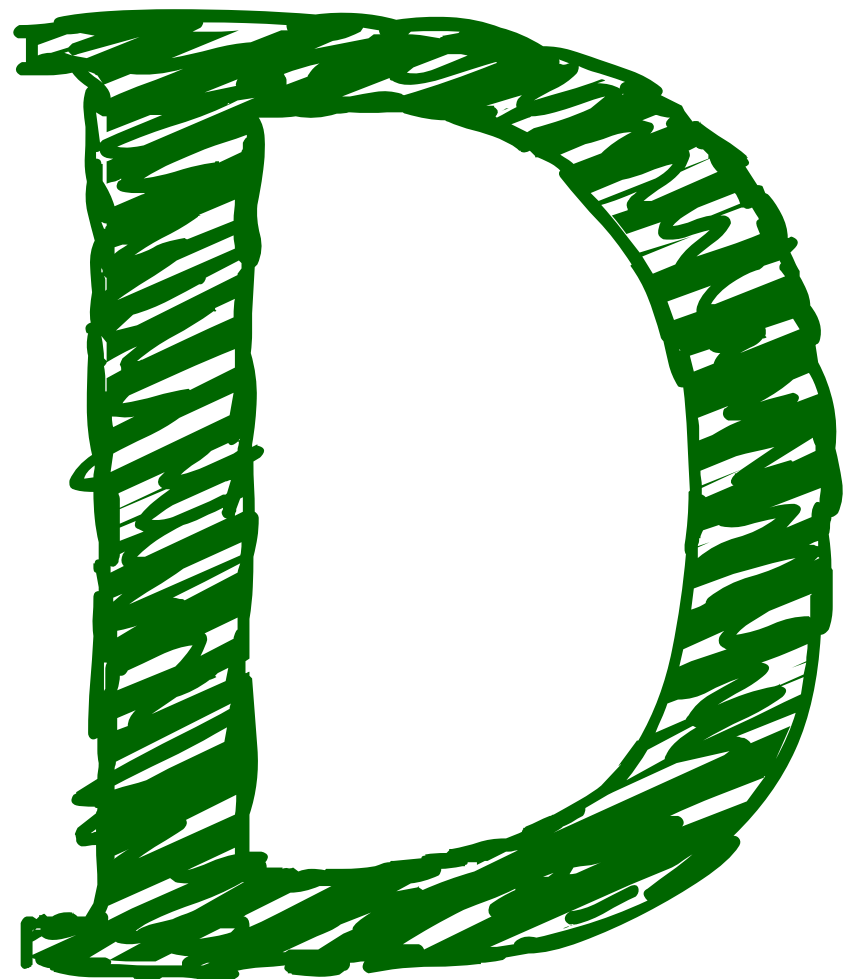
13. The proposed project _____
within the budget of the school.

A. are

B. is appearing

C. are appearing

D. is



14. I thought the task was heavy
for Ikea and _____.

A. me

B. her

C. myself

D. us

Pronoun Chart

	Subject Pronouns	Object Pronouns	Possessive Adjectives	Possessive Pronouns	Reflexive Pronouns
1 st person	I	me	my	mine	myself
2 nd person	you	you	your	yours	yourself
3 rd person (male)	he	him	his	his	himself
3 rd person (female)	she	her	her	hers	herself
3 rd person	it	it	its	(not used)	itself
1 st person (plural)	we	us	our	ours	ourselves
2 nd person (plural)	you	you	your	yours	yourselves
3 rd person (plural)	they	them	their	theirs	themselves

Did you know?

www.GrammarRevolution.com

REFLEXIVE PRONOUNS are objects that refer to the **subject**.

They end in **-self** or **-selves**.

The **boy** kicked **HIMSELF**. *HIMSELF = direct object*

She made **HERSELF** coffee. *HERSELF = indirect object*

You should address the letter to **YOURSELF**. *YOURSELF = object of the preposition*



15. You _____ finished eating
before we arrived.

a. should

b. have been

c. have

d. will have



The Future Perfect Tense is used:

B) with the following time expressions:

Before, till, until, by the time, by, by then, etc.

NOTE!

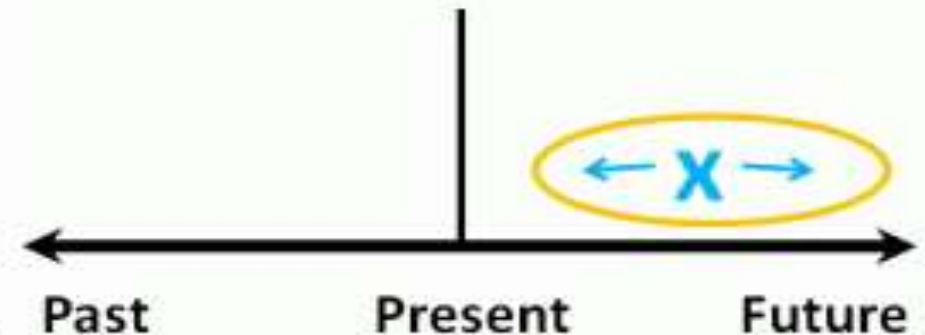
Till/until are mostly used in the negative sentences.

- She will have finished the report **by tomorrow**.
- She won't have completed the report **till/until** 6 o'clock.

Future Perfect

(will) + (have) + (past participle)

(am / is / are) + (going to) + (have) + (past participle)



16. Select the most appropriate for the paragraph:

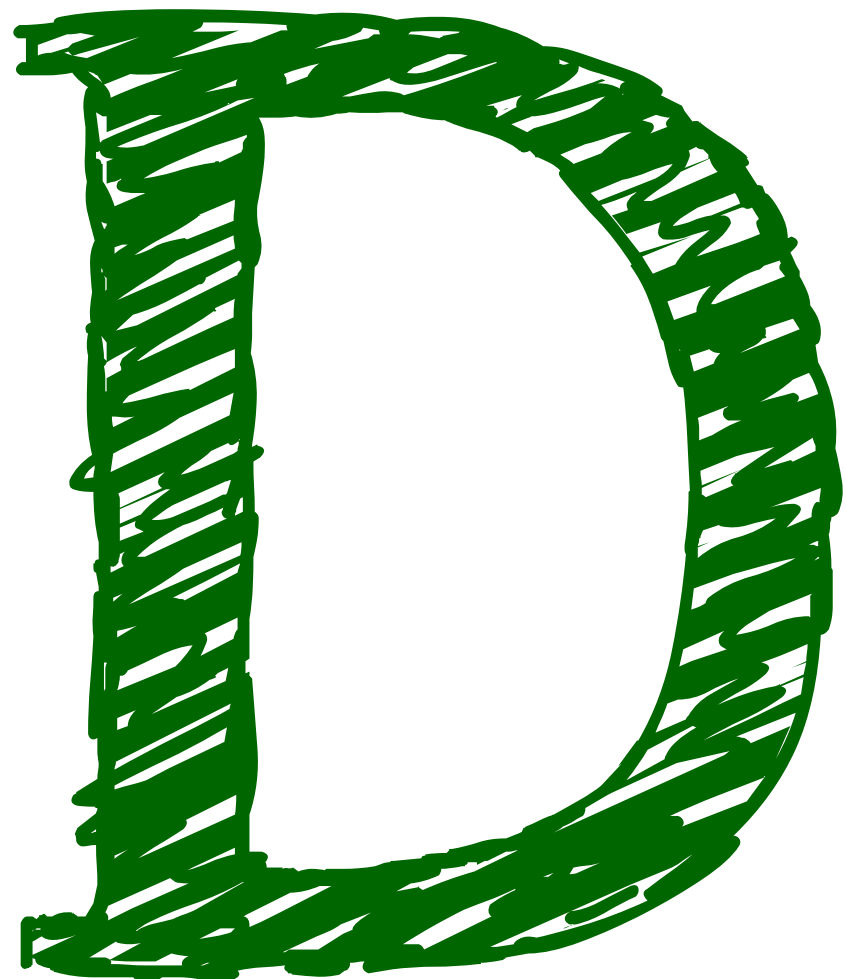
“Stephen Crane was born in 1897 in New Jersey, but his family settled in New York. His father, a Methodist minister, died when Stephen was nine years old. He came to know New York through its slum and street life, particularly in Bowery.

- A. Stephen's family and boyhood
- B. Stephen's genealogy
- C. Stephen as an Orphan
- D. The New York Slums



17. Select the most appropriate heading for the paragraph: Before he has any actual knowledge of war, Stephen Crane wrote the American classic “The Red Badge of Courage,” which made him an overnight success. He wrote “The open Boat” as result of a shipwreck he experienced on his way to Cuba to report a war in progress there”?

- A. A shipwreck
- B. Stephen’s Educational Experience
- C. The Open Boat
- D. Stephen’s Writings



18. The _____ of the story is that honesty is the best policy.

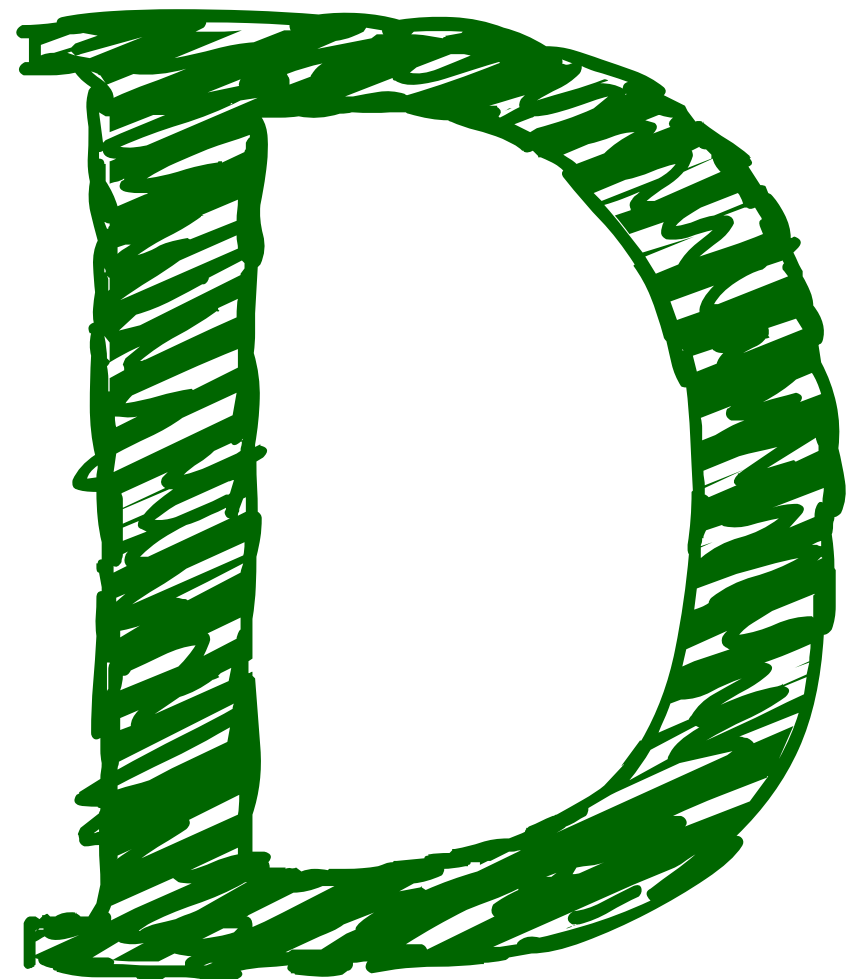
- A. morality
- B. morale
- C. moralism
- D. moral

Morality – the state of right and wrong of an act

Morale - the confidence, enthusiasm, and discipline of a person or group at a particular time.

Moral (adj) concerned with the principles of right and wrong behavior and the goodness or badness of human character.

(n) a lesson, especially one concerning what is right or prudent, that can be derived from a story, a piece of information, or an experience.



19. “Red letter day” is a/an_____ .

A. Idiom

B. Hyperbole

C. simile

D. metaphor

Idiom - is a phrase or a fixed expression that has a figurative, or sometimes literal, meaning.
An idiom's figurative meaning is different from the literal meaning.

Hyperbole is an extreme exaggeration used to make a point. It is like the opposite of “understatement.”

Simile – comparison of two different things using words *such, as, like*

Metaphor – direct comparison of two things



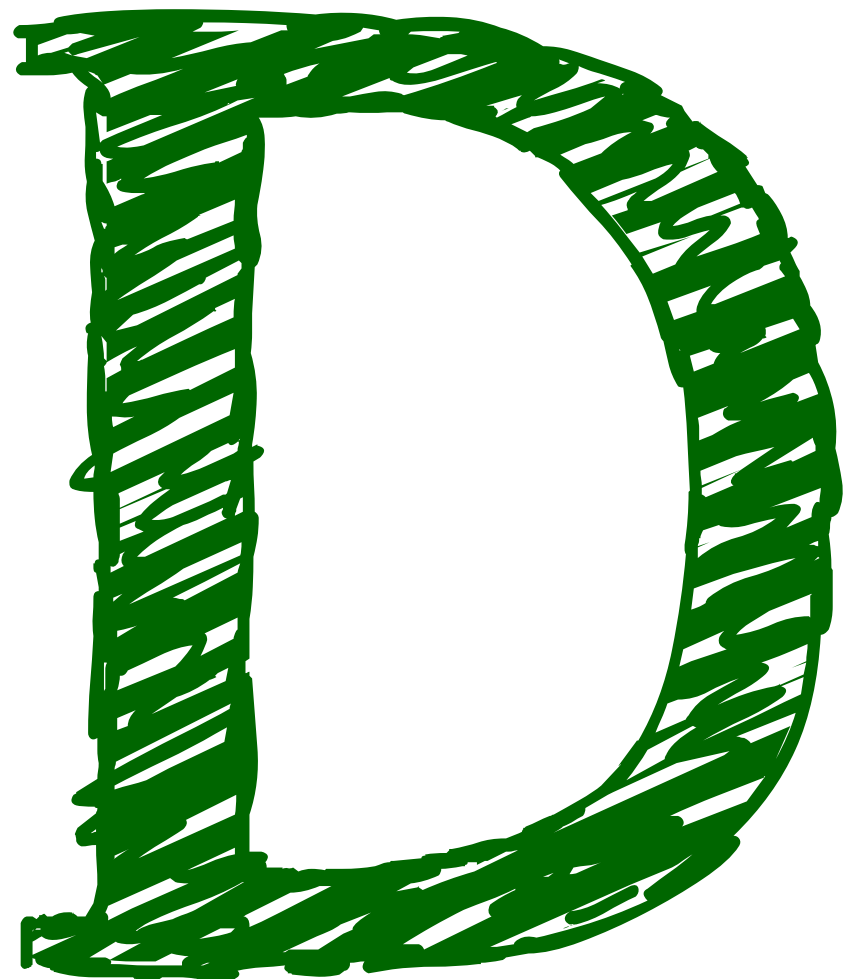
20. For the synecdoche figurative “it was a black tie affair”, what is the equivalent meaning?

- A. Party for men only
- B. Formal gathering
- C. Senior citizen’s meeting
- D. Costume party

B

21. For the synecdoche figurative expression “My mother has a green thumb. I can’t wait to show you her garden.” What is the equivalent meaning?

- A. She grows a vegetable garden
- B. She cultivates a garden
- C. She takes care of flowering plants
- D. She is good at gardening



22. What figure of speech is” “O wild west wind!”

A. Irony

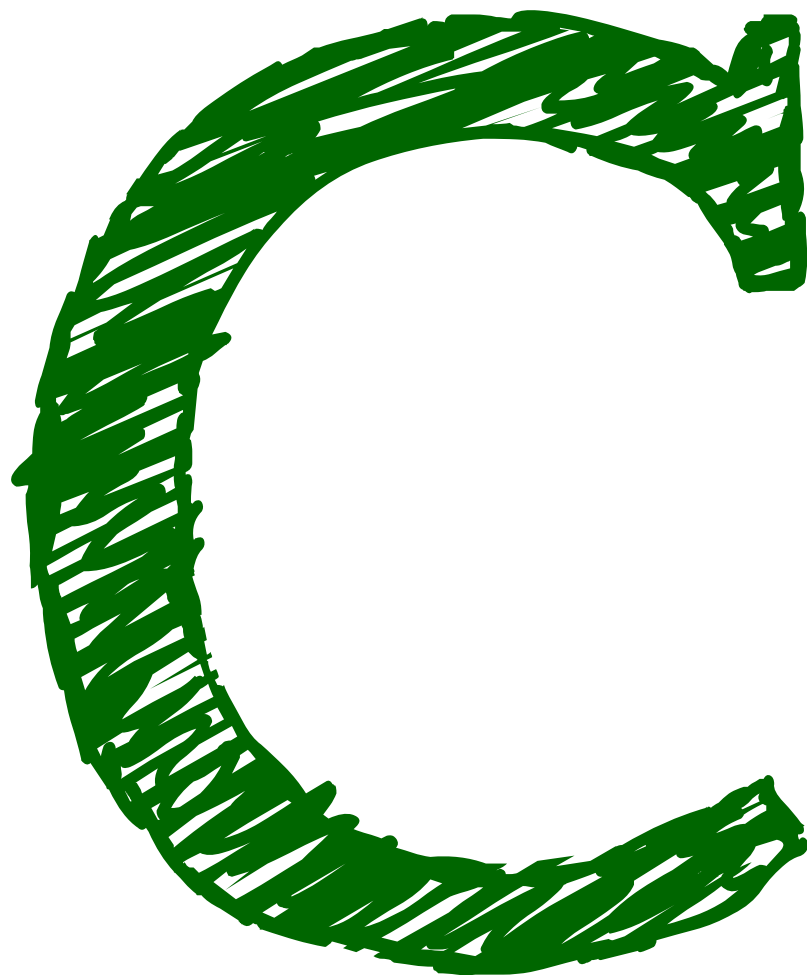
B. Metaphor

C. Apostrophe

D. Hyperbole

Irony – is the expression of one's meaning by using language that normally signifies the opposite, typically for humorous or emphatic effect.

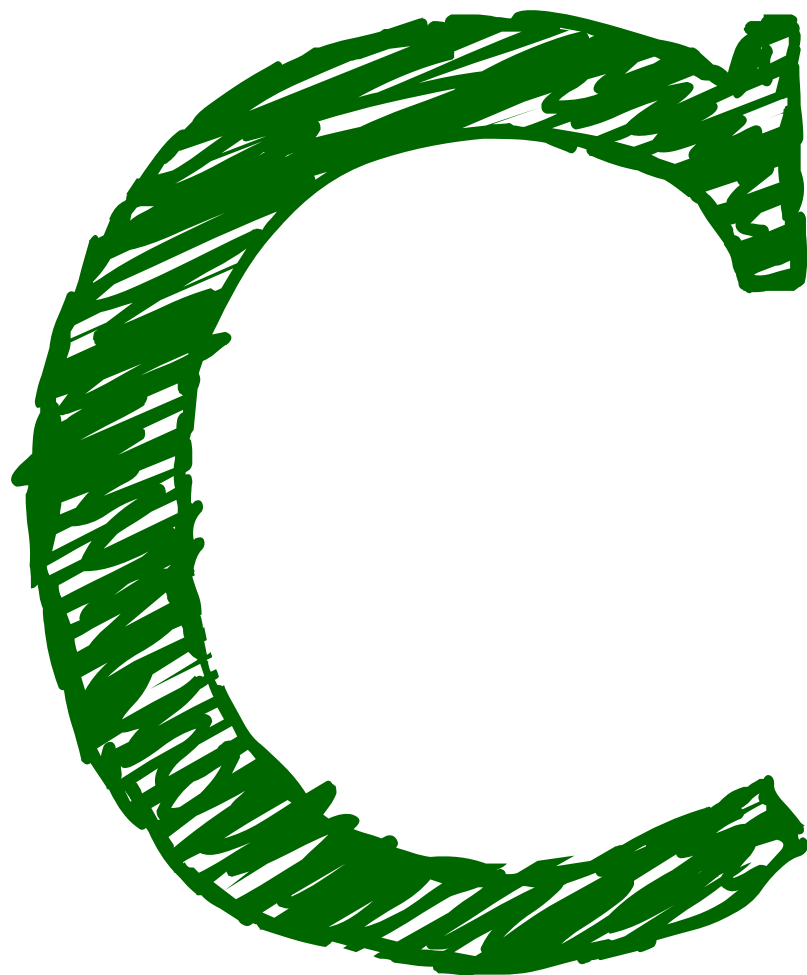
Apostrophe - is a figure of speech sometimes represented by exclamation “O”. A writer or a speaker, using an apostrophe, detaches himself from the reality and addresses an imaginary character in his speech.



23. What role of the women is displayed by the lines in Luis Dato's Spouse: "She holds no joys beyond the day's tomorrow, finds no world beyond her love's embrace, looks upon the form behind the furrow, who is her mind, motion, time and space?"

- A. Work partner
- B. Lady entrepreneur
- C. House wife
- D. Career woman

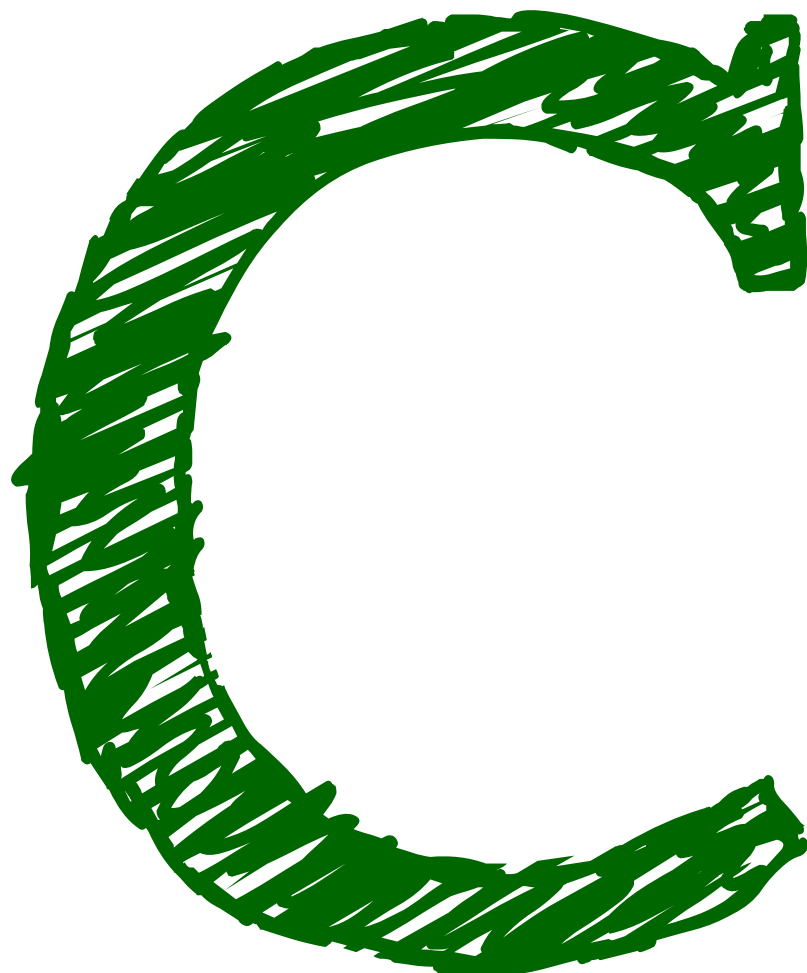
Both man and woman in this poem suffer the curse spoken by God to Adam and Eve. The woman suffers because she is a prisoner and a slave to the desire and devotion that she has for her husband as described in the third stanza. She builds her world around her husband and by doing so, has also set her own limitations in that she eventually refuses any joy that life could have further offered her if only she would, for once, explore the outer world beyond her self-made world with her husband. Because of this, she only looks up on her husband to satisfy her, an expectation that is only bound to be disappointed, also because of the curse upon the man.



24. What became prophetically true in France when Edwin Markham saw the French peasant as “The man with the Hoe,” a plundered slave under the rule of kings and princes?

- A. French revolution
- B. Advent of mechanized agriculture
- C. Cult of Reason in France
- D. Napoleon’s imperial rule

- **Cult of Reason** was a belief system established in France and intended as a replacement for Christianity during the French Revolution
- The poem “The Man with the Hoe,” by Edwin Markham, was the poet’s comment on labor after observing the painting in which a French peasant toils in the fields. The laborer serves as a symbol of a hard deprived life as mankind toils in the field. The man receives no reward for his labor nor does he have much time to rest for being back in the fields. For the man, life is mundane and repetitive. He is a prisoner in the fields by something corrupt that uses him like a ox. The man, who was created in God’s image, falls short he believes of what God had expected for mankind



25. Filipino writer whose stories and poems depict Filipino-Spanish culture beliefs and traditions

- A. Edilberto Tiempo
- B. Bienvenido Santos
- C. N.V.M. Gonzales
- D. Nick Joaquin



- **Edilberto Tiempo** - was a Filipino writer and professor. He established a tradition in excellence in creative writing and the teaching of literacy craft which continues to this day



- **Bienvenido Santos** -as a Filipino-American fiction, poetry and nonfiction writer. He is widely credited as a pioneering Asian – American writer.



- **N. V. M. Gonzales** -was a Filipino novelist, short story writer, essayist and poet. Conferred as the National Artist of the Philippines for Literature in 1997.



- **Nick Joaquin** - was well known as a historian of the brief Golden Age of Spain in the Philippines, as a writer of short stories suffused with folk Roman Catholicism, as a playwright, and as a novelist.



26. “To be or not to be” is the beginning of the soliloquy of _____.

A. Richard III

B. Julius Ceasar

C. Macbeth

D. Hamlet



HAMLET'S SOLILOQUY.

SHAKESPEARE.

To be, or not to be, that is the question : —
Whether 'tis nobler in the mind to suffer
The slings and arrows of outrageous fortune,
Or to take arms against a sea of troubles,
And, by opposing, end them. To die — to sleep ;
No more ; and, by a sleep, to say we end
The heart-ache, and the thousand natural shocks
That flesh is heir to, — 'tis a consummation
Devoutly to be wished. To die — to sleep ;
To sleep ! perchance to dream : ay, there's the rub ;
For in that sleep of death what dreams may come,
When we have shuffled off this mortal coil,
Must give us pause.

27. Kailan nagsimula ang pagkakaroon ng modernisasyon ng wikang pambansa?

A. 1976

B. 1997

C. 1987

D. 1988



Modernisasyon ng Alpabeto ng Wikang Pambansa

- Ang dating 20 letra ay naging 31 na (c, ch, f, j, ll, n, q, rr, u, x, z)
- sa pamamagitan ng DECS Blg. 194 (1976)

28. Ang mga salitang teka, saan, tena, dali ay nagtataglay ng _____.

A. asimilasyon

B. metatesis

C. pagkakaltas

D. tono

Asimilasyon – ang tunog ng isang bahagi ng salita ay itinutulad sa katabi nitong tunog. Hal. Sing+dalas=sindalas; pang+bilang=pambilang = pamilang

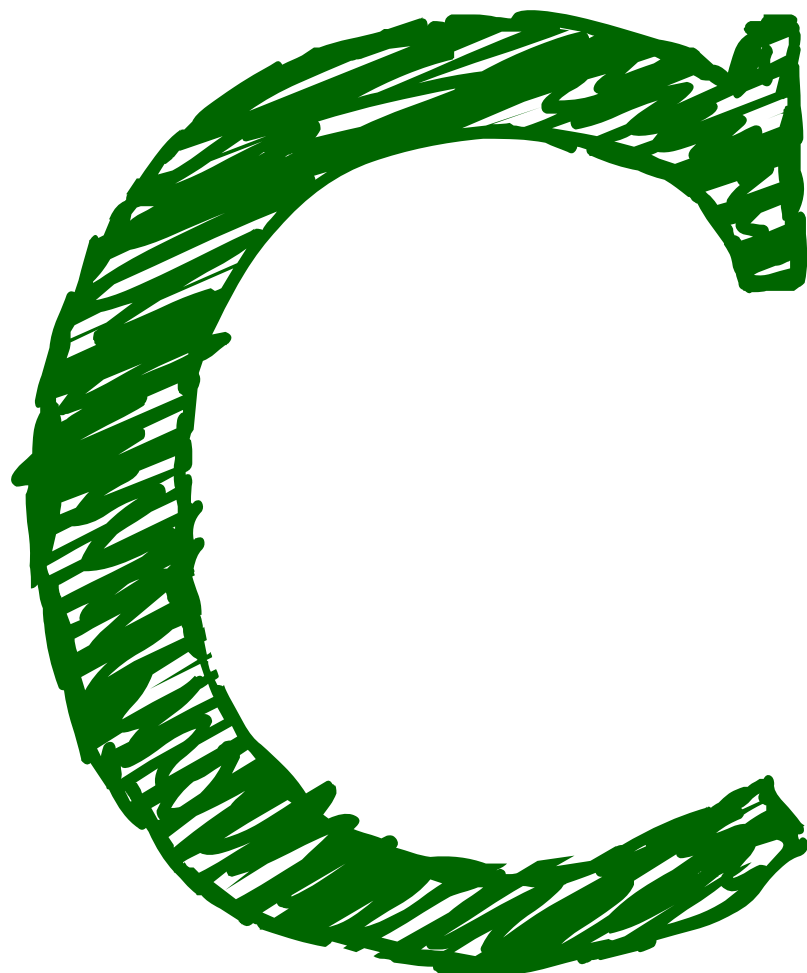
Metatesis- kapag ang salitang ugat ay nagsisimula sa /l/ o /y/ ay ginitlapihan ng [-n], ang /l/ o /y/ ng salitang ugat at ang /n/ ng gitlapi ay nagkakapalit ng posisyon Hal. –in + lipad = linipad = nilipad

Pagkakaltas ng Ponema – may nawawalang ponema sa loob ng salita
Hal. Takip+an=takipan=takpan; dala+han=dalahan = dalhan

Pagpapalit ng Ponema – may mga pagbabago o napapalitang ponema sa pagbubuo ng salita Hal. Ma+dami = madami = marami

Paglilipat-diin – may mga salitang nagbabago ng diin kapag nilalapian. Hal. Basa+hin = basahin

Pag-aangkop – pagsasama ng dalawang salita. May pagkakaltas pa ring kasama ito. Hal. Wika + ko = ikako/kako



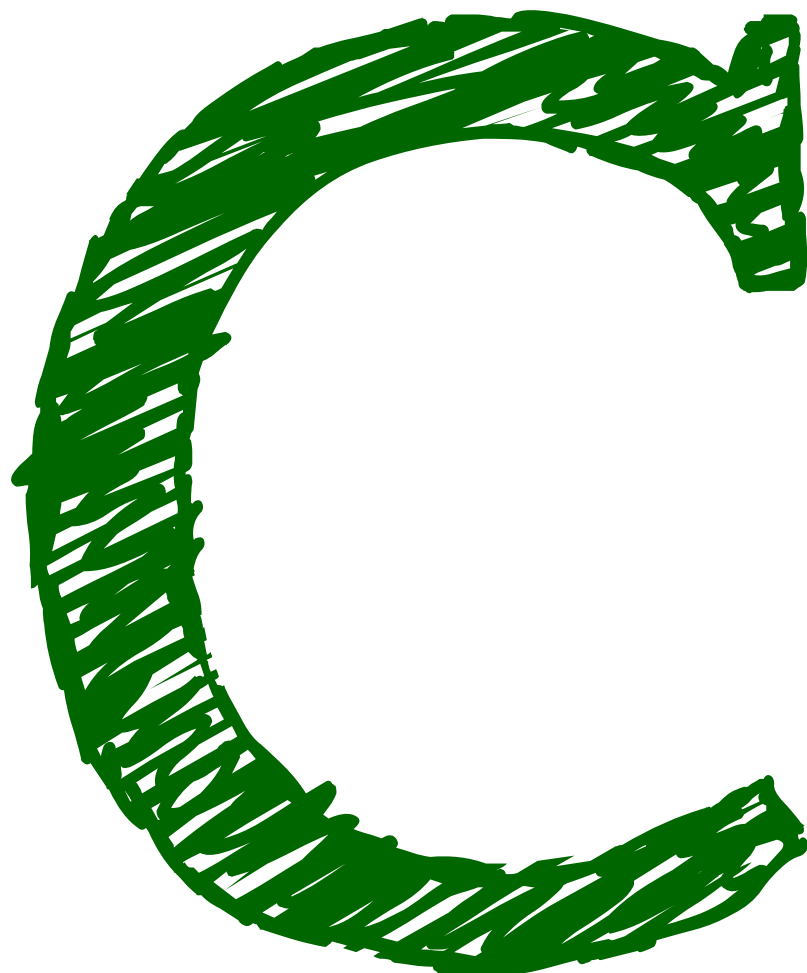
29. Ano ang tumutukoy sa bilang ng pantig sa bawat taludtod ng tula?

A. Diin

B. Tugma

C. Sukat

D. Talinghaga





Elemento ng Tula

Sukat-tumutukoy sa sukat ng taludtod sa 1 saknong.

saknong-grupo ng mga salita sa isang tula.

Tugma-pagkakaparihas ng tunog sa huling bahagi ng isang taludtod.

Karikatan-maririkit na salita na ginagamit upang masiyahan ang mga mambabasa.

Talinhaga-natatagong kahulugan ng tula.

30. Tumutulong ito sa mga salitang nakapag-iisa at may kahulugan. Kilala rin itong salitang-ugat.

A. Panlapi

B. Morpema

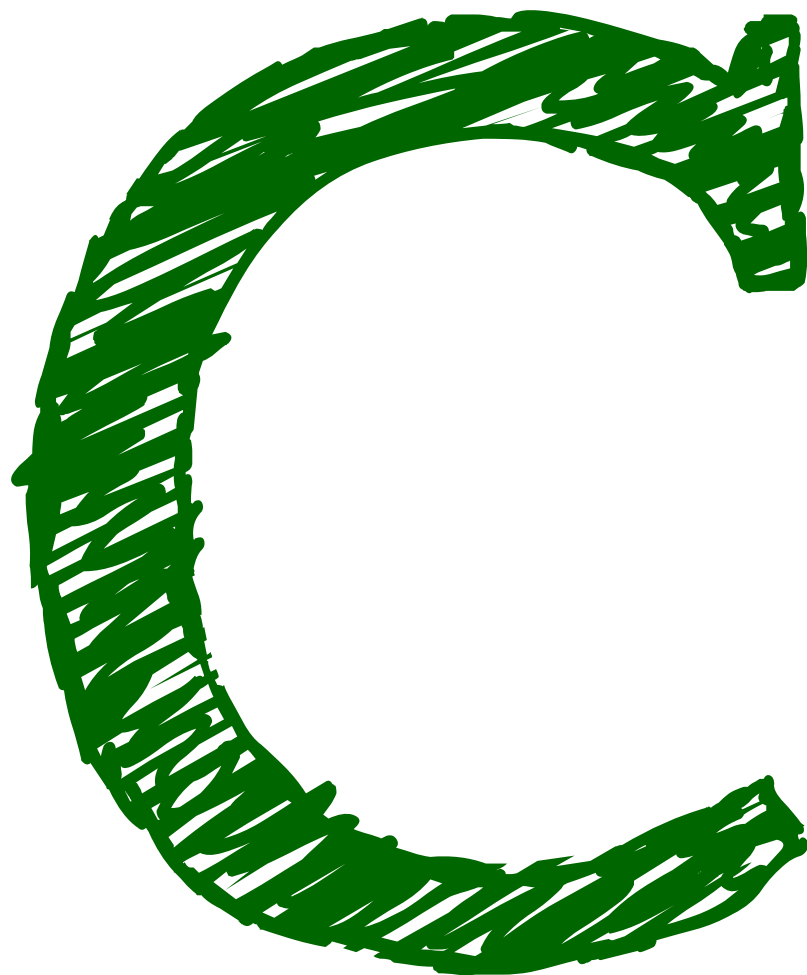
C. Talastasan

D. Dayalek

B

31. Ang pagpapalitan ng mga ideya, opinion, salaysay sa pamamagitan ng mga sagisag at tinatawag na _____.

- A. Pagtuklas
- B. Paglalahad
- C. Talastasan
- D. Pakikinig



32. Wikang nabuo mula sa pangunahing wika ng isang lugar o lalawigan na kadalasang sinasalita sa ibang bayan ng naturang lugar.

A. Etnolek

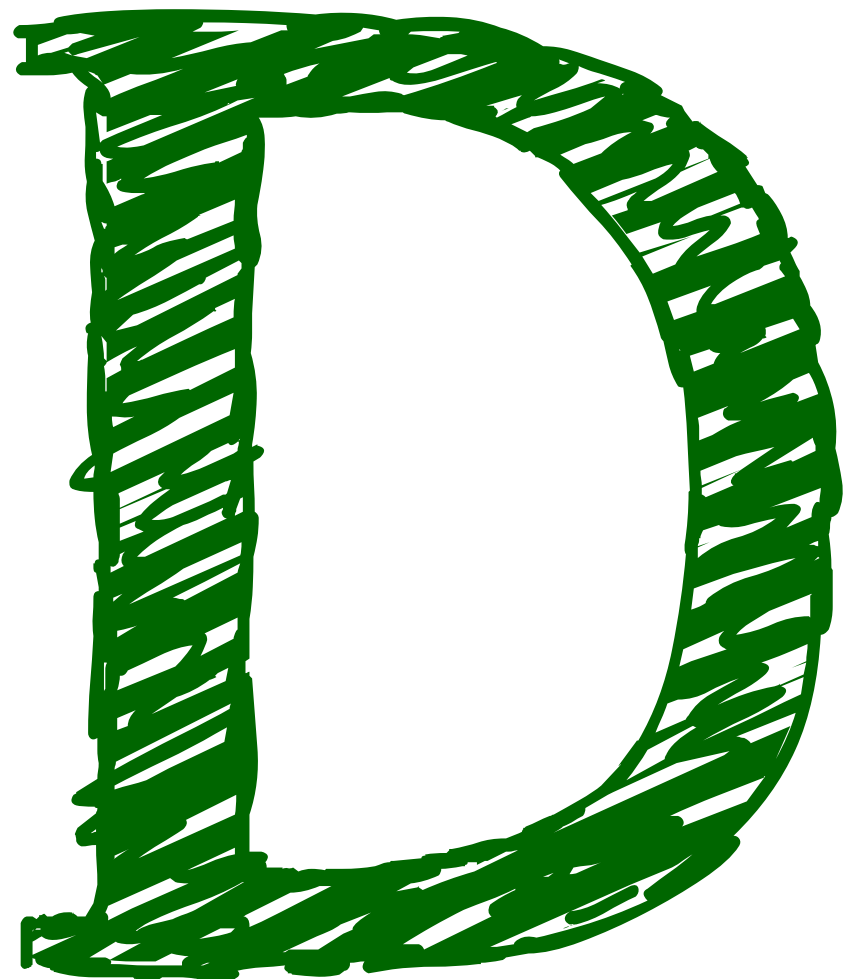
B. Ekolek

C. Sosyolek

D. Dayalek

Varayti ng Wika

- Etnolek – wika ng etnolinggwistikong grupo
- Ekolek – wika sa bahay
- Sosyolek – wikang batay sa katayuan o status sa lipunan
- Dayalek – wikang pekulyar o katangi-tangi sa isang lugar o rehiyon, kasama ang punto, bokabularyo o pagkakabuo
- Idyolek – nakagawiang pamamaraan sa pagsasalita ng isang indibidwal o ng isang pangkat
- Register – wika batay sa uri at paksa ng talakayan, tagapakinig o okasyon



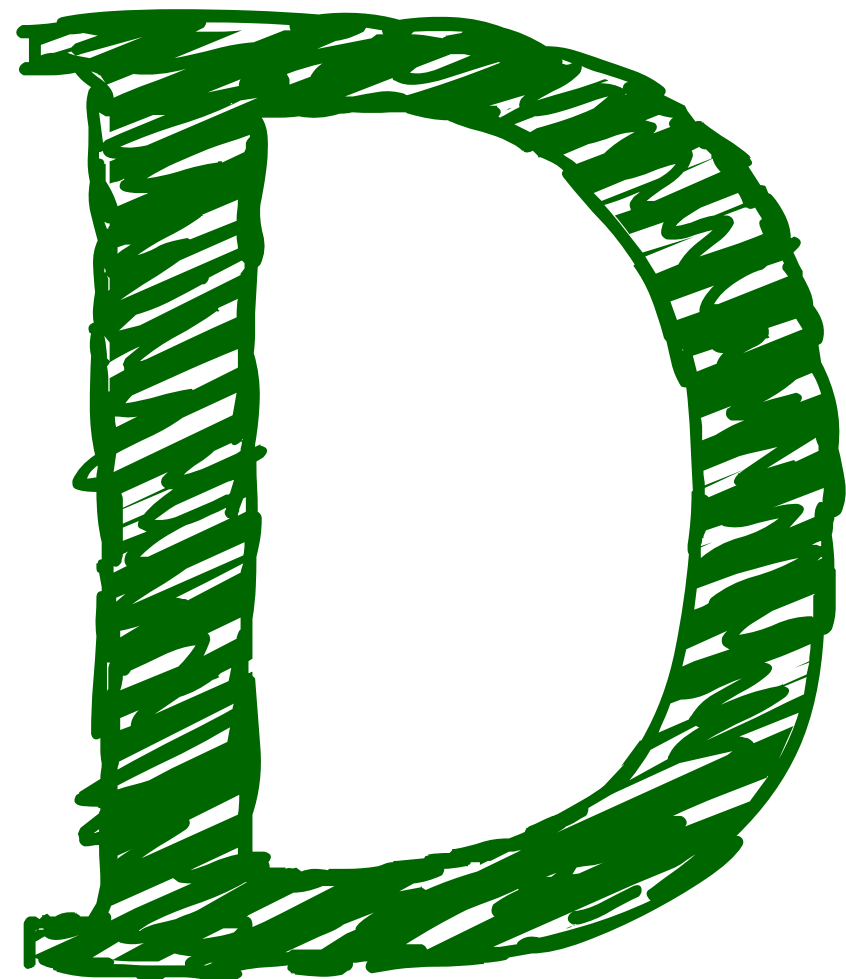
33. Alin sa mga sumusunod ang nararapat sa komunikasyon na pasulat?

- A. Lakas ng boses
- B. Maayos na pagpapalugit
- C. Maliksing mga mata
- D. Pagkibit ng balikat



34. Ang simbolong kumakatawan sa mga bagay at pangyayaring nais ipahayag ng tao sa kanyang kapwa ay _____.

- A. tunog
- B. sining
- C. bokabularyo
- D. wika



35. Ano ang kalabang mortal ng pakikinig?

A. Talinghaga

B. Ingay

C. Okasyon

D. Oras



36. Ano ang pinakapayak na anyo ng salita na walang kahalong panglapi?

- A. Ponema
- B. Laguhan
- C. Gitlapi
- D. Salitang ugat



2a. UNLAPI: panlaping kinakabit sa unahan ng salita.

2b. GITLAPI: panlaping nasa gitna ng salita.

2c. HULAPI: panlaping ikinakabit sa hulihan ng salita.

2d. KABILAAN: panlaping ikinakabit sa unahan at hulihan ng salita.

2e. LAGUHAN: panlaping ikinakabit sa unahan, gitna at hulihan ng salita.

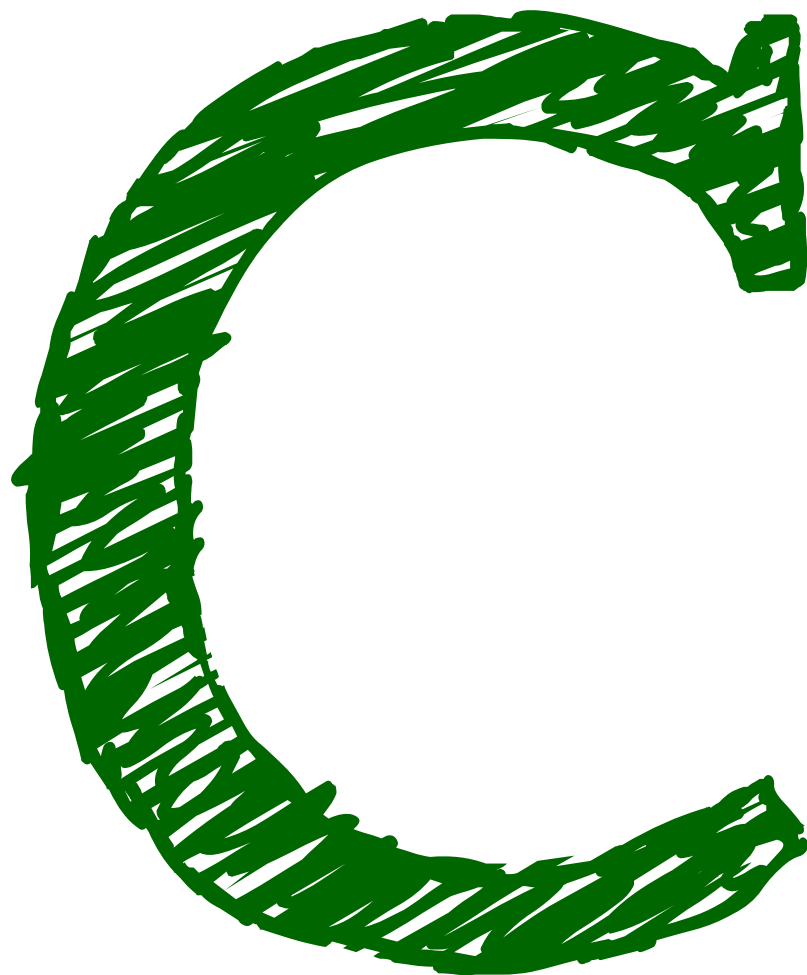
37. Sa pangungusap na “Malakas ang boses mo,”anong bahagi ng pangungusap ang salitang “malakas?”

A. Pandiwa

B. Pangatnig

C. Pang-uri

D. Panghalip



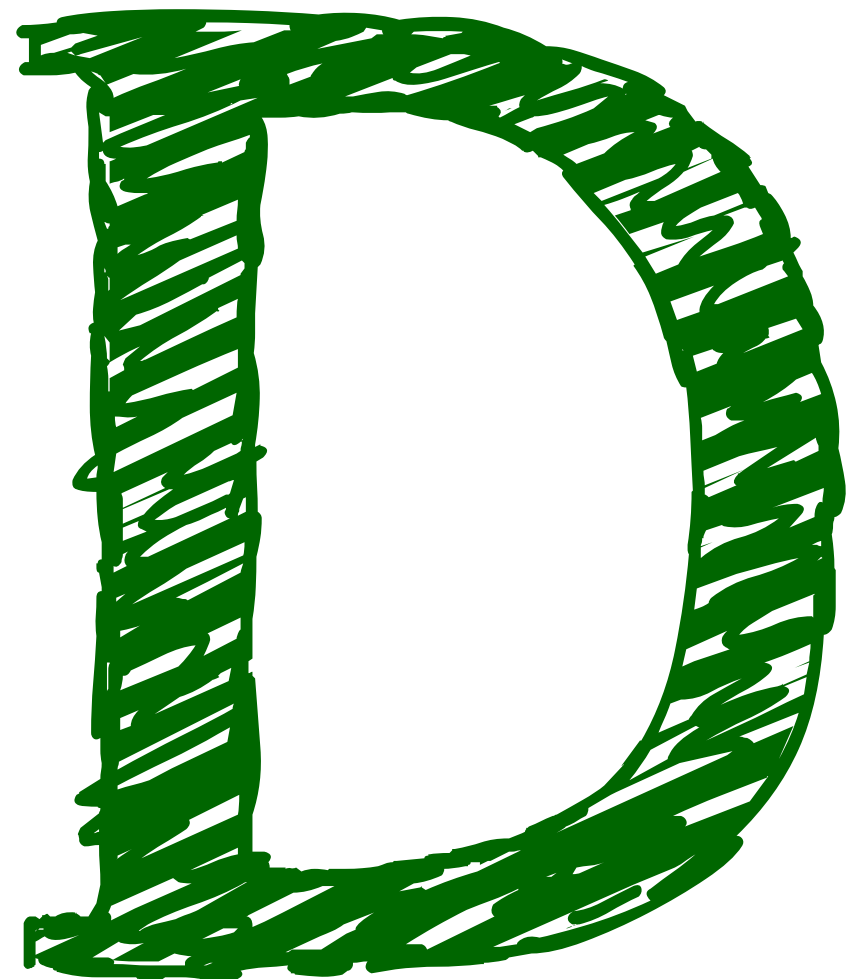
38. Ang katumbas na bigkas ng titik Q sa kasalukuyan ay: _____.

A. ku

B. kuya

C. kuyu

D. kyu



39. Bahaging pananaliksik kung saan matatagpuan ang panimula, kaligiran ng pag-aaral at konseptuwal framework.

- a. Kabanata II
- b. Kabanata I
- c. Kabanata IV
- d. Kabanata III

B

40. Anong paraan ng pagkuha ng balangkas ang ginagamitan ng sunud-sunod na tatlong tuldok para ipakita na may bahaging hindi na sinipi sa talata?

- A. Sinopsis
- B. Elipsis
- C. Abstrak
- D. Sintesis

B

41. Pahayag na pasaklaw na nabubuo sa pamamagitan ng personal na panlasa o pagpili.

- A. Probable generalization
- B. Subjective generalization
- C. Analytical generalization
- D. Categorical generalization

B

42. Dulong pampanitikan na nagbibigay ng diin ng sariling panlasa ng bumabasa. Kilala rin ito bilang reader-response theory.

- A. Antropolohiya
- B. Impresyonista
- C. Patalambuhay
- D. Pansikolohiya

B

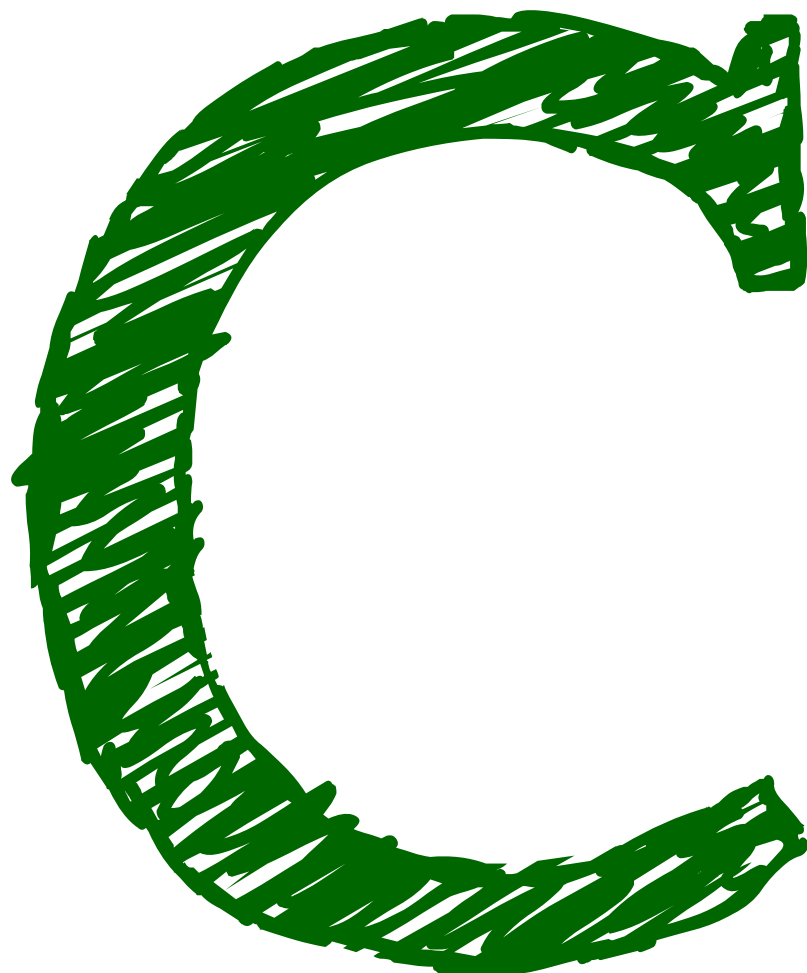
43. Layunin ng tagapagsalita na may malakas na “sense of humor”

- A. manghikayat
- B. manlibang
- C. magpatibay
- D. magkwento

B

44. Proseso ng paghahatid ng saloobin,
opinion, karunungan sa pamamagitan ng
makabuluhang tunog

- A. pakikinig
- B. pagbabasa
- C. pagsasalita
- D. pagtatala



45. Nakikipag-away ka sa speaker.
Ito ay pakikinig na _____ .

A. masusi

B. pasibo

C. kombatib

D. may lugod



46. Paraan ng pagbasa na ginagamit kung ang akda ay mahirap unawain:

- A. pre-view
- B. muling-basa
- C. masusi
- D. kaswal

B

47. Simplify: $x - 3y(x - 8y) - (-6xy + 7x)$

A. $-6x + 3xy + 24y^2$

B. $6x - 3xy - 24y^2$

C. $-6x + 3xy$

D. $-6x - 3xy$

Simplify: $x - 3y(x - 8y) - (-6xy + 7x)$

$$x - 3xy + 24y^2 + 6xy - 7x$$

$$x - 7x - 3xy + 6xy + 24y^2$$

$$**-6x + 3xy + 24y^2**$$



48. A recipe calls for 2 cups of milk for every 7 cups of flour. A chef will use 28 cups of flour. How many cups of milk must he have?

A. 10

B. 8

C. 12

D. 14

A recipe calls for 2 cups of milk for every 7 cups of flour. A chef will use 28 cups of flour. How many cups of milk must he have?

Milk : Flour

$$2 : 7 = X : 28$$

$$\frac{7X : 56}{7 \quad 7}$$

$$X = 8$$



49. How many seconds are there in a 24-hour day?

A. 86,400

B. 1, 690

C. 1, 779

D. 84, 600

49. How many seconds are there in a 24-hour day?

$$24 \text{ Hrs} \times \frac{60 \text{ min}}{1 \text{ Hour}} \times \frac{60 \text{ sec}}{1 \text{ min}}$$

= 86,400 seconds



50. What number must be subtracted from both numerator and denominator of the fraction $\frac{11}{23}$ to give a fraction whose value is $\frac{2}{5}$?

A. 4

B. 3

C. 2

D. 5

$$\frac{11}{23} - \frac{4}{4} = \frac{7}{19}$$

$$\frac{11}{23} - \frac{5}{5} = \frac{6}{18} \text{ or } \frac{1}{3}$$

$$\frac{11}{23} - \frac{3}{3} = \frac{8}{20} \text{ or } \frac{2}{5}$$

$$\frac{11}{23} - \frac{2}{2} = \frac{9}{21} \text{ or } \frac{3}{7}$$



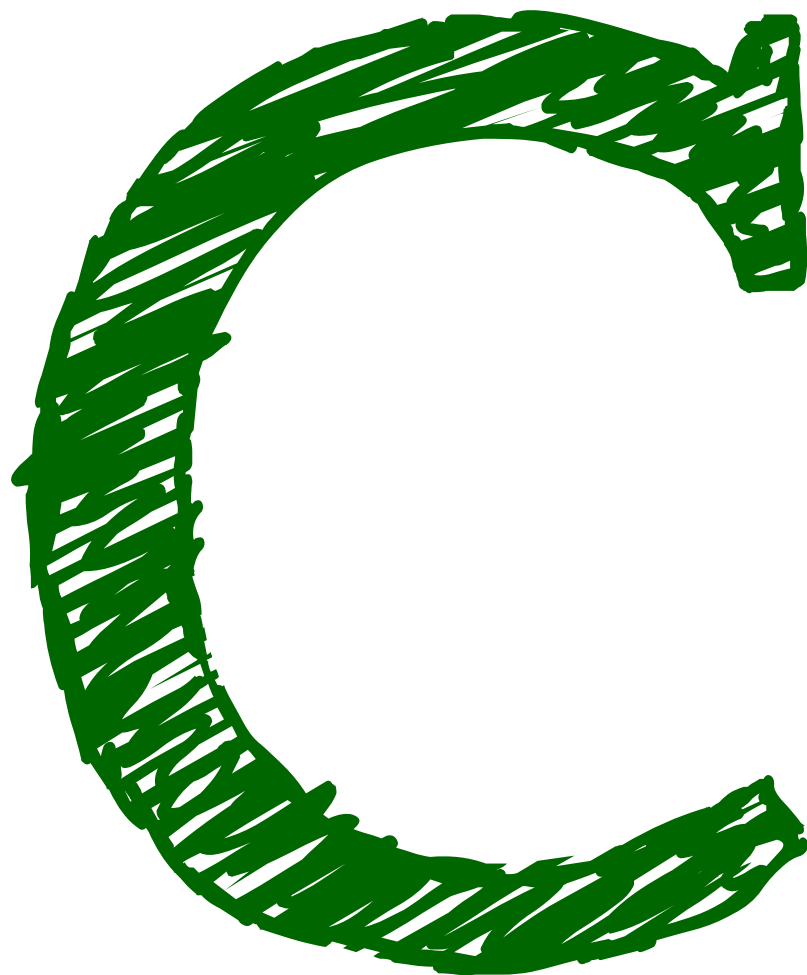
51. Among the given decimals, which is equivalent to 9%?

A. 9.0

B. 0.009

C. 0.09

D. 0.9



52. Simplify: $5 - \{2 - (-4) + 11 - 8\}$

A. 4

B. 9

C. -4

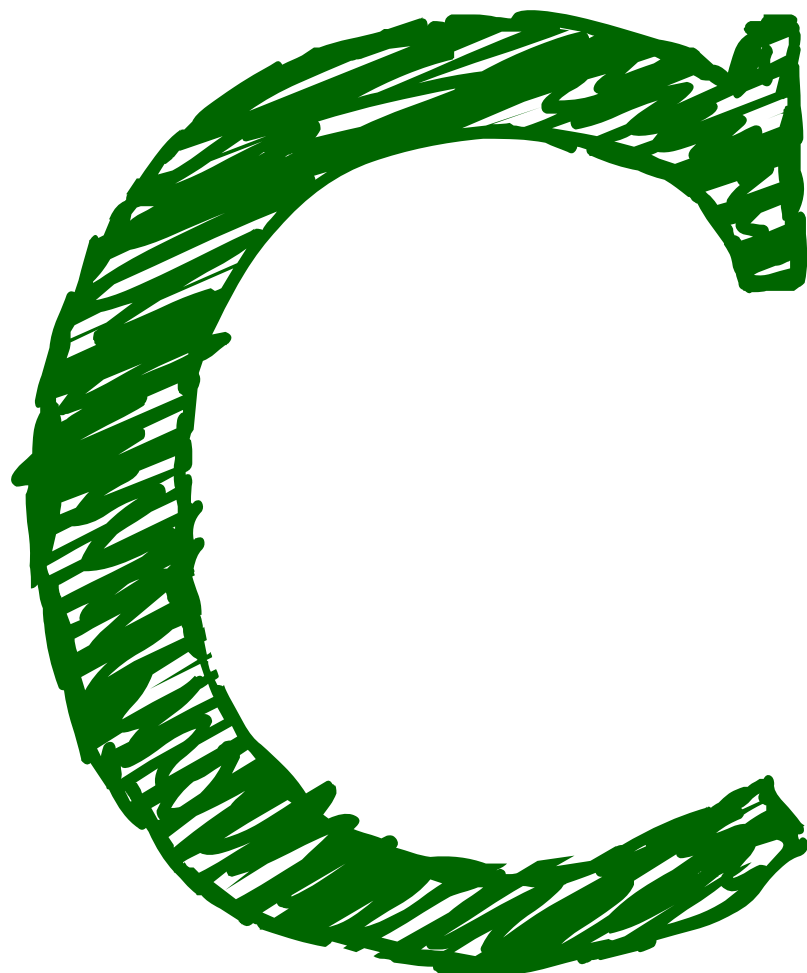
D. 6

52. Simplify: $5 - \{2 - (-4) + 11 - 8\}$

$$5 - \{2 + 4 + 11 - 8\}$$

$$5 - 9$$

$$\textcolor{red}{-4}$$



53. The sum of three consecutive integers is 123. What are the integers?

A. 42, 44, 46

B. 40, 41, 42

C. 39, 40, 41

D. 41, 43, 45

$$x+x+1+x+2 = 123$$

$$3x+3 = 123$$

$$3x = 123-3$$

$$3x = 120$$

$$\frac{3x}{3} = \frac{120}{3}$$

$$x = 40$$

$$x = 40$$

$$x + 1 = 41$$

$$x + 2 = \underline{42}$$

$$123$$

B

54. Total amount after adding 8% interest for 3 months of PhP6,000.00?

A. PhP11,500.00

B. PhP6,120.00

C. PhP10,500.00

D. PhP11,050.00

$$P + (PRT)$$

$$= 6,000 + (6,000 \times 0.08 \times \frac{3}{12})$$

$$= 6,000 + 120$$

$$= \mathbf{6,120}$$

B

55. Subtract $5a-2b$ from the sum of $7a+5b$ and $a+b$

A. $3a+4b$

B. $-3a+4b$

C. $8a+2b$

D. $3a-4b$

$$= \{(7a + 5b) + (a + b)\} - (5a - 2b)$$

$$= \{7a + a + 5b + b\} - 5a + 2b$$

$$= 8a + 6b - 5a + 2b$$

$$= \mathbf{3a + 8b}$$



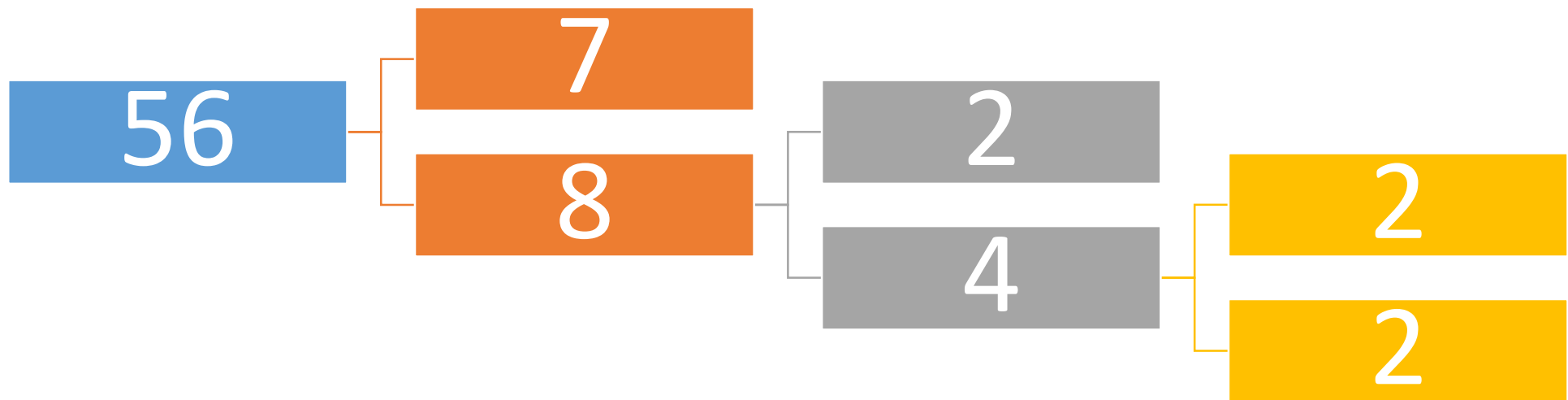
56. Which are the prime factors of 56?

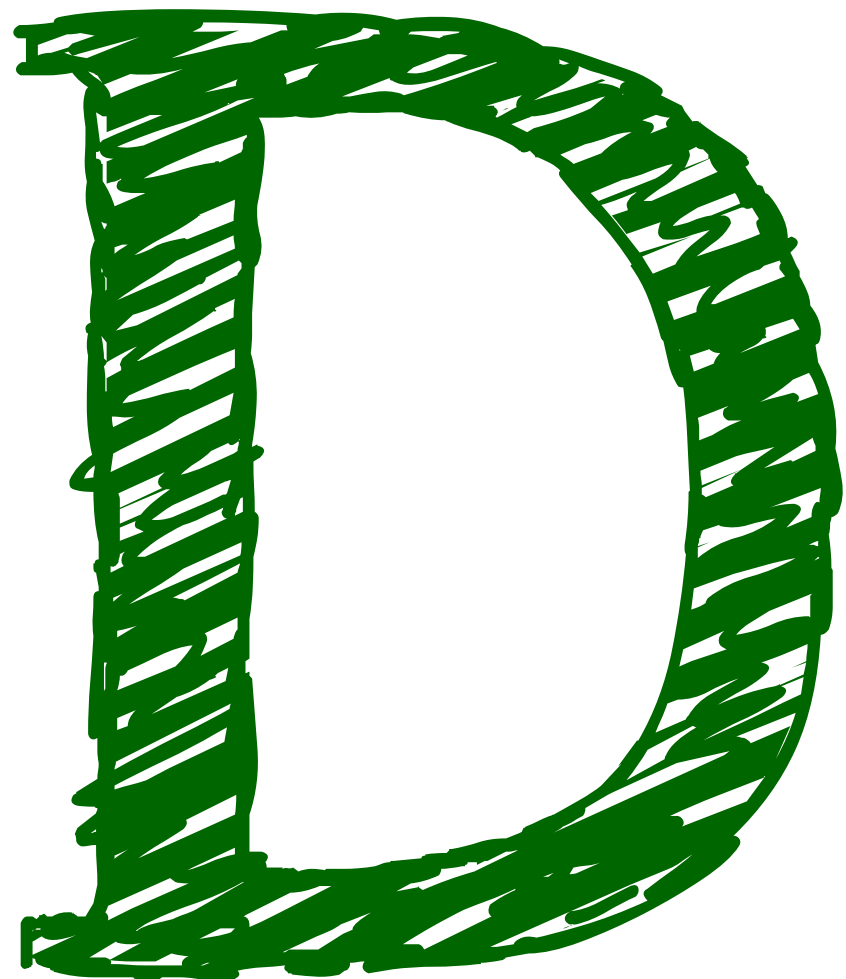
A. 7, 4, 2

B. 14, 2, 2

C. 7, 8

D. 7, 2, 2, 2





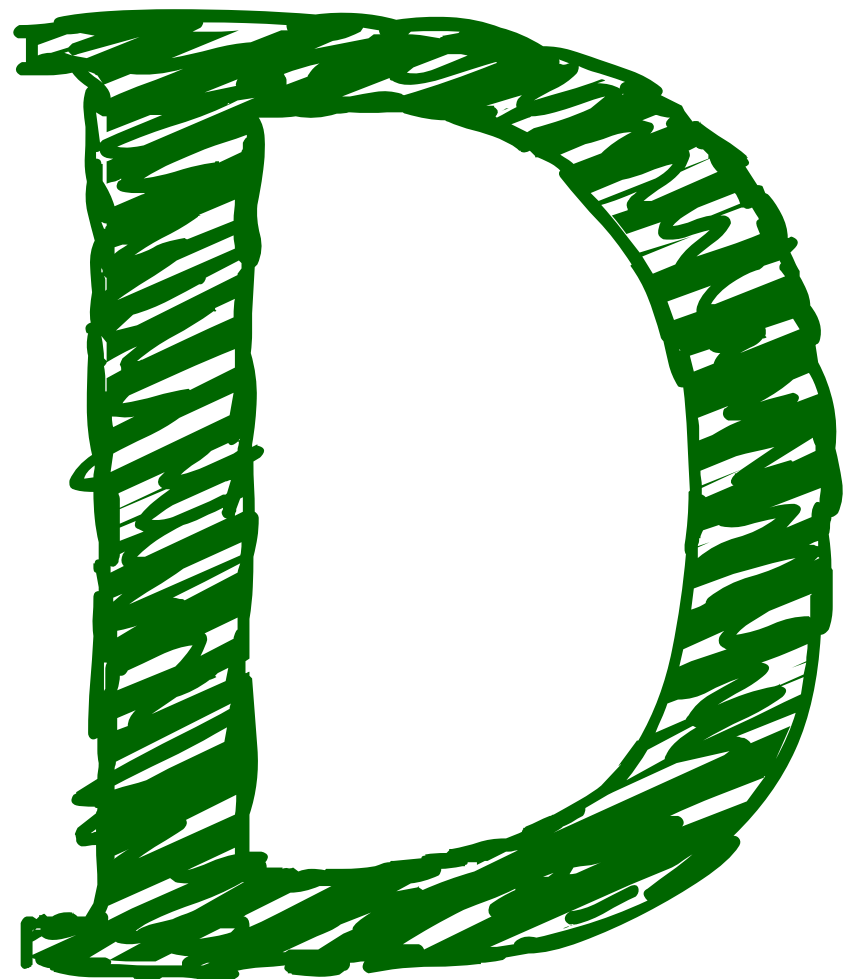
57. Find the least common multiple of $9a^2-1$, $6a^3-2a^2$, $9a+3$

A. $6a^3 (3a+1) (3a-1)$

B. $6a (3a+1)(3a-1)$

C. $6a^3 (3a^2+1) (3a-1)$

D. $6a^2 (3a+1)(3a-1)$



58. If 50% of x is 20, what is 20% of x ?

A. 16

B. 27

C. 8

D. 12

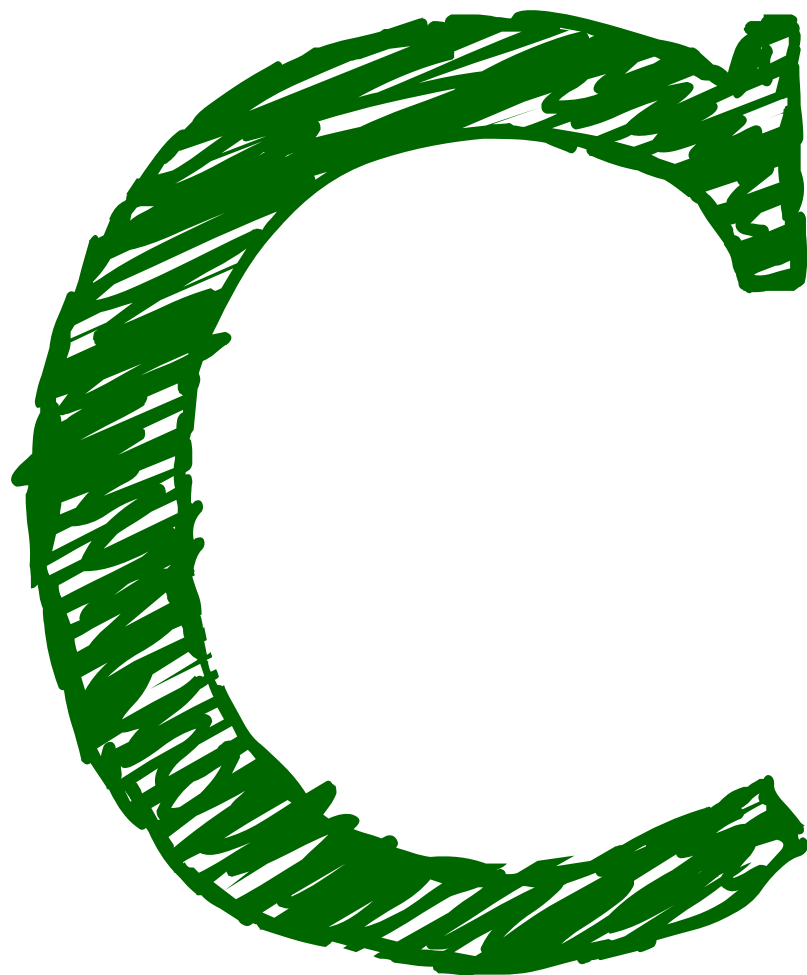
$$.5x = 20$$

$$= 40 \times .20$$

$$\frac{.5x}{.5} = \frac{20}{.5}$$

$$= 8$$

$$X = 40$$



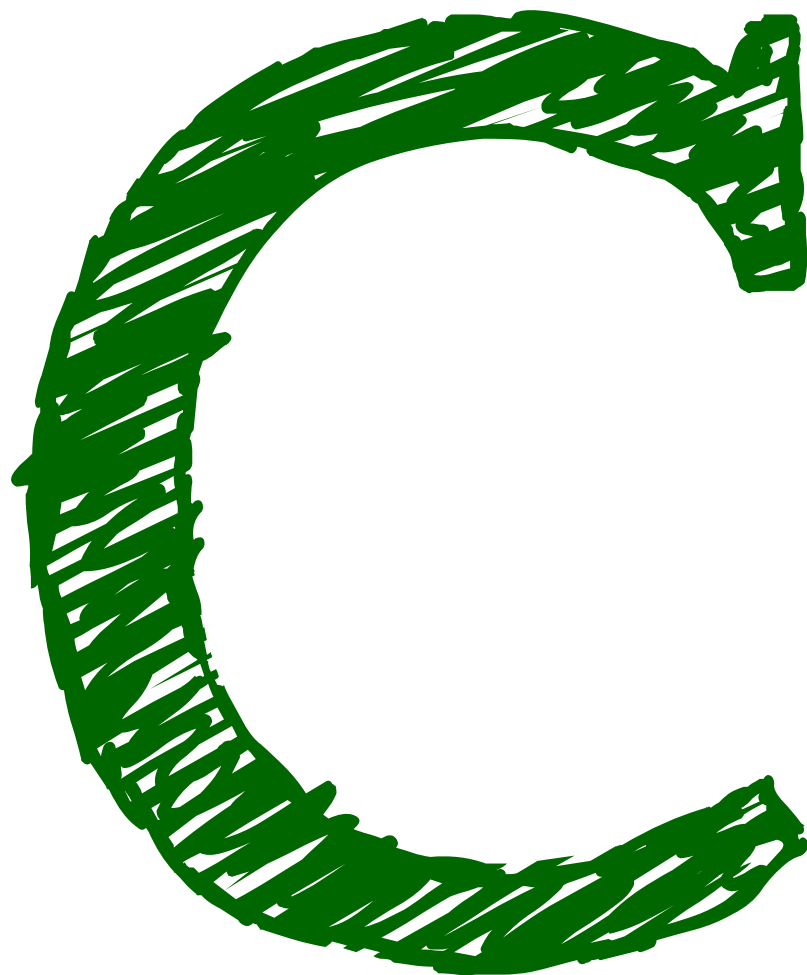
59. Interest at 5% for 3months
on PHP6,000.00

A. PhP150.00

B. PhP120.00

C. PhP75.00

D. PhP250.00



60. The least common multiple (LCM) of 2, 3 and 4 is _____.

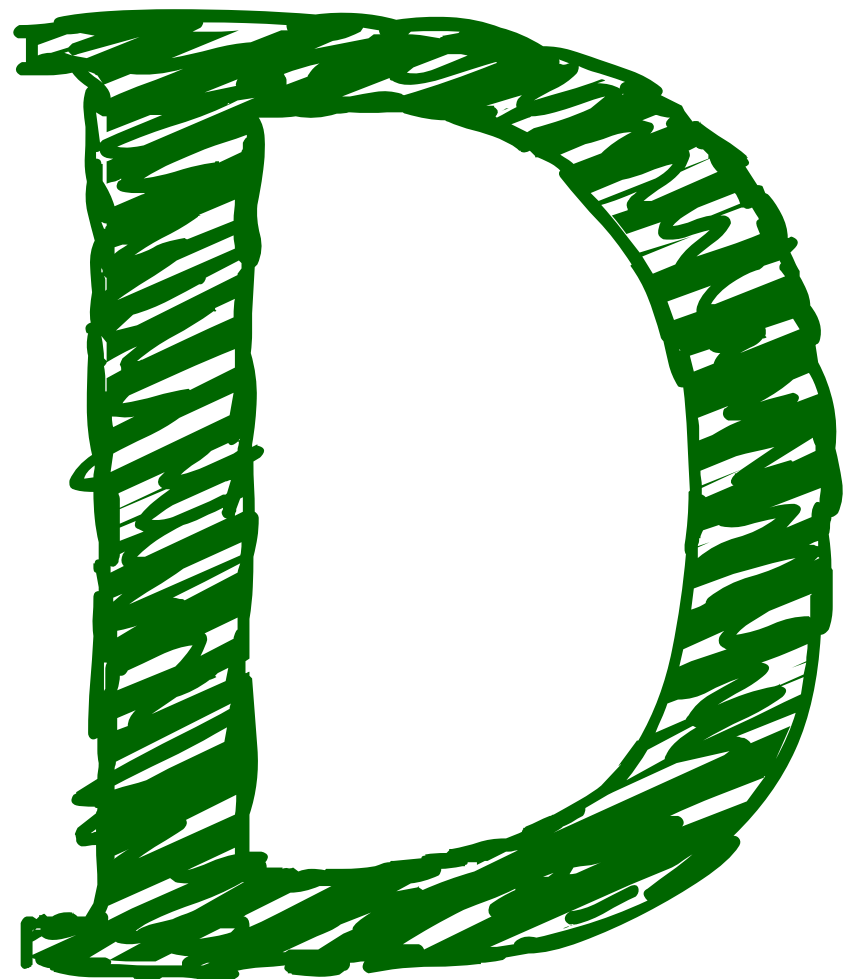
A. 24

B. 14

C. 13

D. 12

2	4	6	8	10	12
3	6	9	12	15	18
4	8	12	16	20	24



61. Simplify: $(8x-24)/(2x^2-x-15)$

A. $8/(2x+5)$

B. $4/(x-13)$

C. $(x-4)/(x+4)$

D. $5/(2x-5)$

$$= \frac{8 (X - 3)}{(2X + 5)(X - 3)}$$

$$\frac{8}{(2X + 5)}$$



62. The Greatest Common Factor (GCF) of 22, 15, 7 is _____.

A. 1

B. 3

C. 2

D. 4

22 {**1**,2,11,22}

15 {**1**,3,5,15}

7 {**1**,7}



63. How many prime numbers are there between 1 and 100?

A. 25

B. 24

C. 23

D. 22



Prime and Composite Numbers

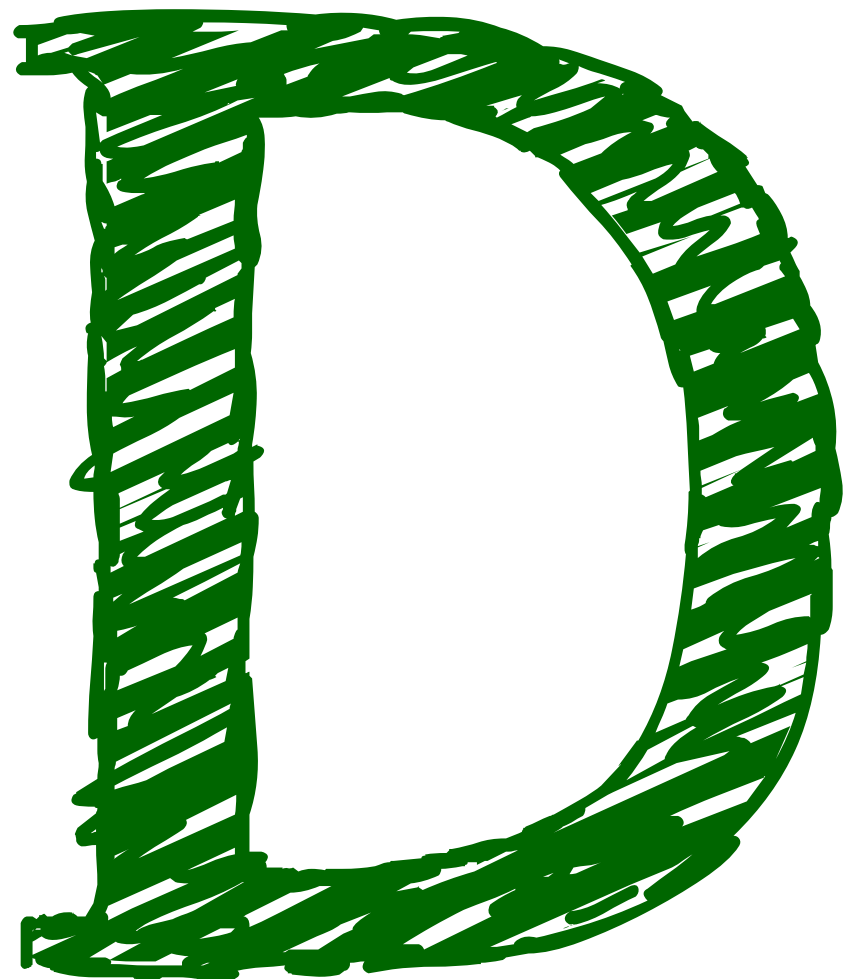
1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

64. A Toyota car travelling at rate of 70km. per hour leaves the house 2 hours after a Kia car has left and overtakes it in 5 hours. At what rate was the Kia car travelling?

- A. 20 kilometers per hour
- B. 30 kilometers per hour
- C. 40 kilometers per hour
- D. 50 kilometers per hour

64. A Toyota car travelling at rate of 70km. per hour leaves the house 2 hours after a Kia car has left and overtakes it in 5 hours. At what rate was the Kia car travelling?

	1	2	3	4	5	6	7
Toyota	70	140	210	280	350		
Kia	50	100	150	200	250	300	350



65. The largest common factor of two or more numbers is called?

- A. Greatest common factor
- B. Prime factor
- C. Least common factor
- D. Prime common factor

Greatest Common Factor

[more ...](#)

The highest number that divides exactly into two or more numbers.

If you find all the factors of two or more numbers, and some factors are the same ("common"), then the largest of those common factors is the Greatest Common Factor.

Abbreviated "GCF". Also called "Highest Common Factor"

Example: the GCF of 12 and 30 is 6, because 1, 2, 3 and 6 are factors of both 12 and 30, and 6 is the greatest.



66. Factor: $2x^2 + 7x - 15$

A. $(x-5)(2x+3)$

B. $(x+5)(2x-3)$

C. $(2x+5)(x-3)$

D. $(2x-5)(x+3)$

B

67. The island of Luzon is estimated to be 100,000 square kilometers. In exponential form can be expressed as _____.

A. 1×10^6

B. 10^5

C. 1×10^4

D. 10^6

B

68. Mr. Conde had incurred the following expenses in his trips to the Mindanao islands: PhP3,200.00; Php2,500.00 and Php1,500.00. What percent of this total monthly budget of PhP40, 000.00 did he spend for this trip?

A. 35%

B. 18%

C. 30%

D. 20%

$$= \frac{3,200 + 2,500 + 1,500}{40,000} \times 100$$

$$= \frac{7,200}{40,000} \times 100$$

$$= 18\%$$

B

69. In a University, the ratio of female professors to the male professors is 8:5. If there are 75 male professors, how many are female professors?

A. 180

B. 120

C. $375/8$

D. 225

F:M

$$8 : 5 = x : 75$$

$$5x = 600$$

$$\frac{5x}{5} = \frac{600}{5}$$

$$**X = 120**$$



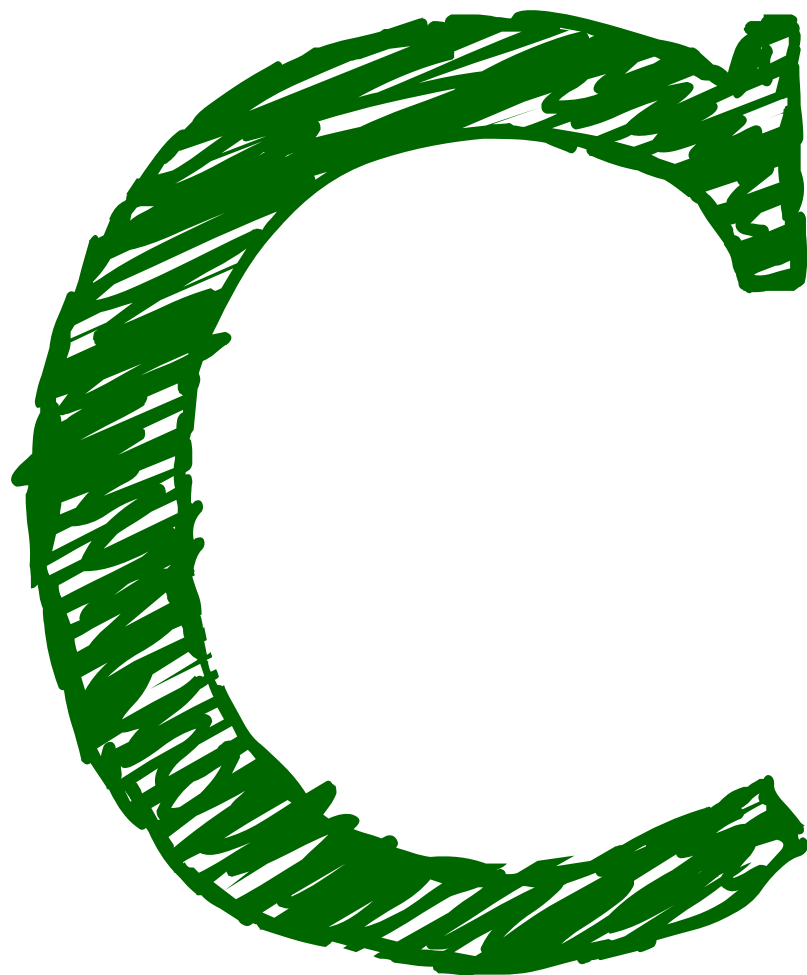
70. Simplify: $(x^2 - y^2)/(x - y)$

A. $-x - y$

B. $x - y$

C. $x + y$

D. $y - x$



71. Simplify: $6 - \{3 - (4) + 11 + 8\}$.

A. 26

B. -26

C. 20

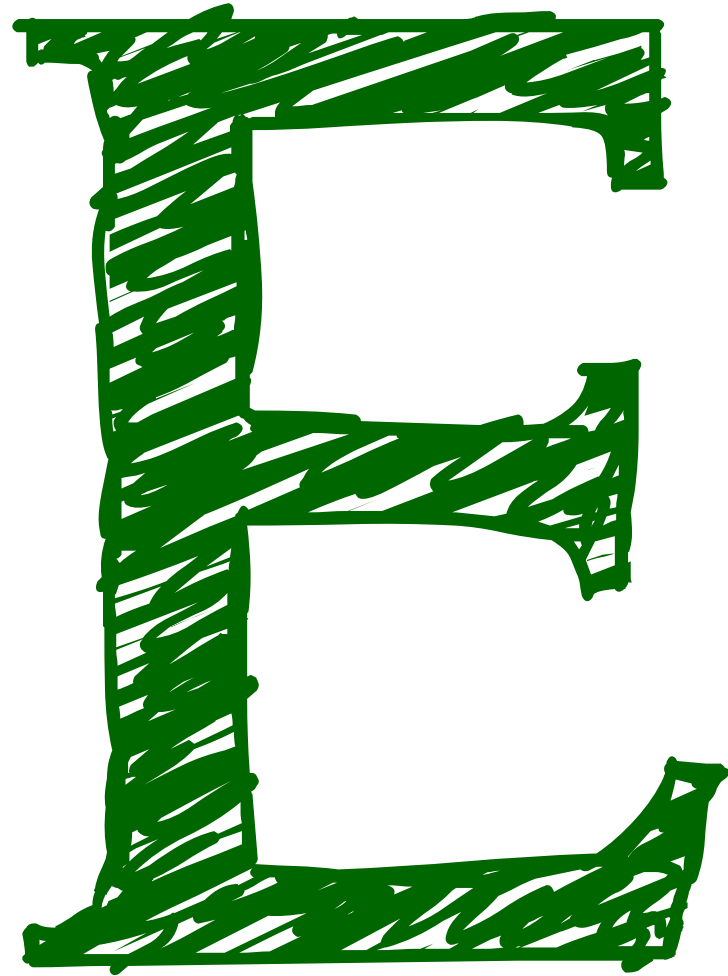
D. -20

71. Simplify: $6 - \{3 - (4) + 11 + 8\}$.

$$6 - 3 + 4 - 11 - 8$$

$$10 - 22$$

$$\mathbf{-12}$$



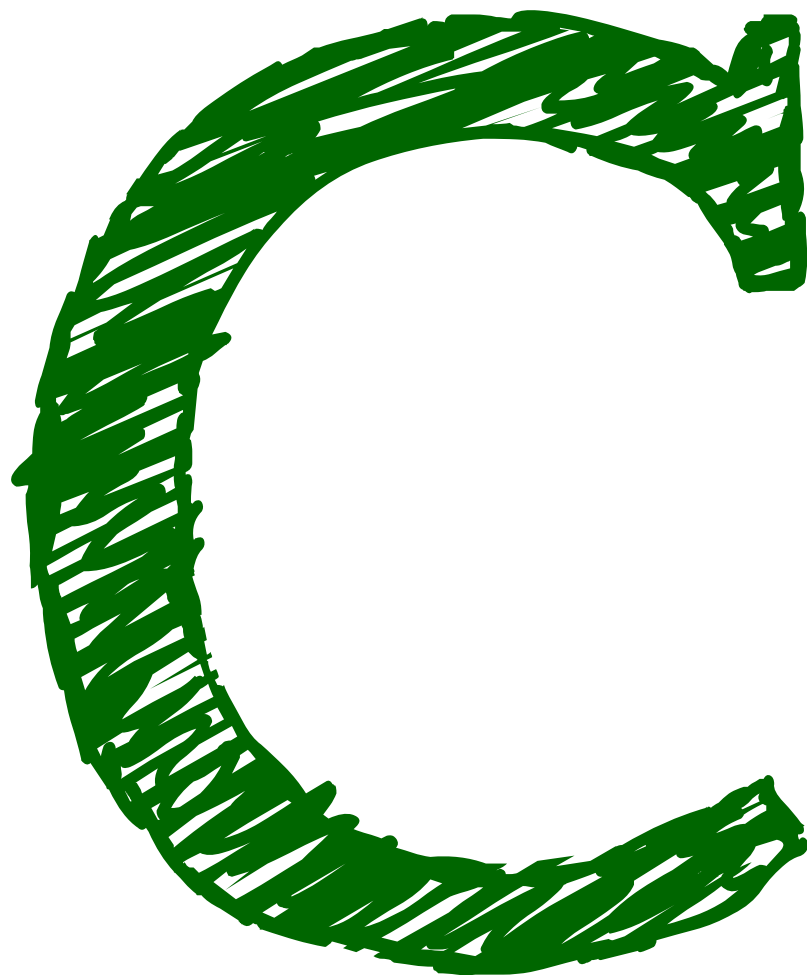
72. Which are the prime factors of 63?

A. 7, 9

B. 3, 21

C. 7, 3, 3

D. 7, 3, 2



73. The grades in Mathematics of the students in section A are as follows: 80, 75, 60, 95, and 100. What is the population variance of their group?

A. 206

B. 260

C. 216

D. 224

What is Variance?

- The difference between an expected and actual result such as between a budget and actual expenditure.
- Statistics: The arithmetic mean of the squares of the deviations of all values in a set of numbers from their arithmetic mean.

80	-2	4
75	-7	49
60	-22	484
95	13	169
100	18	324
82		206



74. Which among the measures of central tendency is not influenced by outliers?

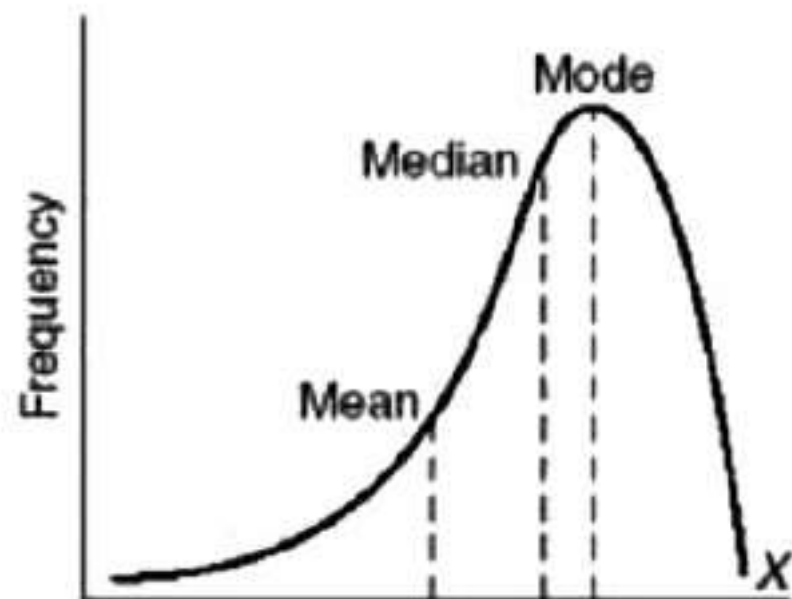
A. Median

B. Weighted Mean

C. Mean

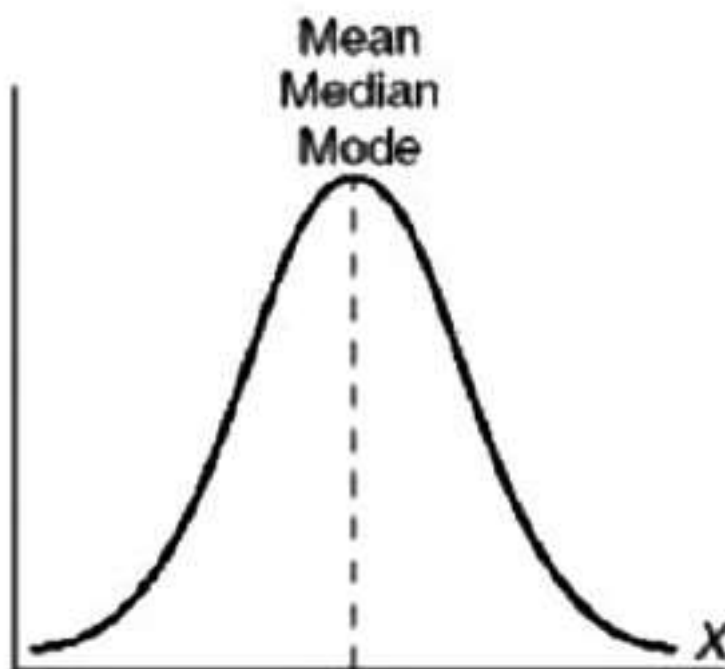
D. Mode

Negatively skewed



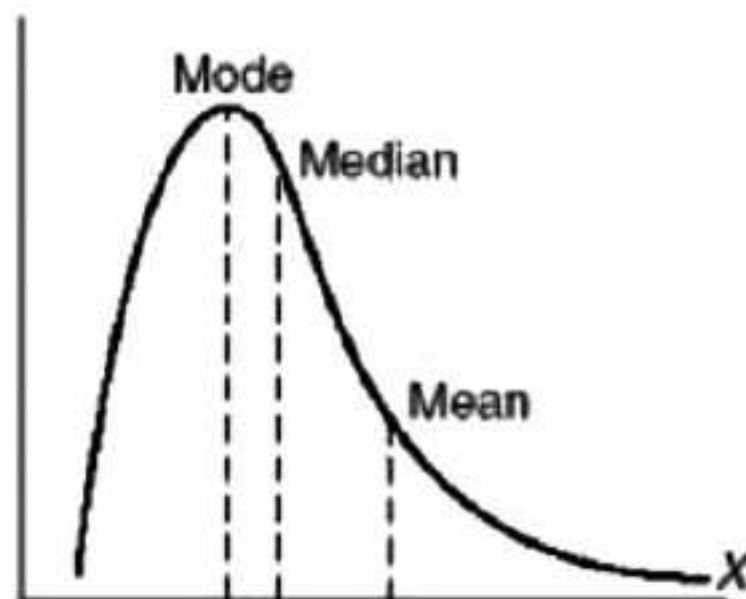
Negative direction

Normal (no skew)



The normal curve
represents a perfectly
symmetrical distribution

Positively skewed



Positive direction



75. The grades in mathematics of the students in section A are as follows scores: 12, 10, 13, 11, 15, 20, 19, and 17. What is the population standard deviation of their group?

A. 3.50

B. 3.48

C. 3.51

D. 3.49

$$s_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

n $\equiv$ The number of data points

$\bar{x}$ $\equiv$ The mean of the x_i

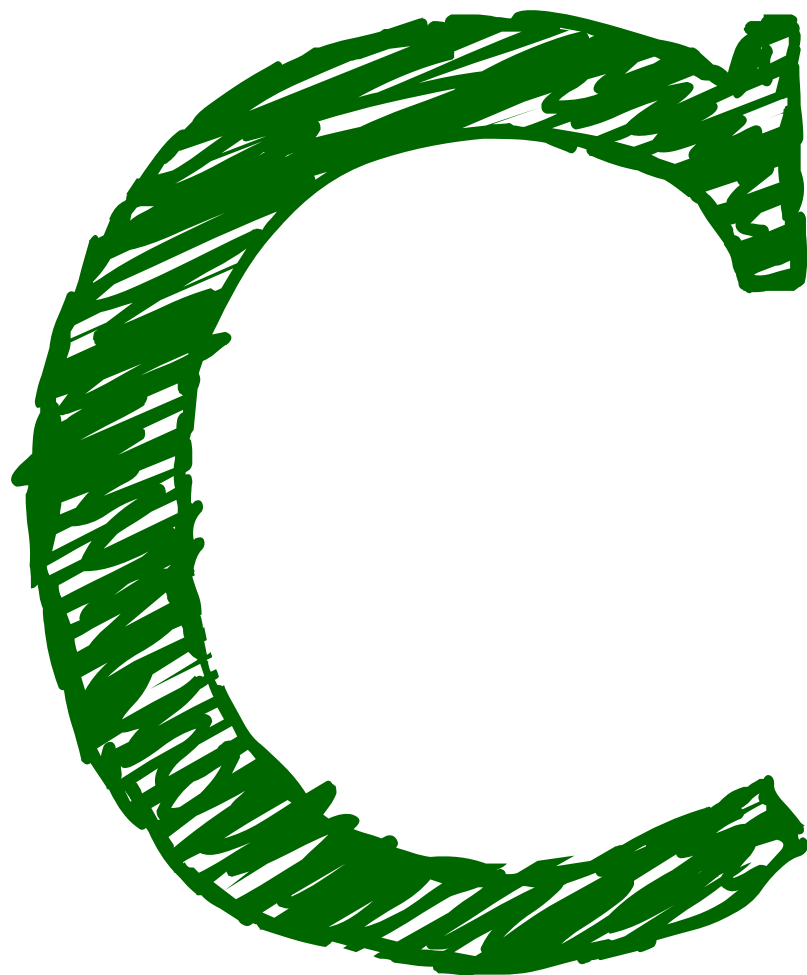
x_i $\equiv$ Each of the values of the data

12	-2.625	6.890625
10	-4.625	21.39063
13	-1.625	2.640625
11	-3.625	13.14063
15	0.375	0.140625
20	5.375	28.89063
19	4.375	19.14063
17	2.375	5.640625
14.625		97.875

$$SD = \sqrt{\frac{97.875}{7}}$$

$$SD = \sqrt{13.9821}$$

$$SD = 3.74$$



76. In how many ways can 5 girls be seated in a row of 5 seats?

A. 95

B. 100

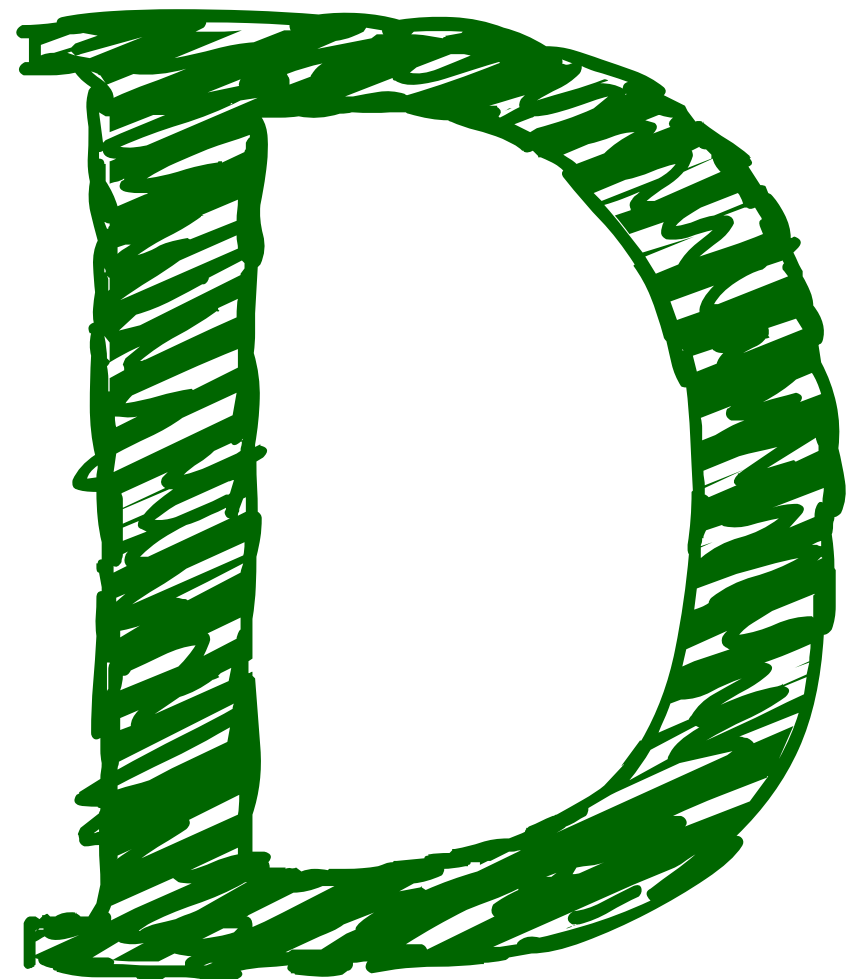
C. 105

D. 120

$$= 5P5$$

$$= 5!$$

$$= 120$$



77. If a die is rolled, what is the probability of getting a number divisible by 2?

A. $\frac{1}{6}$

B. $\frac{1}{4}$

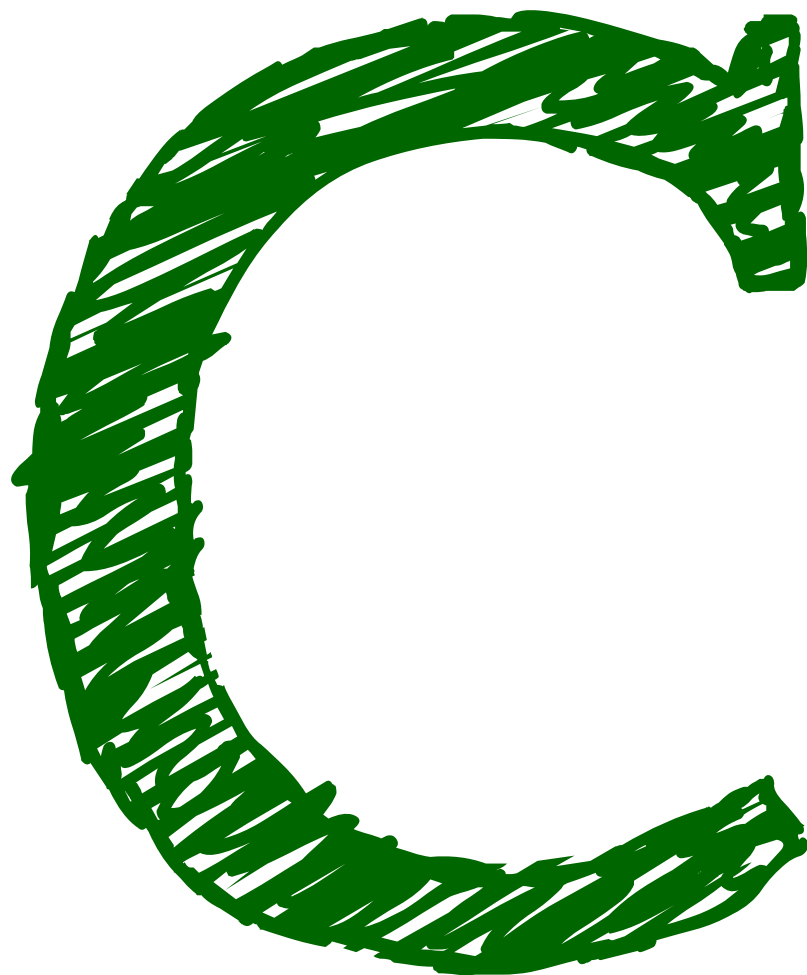
C. $\frac{1}{2}$

D. $\frac{1}{3}$

Possible Outcomes

$\{1, \underline{2}, 3, \underline{4}, 5, \underline{6}\}$

$$\frac{3}{6} = \frac{1}{2}$$



78. Which of the following is a product of 13 and an integer?

A. 1326

B. 1323

C. 1343

D. 131



79. In an English test, eight students obtained the following scores: 12, 10, 13, 11, 15, 20, 19, 17. What is the median score?

A. 15.5

B. 14

C. 16.5

D. 17

79. In an English test, eight students obtained the following scores: 12, 10, 13, 11, 15, 20, 19, 17. What is the median score?

10, 11, 12, 13, 15, 17, 19, 20

$$13+15 = 28/2 = 14$$

B

80. A rectangular block of steel has dimensions of 5 meters x 10 meters x 15 meters and weights, 1,000 N. How should this block be placed on a surface to exert the least pressure on the surface?

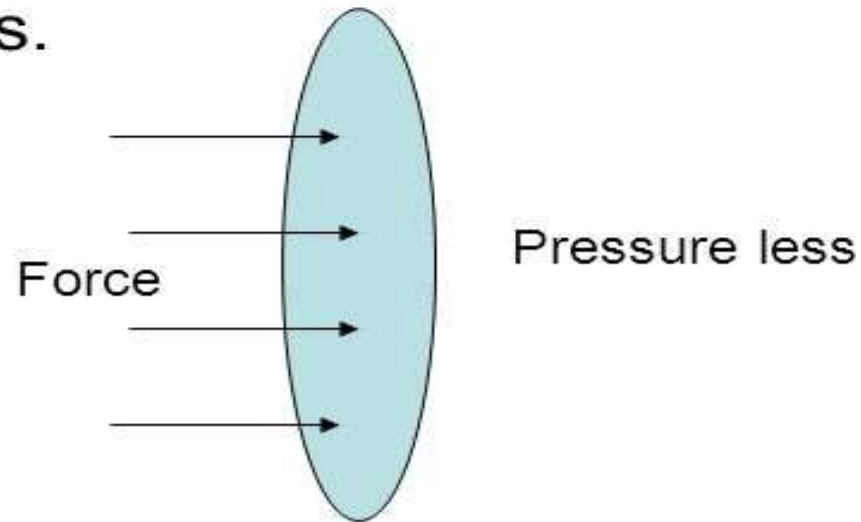
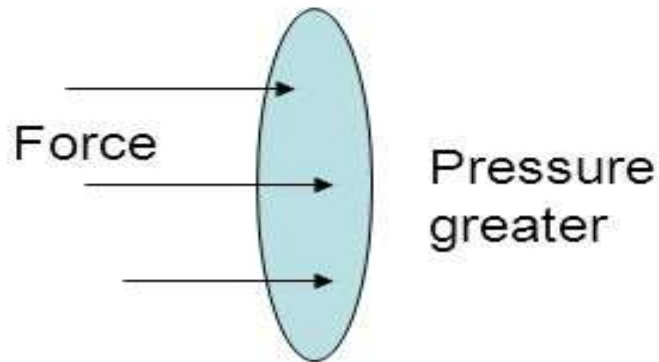
- A. On the 10 meters by 15 meters side
- B. On the 5 meters by 15 meters side
- C. On the 5 meters by 10 meters side
- D. all sides have equal pressure

Pressure

Pressure is a force exerted over an area on the surface of an object.

Force and Pressure are not the same thing:

Pressure decreases as the area over which a force is applied increases.





81. If a student has an average of 76% on his first two tests, and has an average of 85% on the next four tests, what is his final average on all six tests?

A. 82.5%

B. 80.5%

C. 81.3%

D. 82.0%

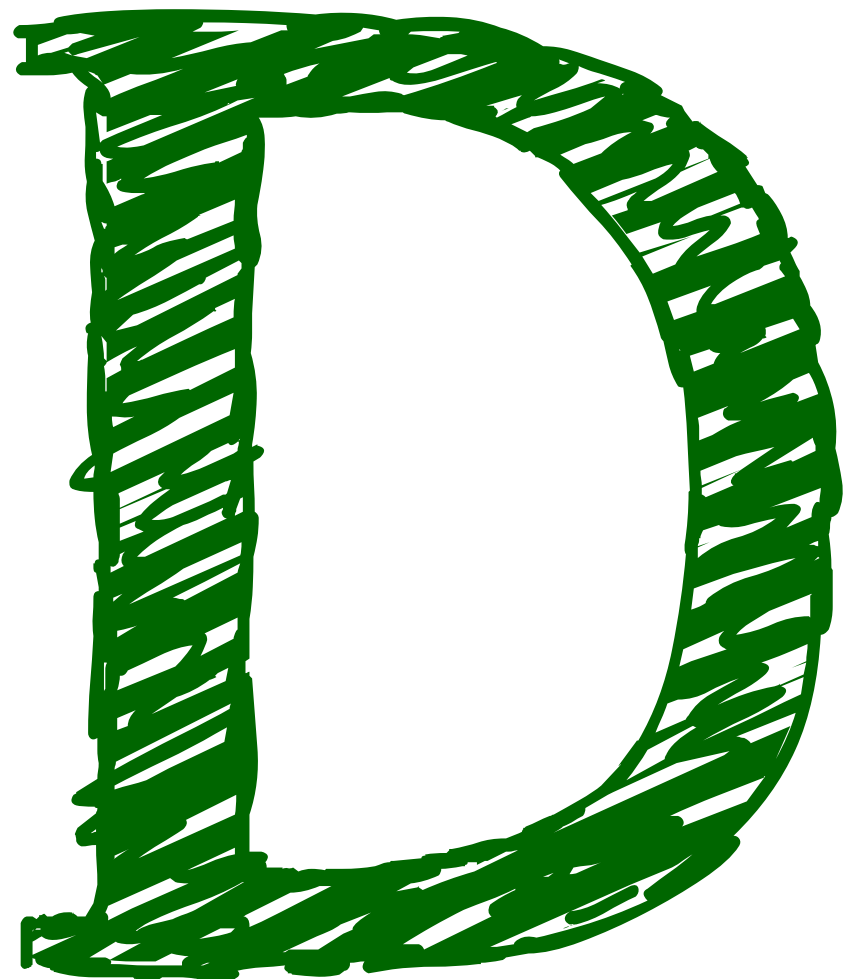
81. If a student has an average of 76% on his first two tests, and has an average of 85% on the next four tests, what is his final average on all six tests?

$$= 76 (2) + 85 (4)$$

$$= 152 + 340$$

$$= 492 / 6$$

$$= 82$$



82. What percent of 75 is 15?

A. 20%

B. 30%

C. 38%

D. 40%

82. What percent of 75 is 15?

$$= \frac{15}{75} \times 100$$

$$= 20\%$$



83. The grades in Mathematics of the students in section A are as follows: 80, 75, 60, 95, 100. What is the mean absolute deviation of this group?

A. 12.4

B. 13.2

C. 14.61

D. 11.7

Mean Absolute Deviation!

the average amount that each number is away from the mean of the whole set.

Step 1: Find the mean (the average)

Step 2: Find the *deviations*: subtract each number and the mean.

Step 3: Find the average of all of the "deviations"

(make the answers positive in step 2, add them all ,then divide)

Example: Find the mean absolute deviation of:

80, 85, 81, 0, 85, 90, 87, 92

Step 1:

Find the average:

$$80 + 85 + 81 + 0 + 85 + 90 + 87 + 92 = 600$$

$$600/8 = 75$$

Step 2:

Find the deviations

$$80 - 75 = 5$$

$$85 - 75 = 10$$

$$81 - 75 = 6$$

$$0 - 75 = -75$$

$$85 - 75 = 10$$

$$90 - 75 = 15$$

$$87 - 75 = 12$$

$$92 - 75 = 17$$

Step 3:

Find the average of the absolute value of the deviations:

$$5 + 10 + 6 + 75 + 10 + 15 + 12 + 17 = 150$$

$$150/8 = 18.75$$

80	2
75	7
60	22
95	13
100	18
82	12.4



84. A ball is drawn at random from a box containing 6 red balls, 4 white balls and 5 blue balls. Find the probability that it is blue.

A. $\frac{1}{3}$

B. $\frac{2}{5}$

C. $\frac{1}{15}$

D. $\frac{4}{5}$



85. The altitude of a triangle is 5 meters and the base is 20 meters. What is the area of the triangle?

- A. 50 square meters
- B. 20 square meters
- C. 24 square meters
- D. 60 square meters

85. The altitude of a triangle is 5 meters and the base is 20 meters. What is the area of the triangle?

$$\begin{aligned} A &= \frac{1}{2} bh \\ &= \frac{1}{2} 20 \times 5 \\ &= \frac{100}{2} \\ &= 50 \end{aligned}$$

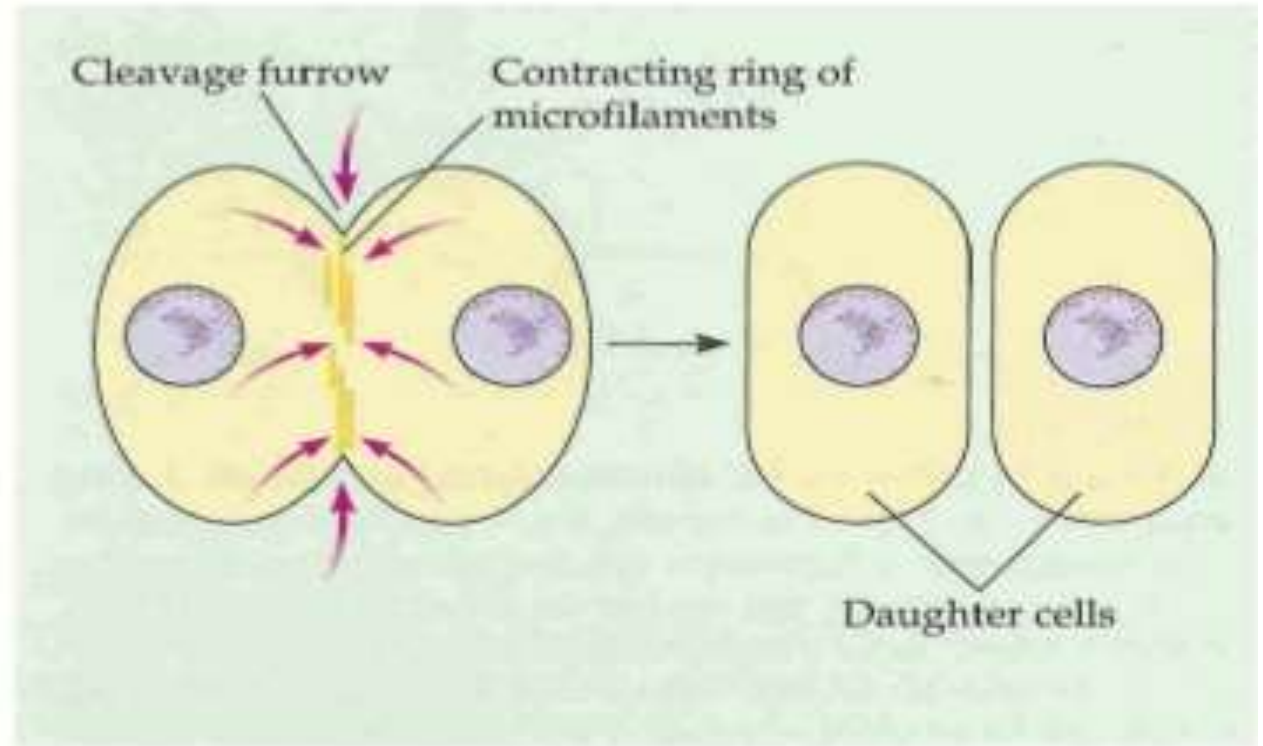


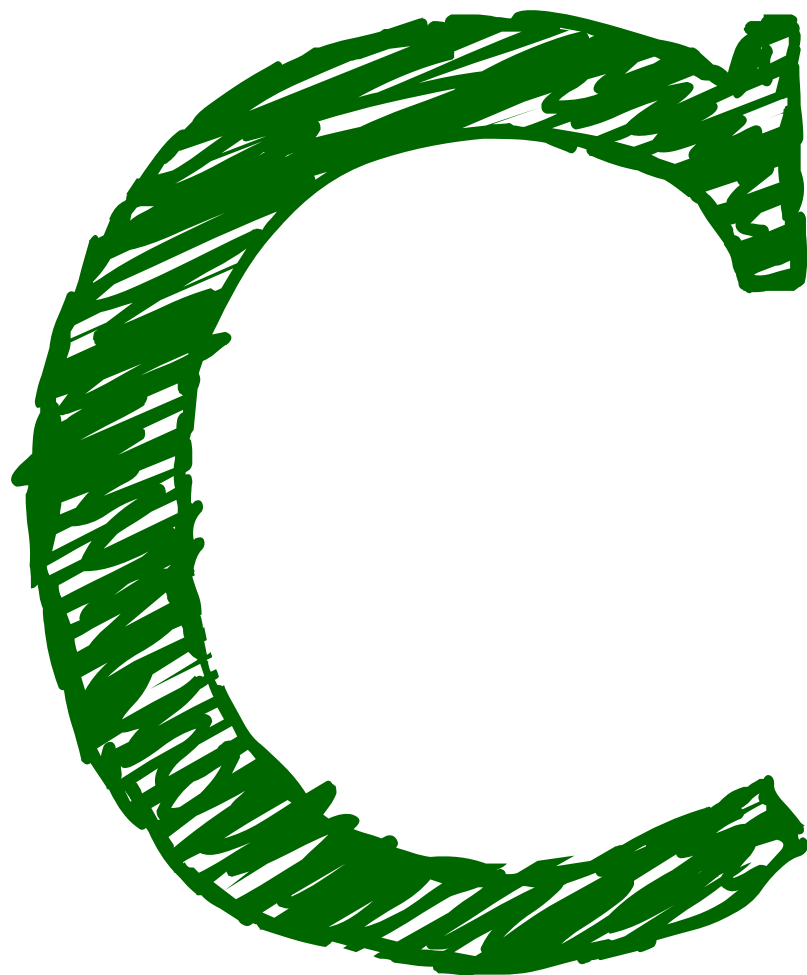
86. What thin structures are essential for cytokinesis, amoeboid movement and changes in cell shape?

- A. Monofilaments
- B. Pseudo filaments
- C. Microfilaments
- D. Carbohydrates

Cytokinesis in Animal Cells

- ✿ Cytokinesis occurs when a ring of microfilaments develop around the cell.
- ✿ These microfilaments are attached to the plasma membrane.
- ✿ The microfilaments contract inwards making a cleavage furrow.
- ✿ They continue to contract until the cell is pinched in two.





87. What is the classification for polysaccharides, triglycerides, polypeptides and nucleic acids?

- a. Acids and bases
- b. Macromolecules
- c. Phospholids
- d. Carbohydrates

Macromolecules

- Biological molecules that are very large on a molecular scale are called macromolecules.
- Macromolecules include:
 - Polysaccharides (complex sugars)
 - Triglycerides (a type of fat molecule)
 - Polypeptide chains and proteins
 - Nucleic acids (DNA and RNA)
- The polymer structures of some macromolecules are synthesized from repeating units of smaller molecules known as monomers.
- A polymer is like a beaded necklace consisting of many individual beads.

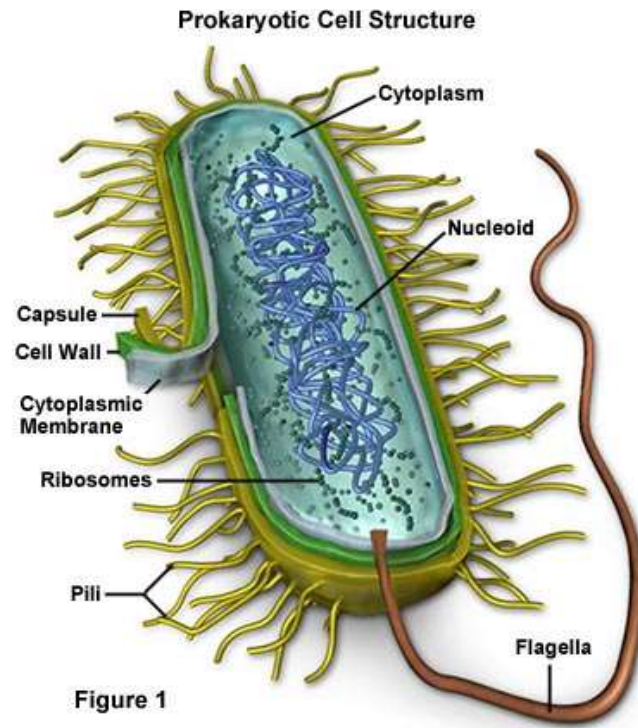


88. Of the following organism, which are included under Kingdom Plantae?

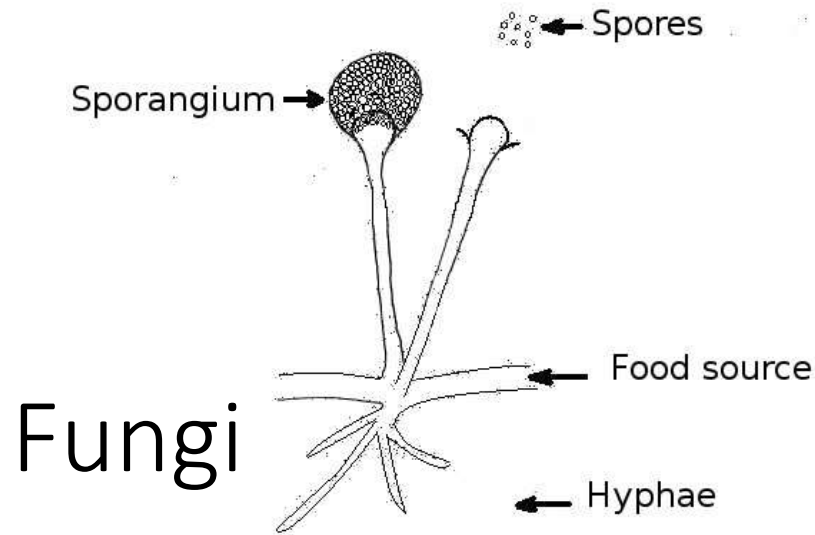
- a. Fungi
- b. Magnoliidae
- c. Prokaryotes
- d. Protozoa



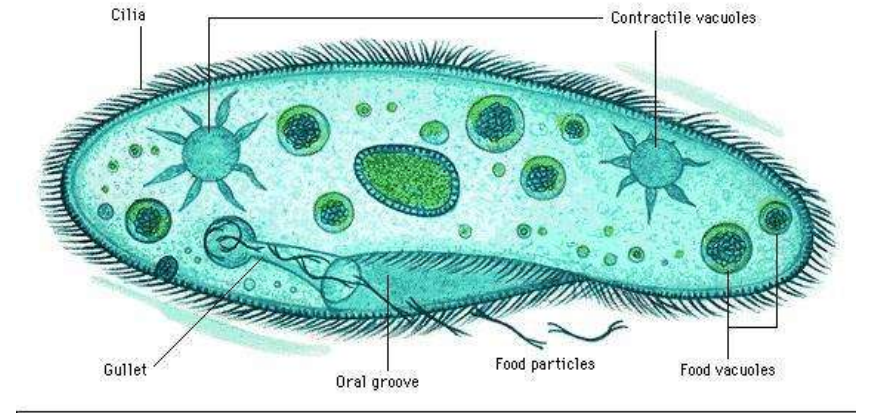
Magnoliidae



Prokaryotes



Fungi



Protozoa



89. Which of the following is the lowest rank of coal and is otherwise called brown coal?

A. Bituminous

B. Lignite

C. Anthracite

D. Charcoal

Ranks of Coal

Lignite

7000 BTU/lb



Subbituminous

9,000 BTU/lb



Bituminous

12,000 BTU/lb



Anthracite

15,000 BTU/lb



Formation Conditions

Low T, P

High T, P

Time of Formation

% C

Lowest Grade
U.S. Coal

Highest Grade
U.S. Coal



90. If carbohydrates are hydrophilic-organic compounds, lipids are _____ organic compounds?

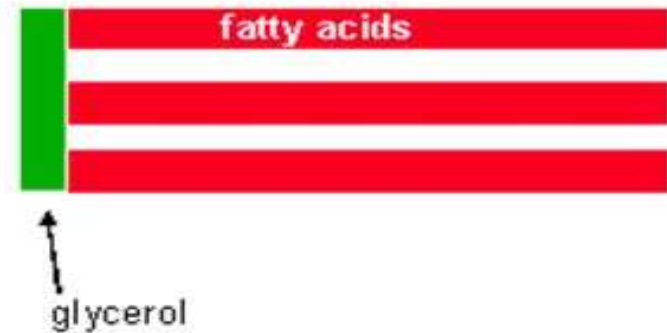
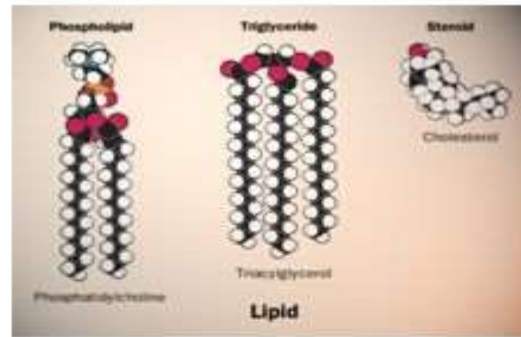
- A. hydrophobic
- B. simple
- C. polypeptido
- D. biomolecules

5.3: Lipids

- Waxes or oils or fats
- HYDROPHOBIC!
- Also made of C, H & O, but lower proportion of oxygen
- **Lipids store more energy than carbohydrates** because they have twice as many energy rich (C-H) bonds than carbohydrates.
- **1 Lipid = 3 fatty acids + 1 glycerol molecule (monomers)**

Example: Triglyceride (polymer):

Composed of 3 fatty acid molecules joined to 1 Molecule of alcohol glycerol.



FUNCTIONS

1. Store energy
2. Chemical signals to cells
3. Boundary to water in cells (IN CELLMEMBRANE)
4. Cushion organs



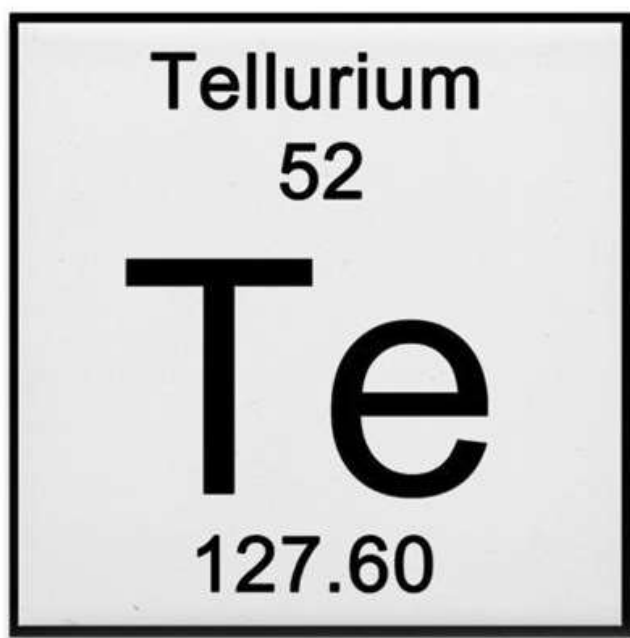
91. Which element has Te as its Symbol?

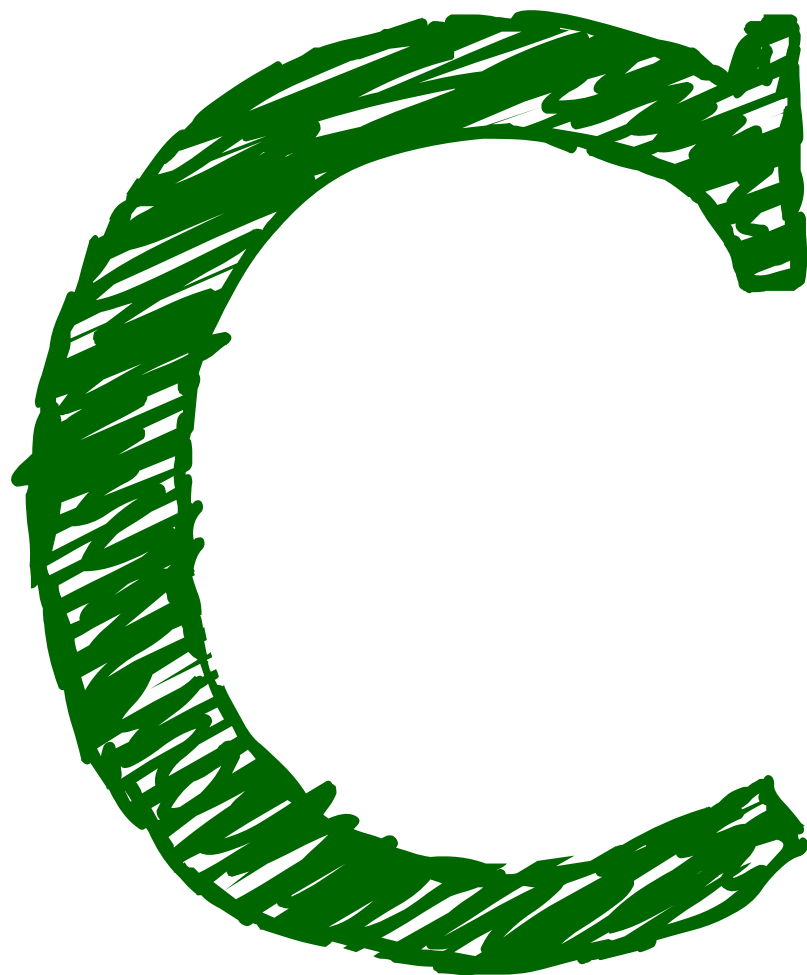
A. Terbium

B. Titanium

C. Tellurium

D. Technetium



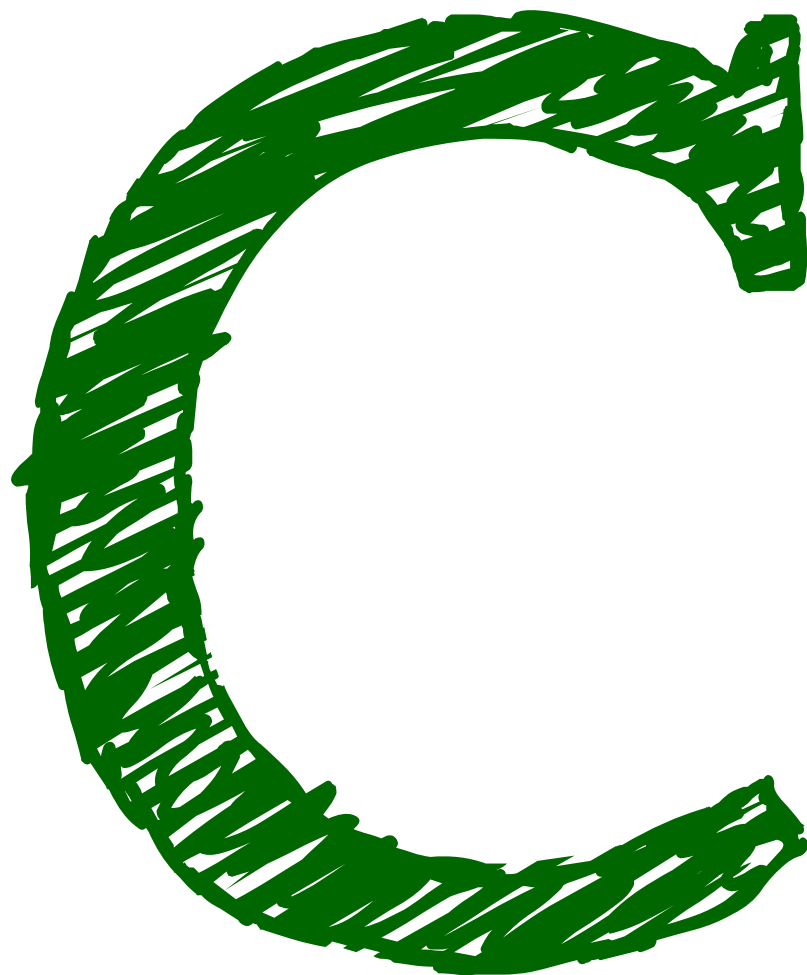


92. What is the term for the slowing-down response by the frog's circulatory during the cold season?

- A. Cold-blooded reaction
- B. Temperature response
- C. Hibernation
- D. Estimation

When the cold weather comes during winter, frogs, toads, and river turtles bury themselves in mud or soil and “hibernate”. In the spring when the sun warms the mud, out they come to absorb the heat of the Sun.





93. Which of the following is a vitamin?

A. Niacin

B. Mucin

C. Pepsin

D. Trypsin

Types of Vitamins

- Water-soluble vitamins

- Thiamine
- Riboflavin
- Niacin
- Pyridoxine
- Pantothenic acid
- Biotin
- Choline
- Folic acid
- Vitamin B₁₂
- Vitamin C



B Complex Vitamins



94. Which element has Ca as its symbol?

- A. Cadmium
- B. Californium
- C. Carbon
- D. Calcium

48

Cd

Cadmium

112.411

98

Cf

Californium

251.080

6

12.01

C

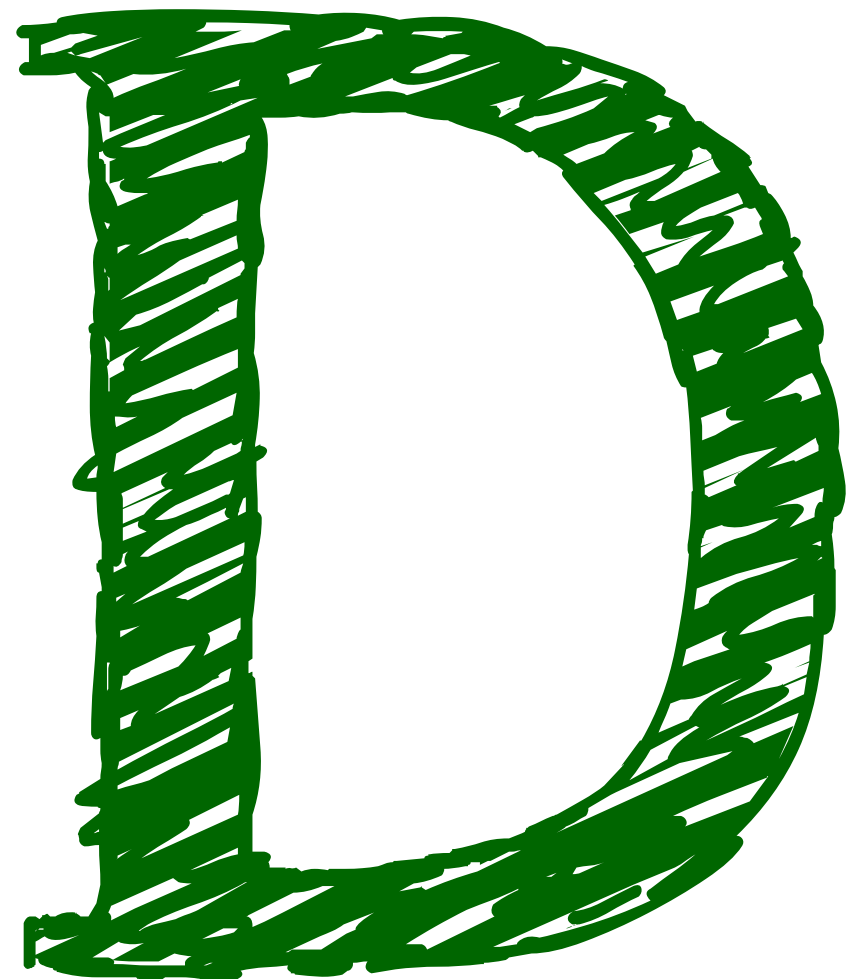
CARBON

20

Ca

Calcium

40.078



95. A generalized image of an object is _____.

- A. concept
- B. definition
- C. impression
- D. mental picture

CONCEPT

1: something conceived in the mind : THOUGHT, NOTION

2: an abstract or generic idea generalized from particular instances



96. Which of the following processes involves chloroplast?

- A. Conversion of light energy to chemical energy
- B. Formulation of reproductive cells
- C. Cell division
- D. Protein synthesis

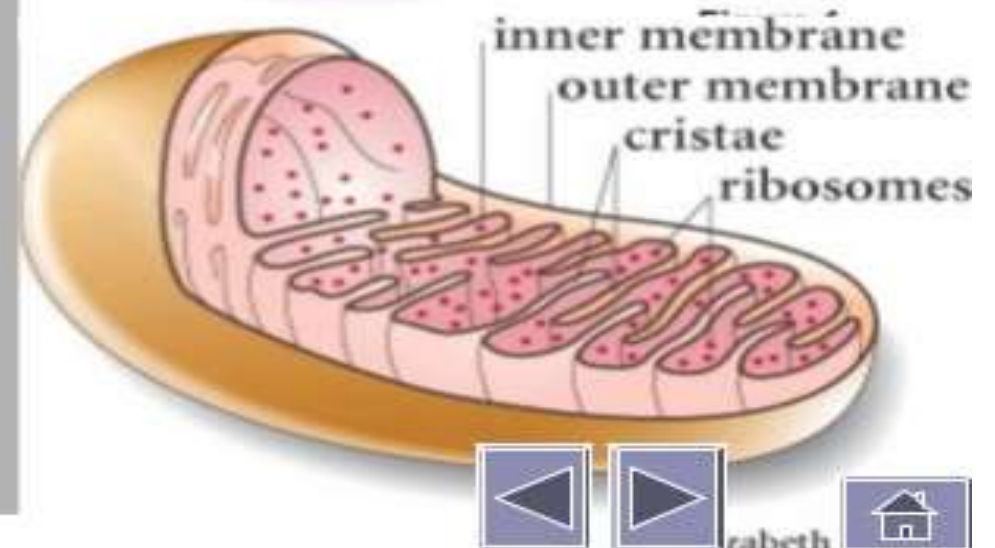
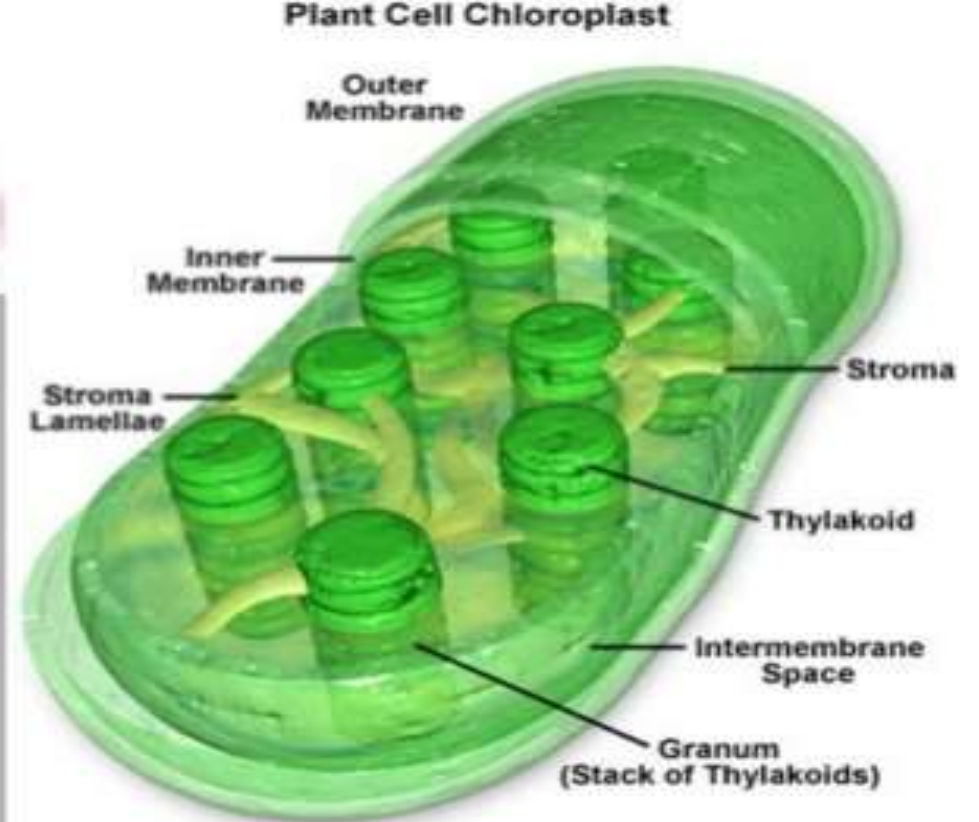
7-2 Cell Structure Vocabulary

■ Chloroplast-

organelle found in cells of plants and some other organisms that uses energy from sunlight to make energy-rich food molecules by photosynthesis.

■ Mitochondrion-

cell organelle that releases energy from stored food molecules.



Elizabeth





97. Besides solid, liquid and gas, what is the fourth form of matter?

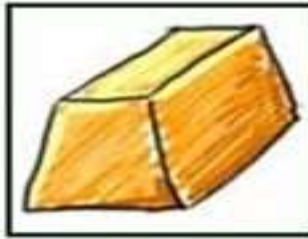
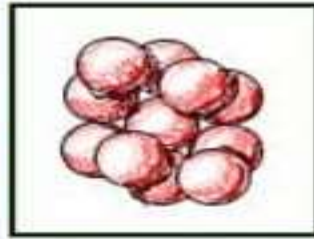
A. Plasma

B. Gel

C. Foam

D. Colloid

5 States



- All matter exists in some sort of physical form or **a state of matter.**
- There are 5 states of matter:
 1. **Solid**
 2. **Liquid**
 3. **Gas**
 4. **Plasma**
 5. Bose-Einstein Condensate (**BEC**)



98. Which of the following is a heterotrophy?

A. Moss

B. Grasshopper

C. Algae

D. Fern

B

99. Which type of reaction does the general equation ($AB + \text{energy} \rightarrow A + B$) show?

- A. Endothermic synthesis
- B. Exothermic decomposition
- C. Exothermic synthesis
- D. Endothermic decomposition



In an exothermic reaction, energy is released into the surroundings as heat. As a result, the temperature of the surroundings increases.



In an endothermic reaction, energy is absorbed from the surroundings. As a result, the temperature of the surroundings drops.

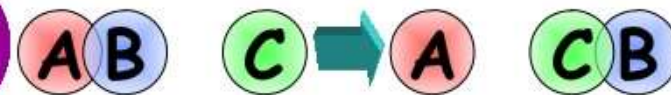
Synthesis



Decomposition

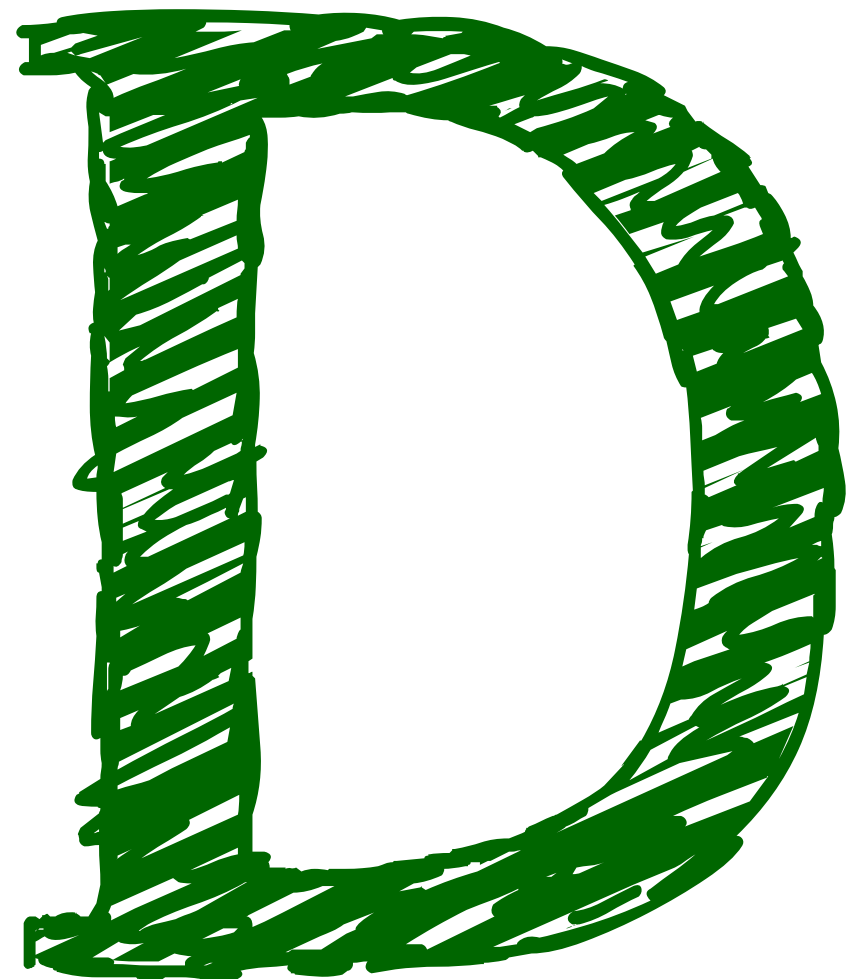


Single Replacement



Double Replacement

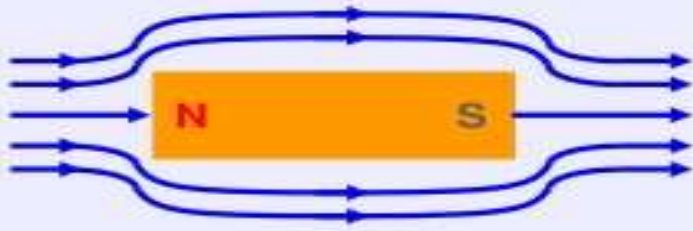
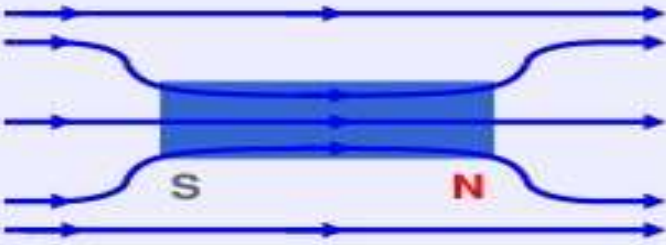
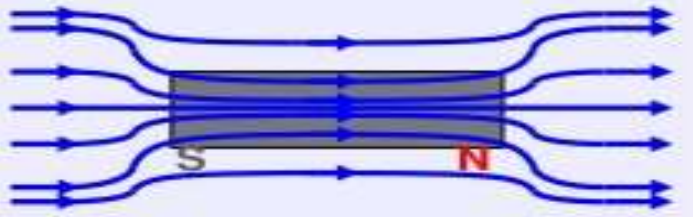




100. Iron is what type of magnetic material?

- A. Non-magnetic
- B. Ferromagnetic
- C. Paramagnetic
- D. Diamagnetic

Comparison of Dia, Para and Ferro Magnetic materials:

DIA	PARA	FERRO
1. Diamagnetic substances are those substances which are feebly repelled by a magnet. Eg. Antimony, Bismuth, Copper, Gold, Silver, Quartz, Mercury, Alcohol, water, Hydrogen, Air, Argon, etc.	Paramagnetic substances are those substances which are feebly attracted by a magnet. Eg. Aluminium, Chromium, Alkali and Alkaline earth metals, Platinum, Oxygen, etc.	Ferromagnetic substances are those substances which are strongly attracted by a magnet. Eg. Iron, Cobalt, Nickel, Gadolinium, Dysprosium, etc.
2. When placed in magnetic field, the lines of force tend to avoid the substance. 	The lines of force prefer to pass through the substance rather than air. 	The lines of force tend to crowd into the specimen. 

B

101. Humans reproduce when an ovum is combined with a sperm leading to the development of an embryo. This form of reproduction is called _____.

- A. Binary fission
- B. Asexual reproduction
- C. Budding
- D. Fertilization

Asexual Reproduction

Fission

Binary fission Amoeba Multiple fission - plasmodium

Budding

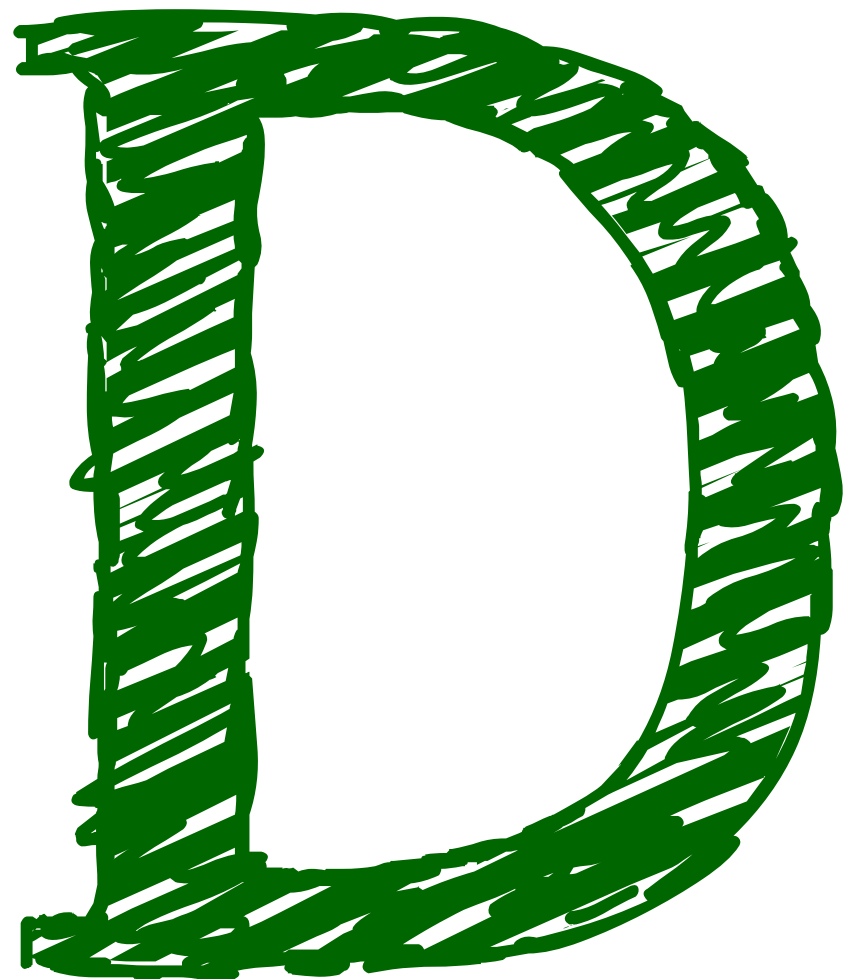
Yeast external nucleus divides,
Internal budding gemules- archaeocytes-
sponges Hydra regeneration

Spore formation

Fungi, Recap algae- fragmentation

Vegetative propagation

- In many plants
-



102. Of the following which is an example of physiological adaptation that allow survival among plants or animals?

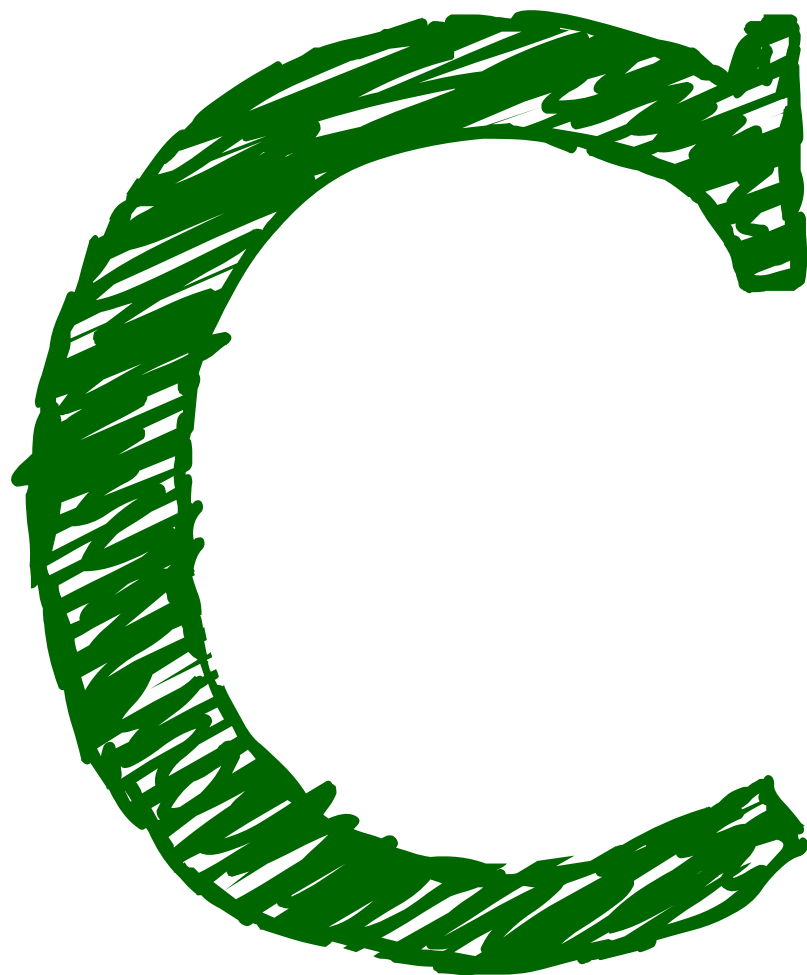
- A. Frog's powerful hind legs to jump away from predators
- B. Bright colors of insect-pollinated flowers
- C. Chameleon's ability to change color with its environment
- D. Walking stick's unique shape and color

Types of Adaptations

- **Structural adaptations:** physical features of an organism, e.g. presence of wings for flight.
- **Behavioral adaptations:** the way an organism acts, e.g. mantid behavior when seeking, capturing, and manipulating prey.
- **Functional (physiological) adaptations:** those involving physiological processes, e.g. the female mantid produces a frothy liquid to surround and protect the groups of eggs she lays.



Praying mantis



103. Which of the following activities help reduce water pollution?

A. Use of soap and detergents

B. Desalination of sea water to potable water

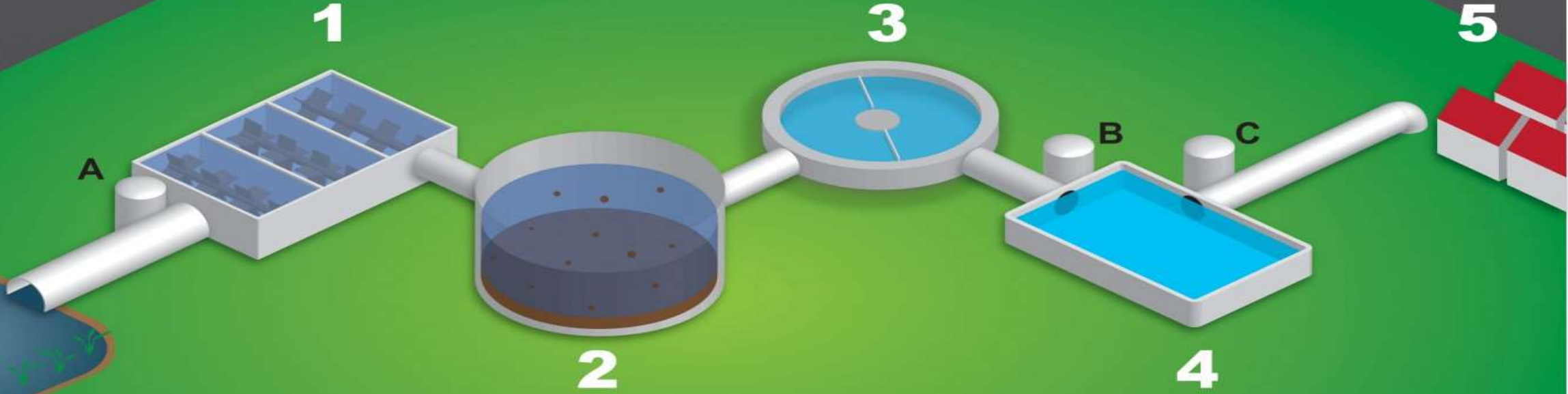
C. Setting up flood control systems

D. Treating waste water prior to discharge

WATER TREATMENT PROCESS

An overview of how water is cleaned and treated before distributing to the population.

(Methods vary between countries depending on water standards)



1

Coagulants are added to the water (A) and then mixed together. Heavy particles in the water then begin to stick together and form large clumps.

2

Water is allowed to settle in a tank and the heavy particles drop to the bottom. These particles are then scraped away to be used as fertiliser.

3

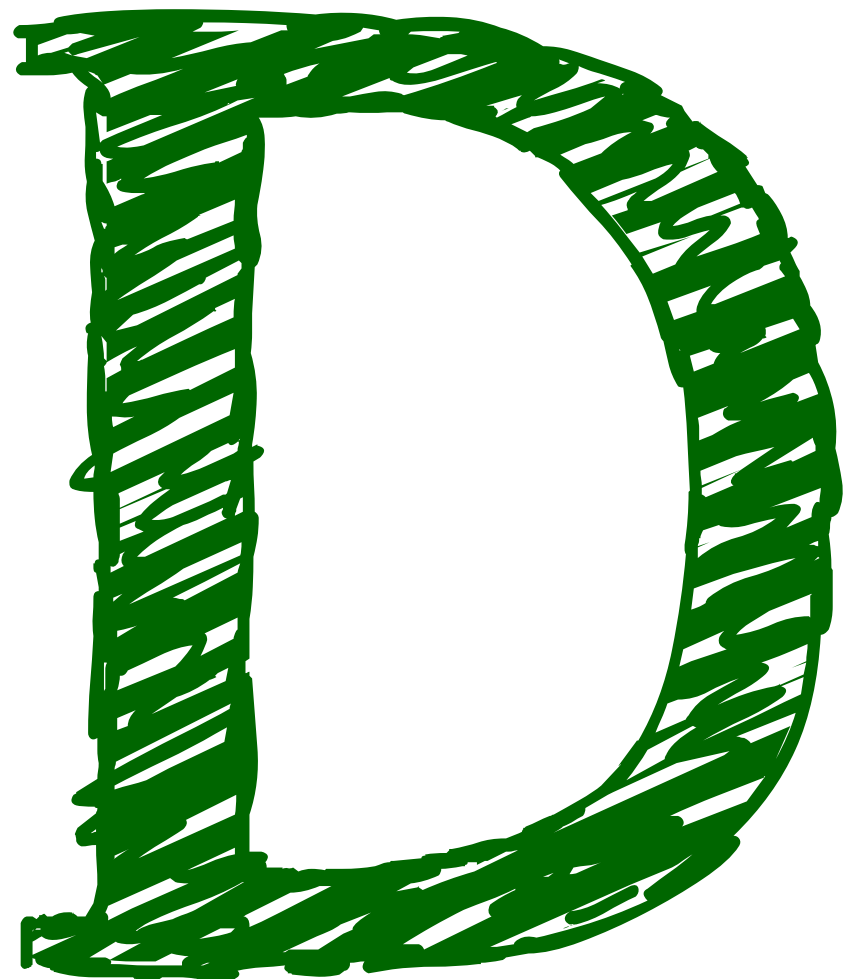
Water is then filtered through layers of fine granulated materials. As particles are removed, turbidity diminishes and clear water emerges.

4

To protect against any bacteria, viruses and other microbes that might remain, disinfectant (B) is added before the water flows into underground reservoirs.

5

pH is maintained by adding alkaline substances (C) to reduce corrosion in the distribution system and the plumbing in your home.



104. What is the main organ for respiration?

- A. Esophagus
- B. Lungs
- C. Nose
- D. Heart

PROCESS OF RESPIRATION

- **Respiration** in the human body involves the flow of gas in outward (*expiratory*) and inward (*inspiratory*) cycles. The flow of gas is the result of pressure changes in the lungs related to the size of the **thorax** (*chest cavity*).
- **Expiration** occurs when the size of the thorax is reduced and the lung is compressed causing air to be pushed out.
- **Inspiration** results in the increase in the size of the thorax (chest cavity).
- **The lungs** expand in volume and lung pressure is below atmospheric pressure. As a result, air rushes (flows) into the expanded lung.

Larynx --Air passes over the larynx and down the trachea

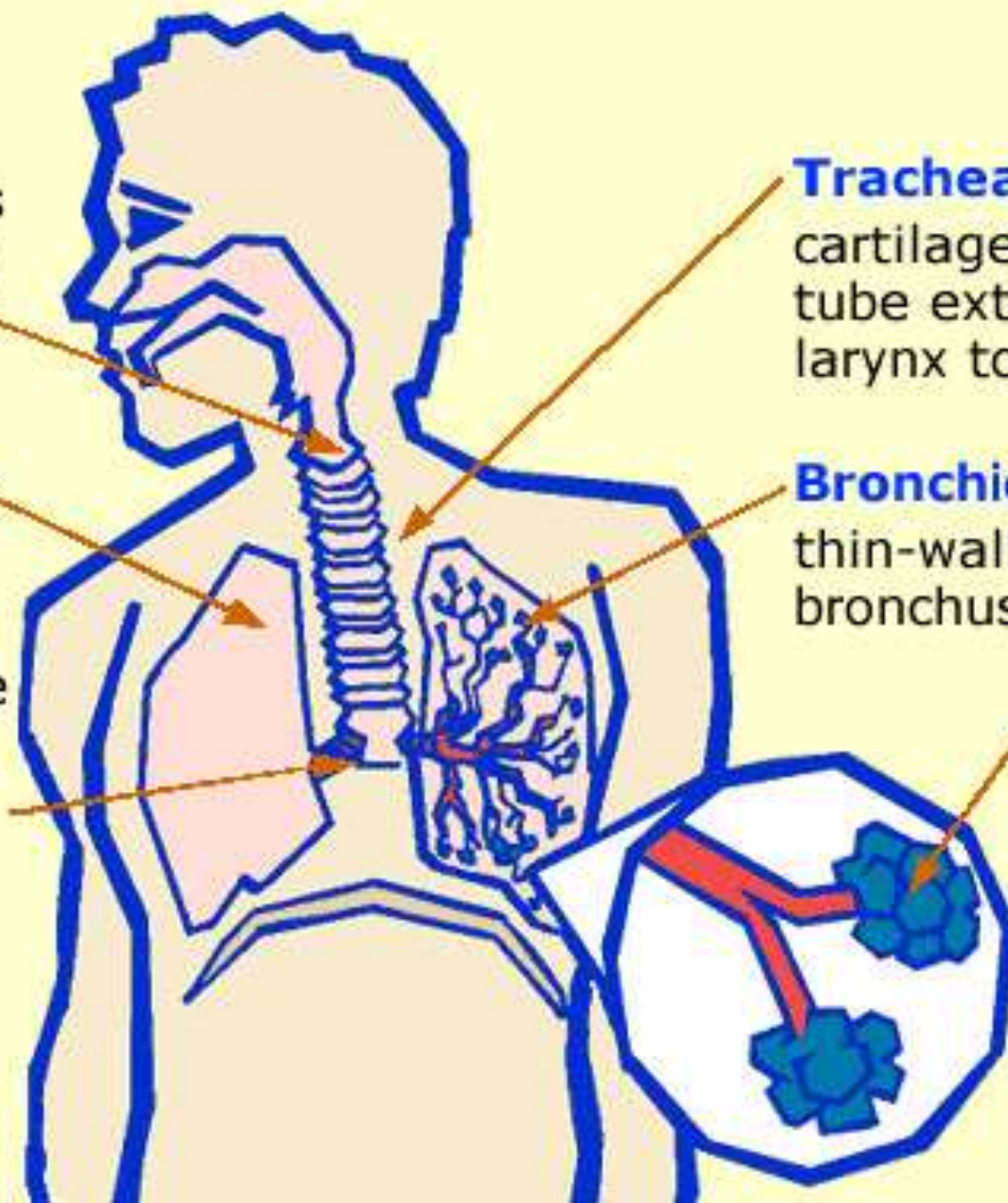
Lung --Organ specialized for the exchange of gases between the blood and the atmosphere

Bronchus --One of the two primary divisions of the trachea that leads into the lungs

Trachea --Windpipe; cartilage-reinforced tube extending from larynx to bronchi

Bronchiole --A thin-walled branch of a bronchus

Alveoli --Clusters of air sacs that are the site of gas exchange for the human body





105. What technology technique is employed during breeding in captivity in order to restore the populations of endangered and threatened species to safe levels?

A. Artificial insemination

B. Cobalt radiation

C. Cross pollination

D. Genetic engineering

Artificial Insemination



Advantages

- *AI helps prevent the spread of infectious or contagious diseases*
- *Fast increase of genetic development and production gain*
- *It enables breeding between animals in different geographic locations*
- *AI is a powerful tool when linked to other reproductive biotechnologies, such as: Estrus Synchronization, Semen Sexing and Multiple Ovulation Embryo Transfer (MOET)*



106. What is the molecule that allows plants to capture energy from sunlight?

A. ATP

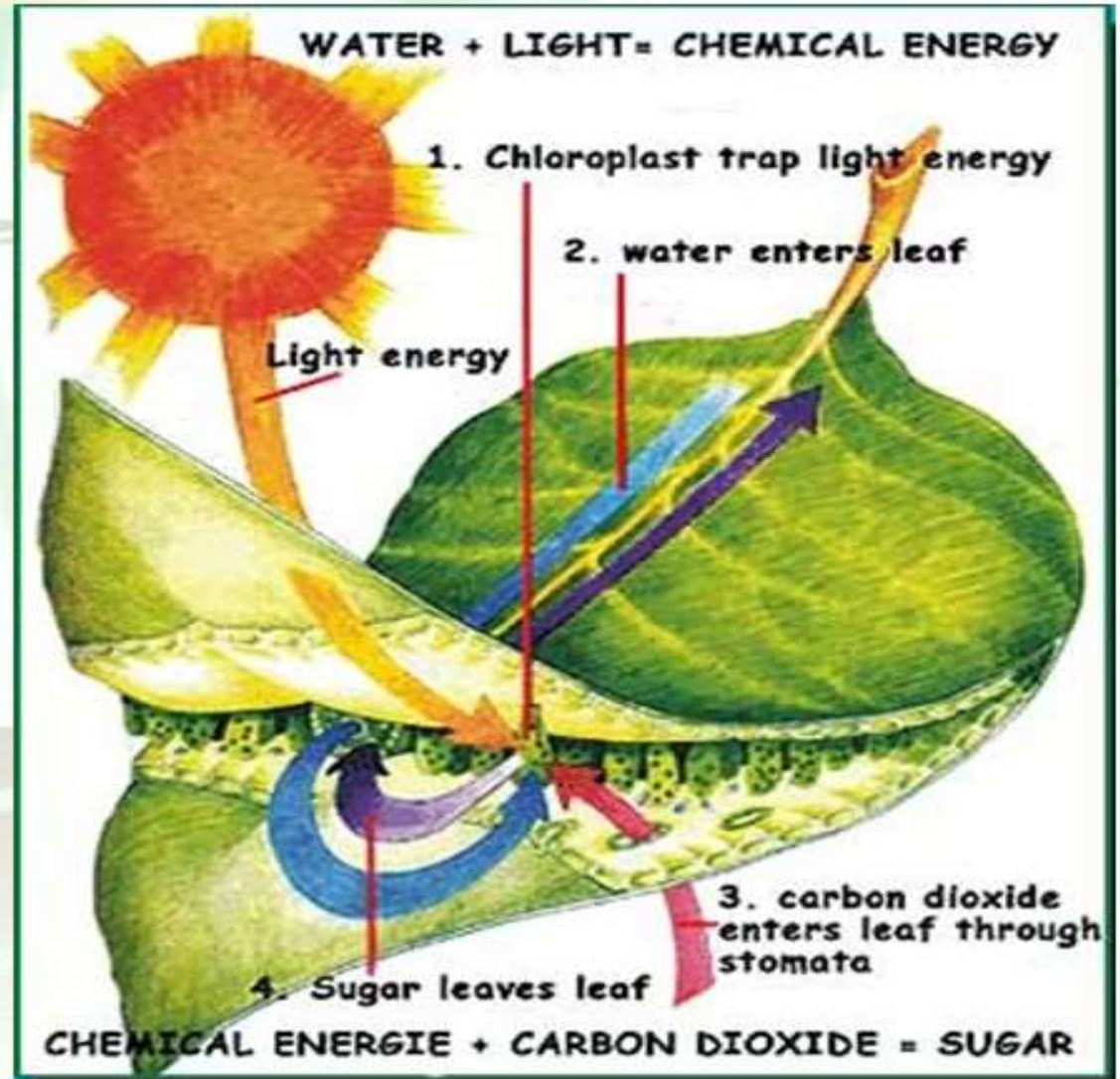
B. Oxygen

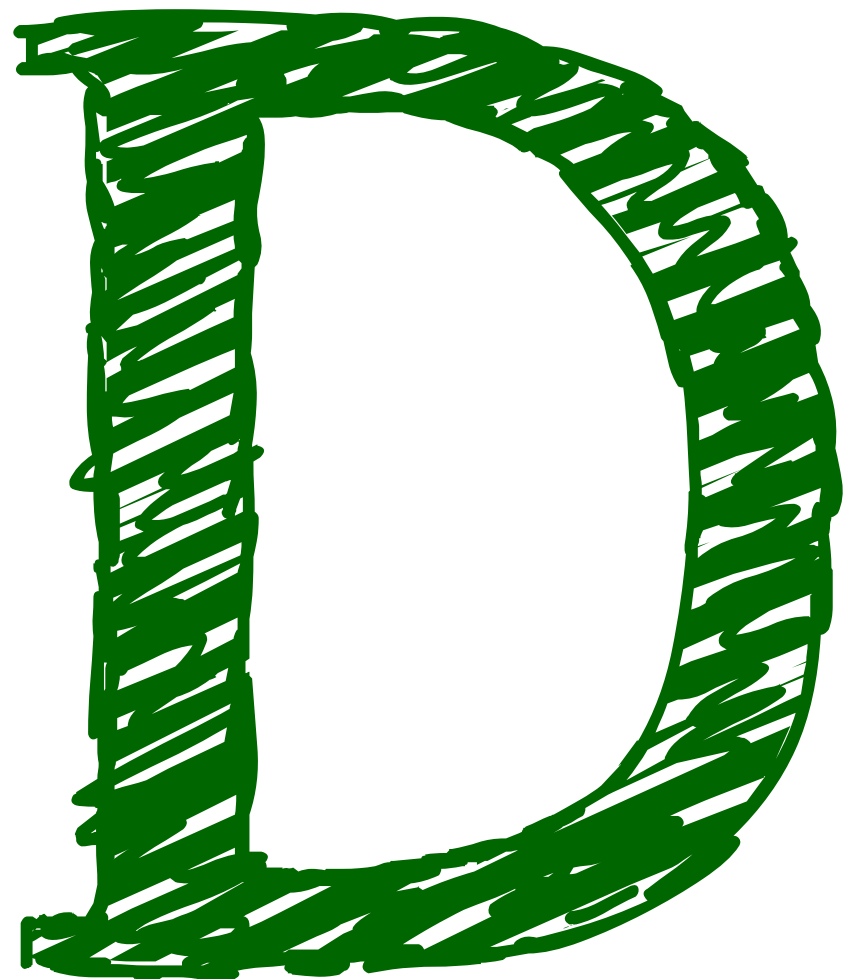
C. Carbohydrates

D. Chlorophyll

How Photosynthesis Works

- CO_2 enters plant via **pores- stomata** in leaves
- water-absorbed by **roots** from soil
- membranes in chloroplasts provide sites for reactions of photosynthesis
- chlorophyll molecules in thylakoids capture energy from sunlight
- chloroplasts rearrange atoms of inorganic molecules into sugars & other organic molecules





107. Dengue is an infectious disease transmitted by an organism classified under which phylum?

- A. Chordata
- B. Arthropoda
- C. Hemichordata
- D. Echinodermata

Invertebrate: Mosquitoes

Scientific name: *Anopheles gambiae*

Genus: Culicidae

Species: Culiseta

Kingdom: Animalia

Phylum: Arthropoda



Interesting Fact: An adult female mosquito consumes about 5-millionths of a liter in a single blood meal.

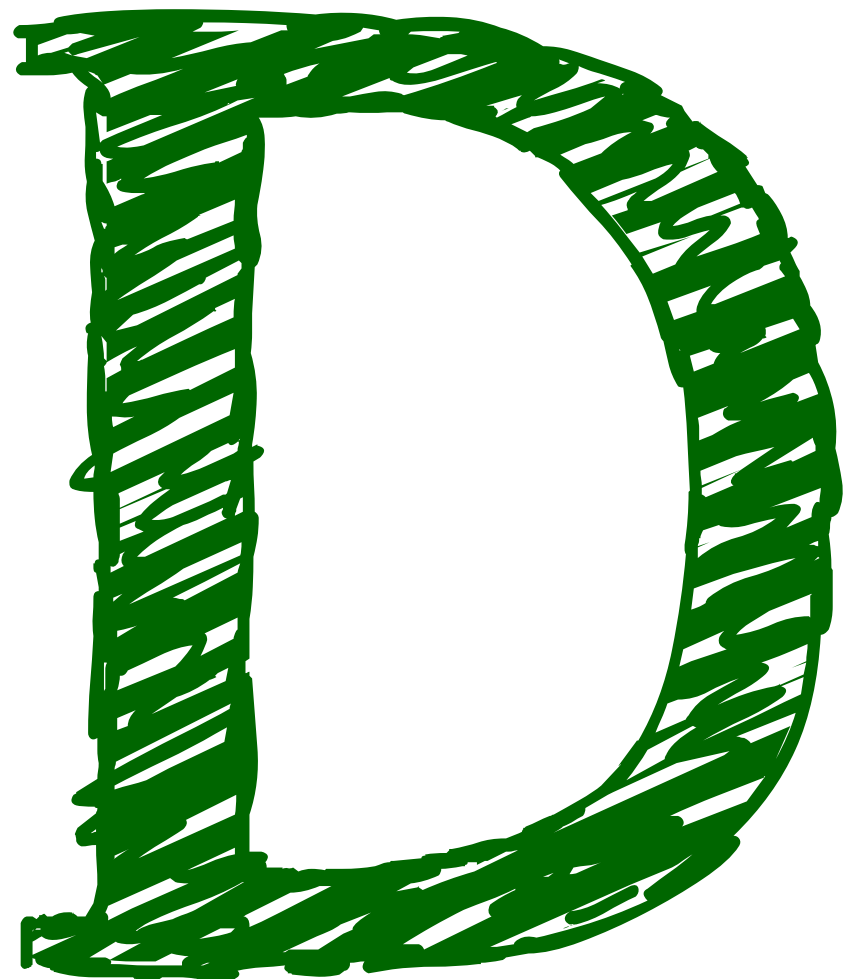


108. The source of water pollution is when water from a river is mixed with the contents of a septic tank, this called _____.

- A. water condensation
- B. hydraulic theory
- C. water manipulation
- D. hypertrophication

- **Eutrophication** (Greek: *eutrophia*—healthy, adequate nutrition, development; German: *Eutrophie*) or more precisely **hypertrophication**, is the ecosystem response to the addition of artificial or natural substances, mainly phosphates, through detergents, fertilizers, or sewage, to an aquatic system. One example is the "bloom" or great increase of phytoplankton in a water body as a response to increased levels of nutrients. Negative environmental effects include hypoxia, the depletion of oxygen in the water, which may cause death to aquatic animals.



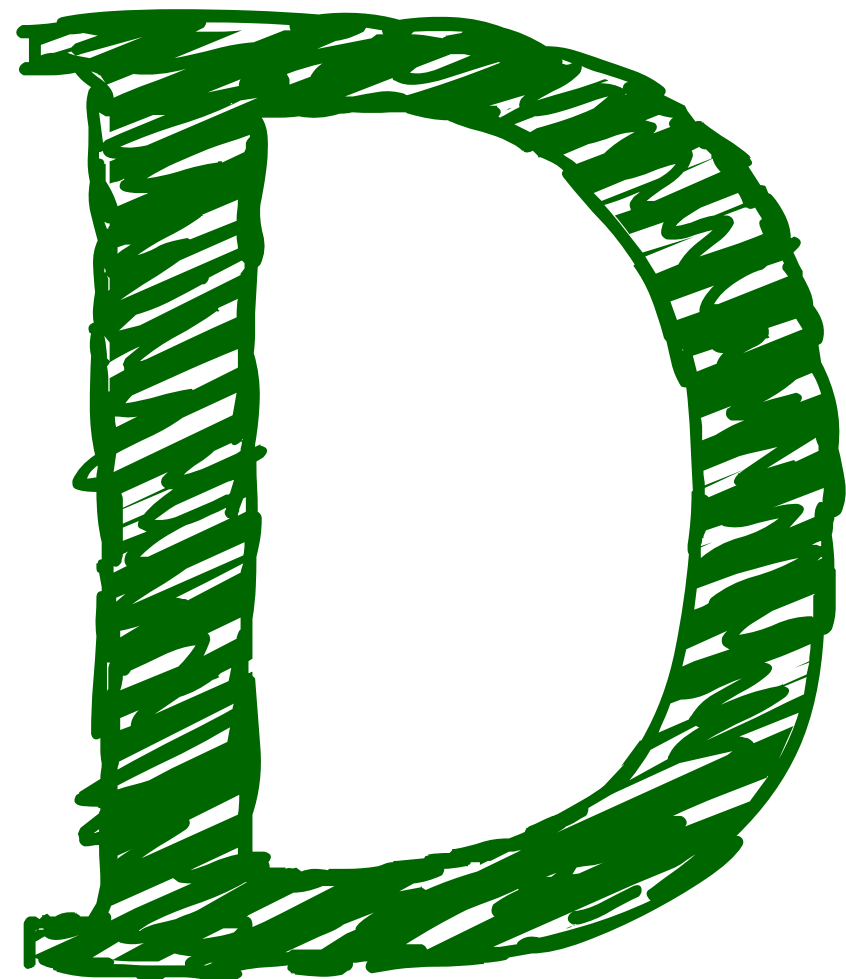


109. What is unique feature of fungi?

- A. They contain mitochondria
- B. They contain genetic material
- C. Some are capable of bioluminescence
- D. They combined chitins and glucans in their cell wall

Fungal Characteristics

- Eukaryotic
- Multicellular
- Heterotrophic
- Absorb nutrients - may be saprobes (absorb from dead material), parasites, or mutualistic symbionts (with algae make lichen).
- Secrete powerful hydrolytic enzymes
- Cell walls contain chitin, an amino sugar polysaccharide also found in arthropod exoskeletons
- Lack flagella

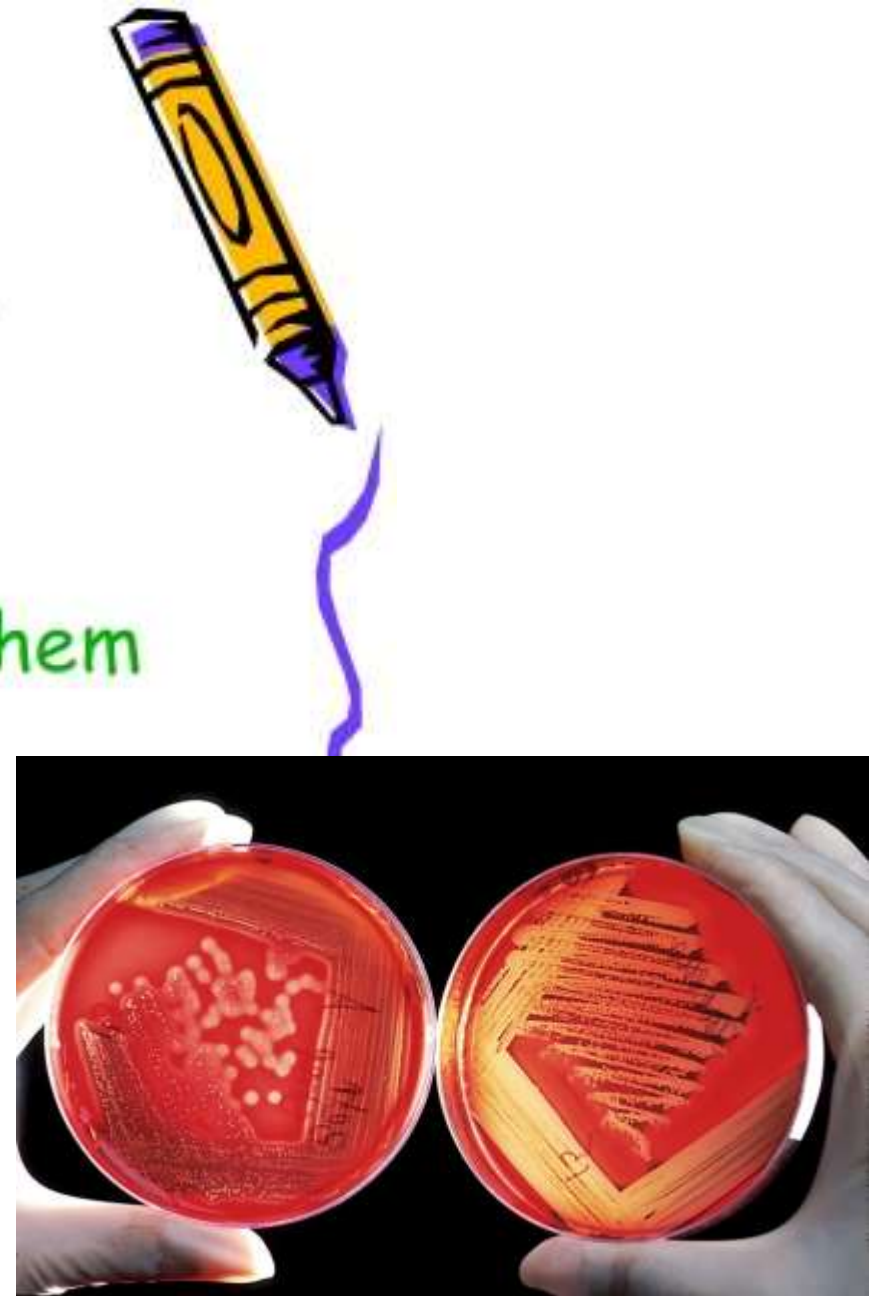


110. What diagnostic test allows microorganism to multiply in a medium under controlled laboratory condition in order to determine the cause of an infectious disease?

- A. Microbial culture
- B. Chest x-ray
- C. Urinalysis
- D. Complete blood count

Microbiological culture

Method of multiplying microbial organisms by letting them reproduce in predetermined culture media under controlled laboratory conditions





111. What organism would most likely be in an arctic environment?

A. Maya bird

B. Turtle

C. Walrus

D. Crocodile



walrus



seal



polar bear



Arctic fox



reindeer



moose



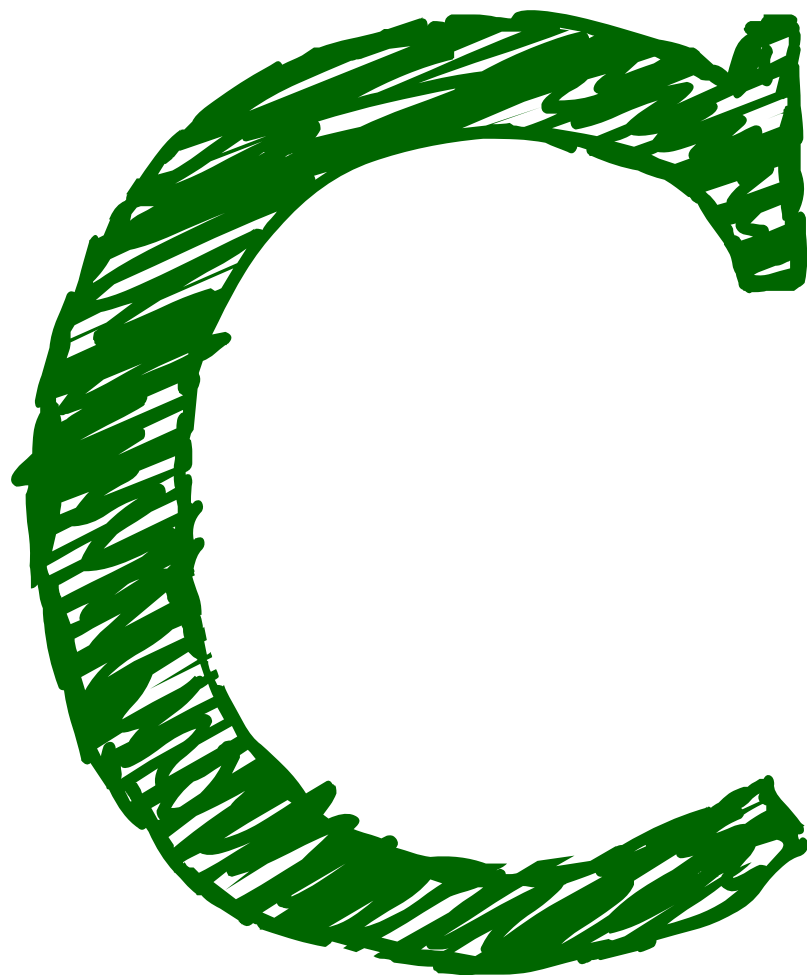
penguin



snowy owl



Arctic dog

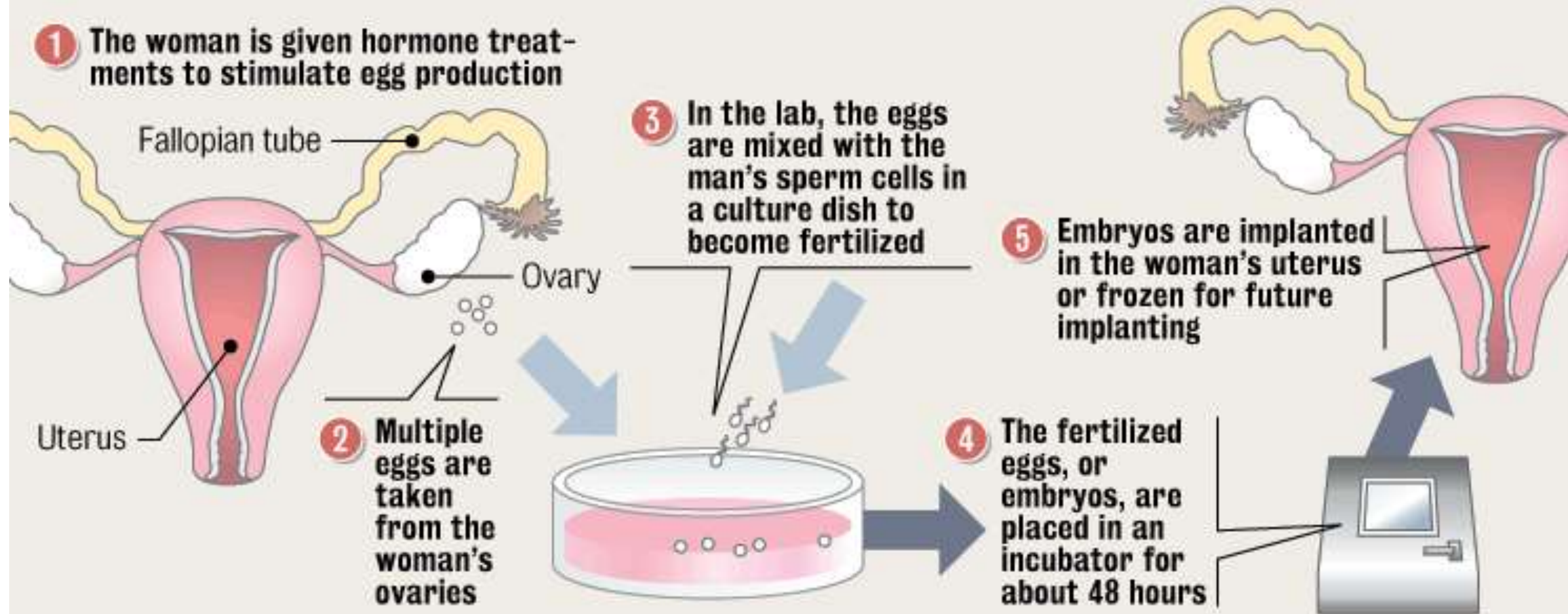


112. The union of an ovum and sperm inside a test tube is _____.

- A. embryology
- B. in vitro fertilization
- C. in vivo fertilization
- D. pollination

THE IN VITRO FERTILIZATION PROCESS

Used as a remedy for infertility, a woman's egg cells are combined with a man's sperm cells outside the uterus. The fertilized egg is then implanted in the woman's uterus and, if successful, begins the pregnancy cycle. The first baby realized from this fertilization method was born in 1978.





113. What are the small hair-like structures that serve a sensory function in eukaryotic cells?

A. Cilia

B. Centromere

C. Actin filaments

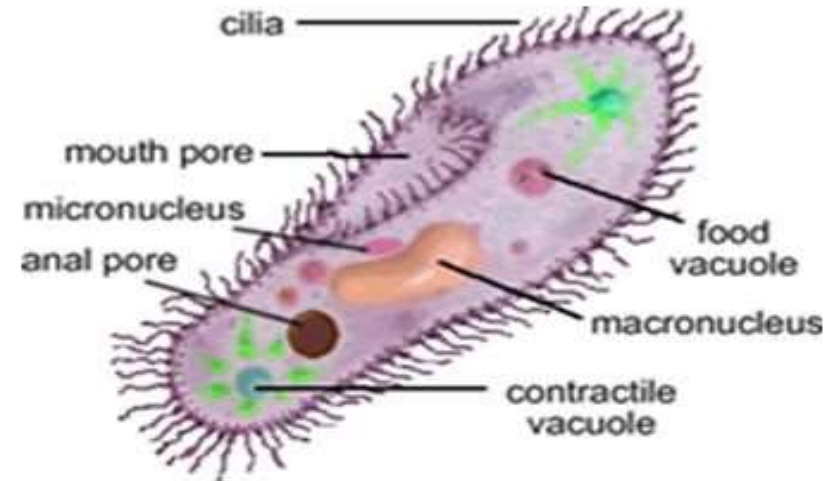
D. Plasma membrane

Movement

<https://www.youtube.com/watch?v=QGAm6hMysTA>

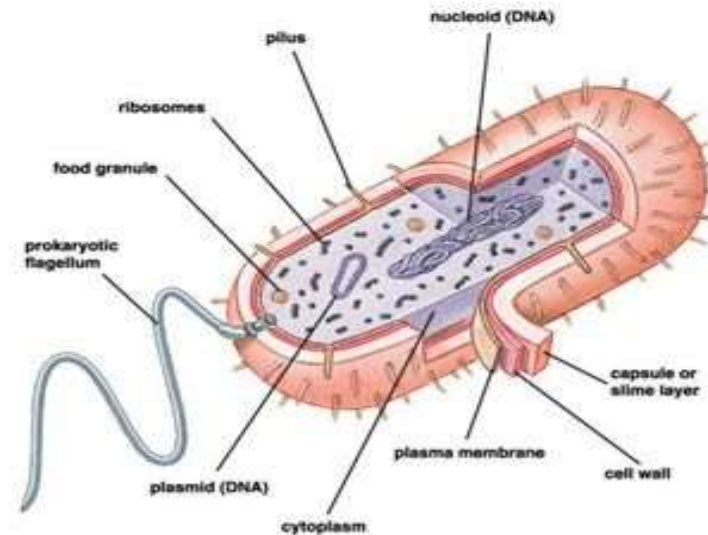
■ Cilia

- Many hair like structures
- Often used for movement
- Non-motile cilia serve as sensory organelles



■ Flagella

- Single, whip like tail used for movement
- Found in: bacteria, protists, specialized plant, animal and fungi cells





114. Which of the following process best explains the evolution of life forms or gradual transformation of species?

A. New combination of genes

B. Natural selection

C. Isolation

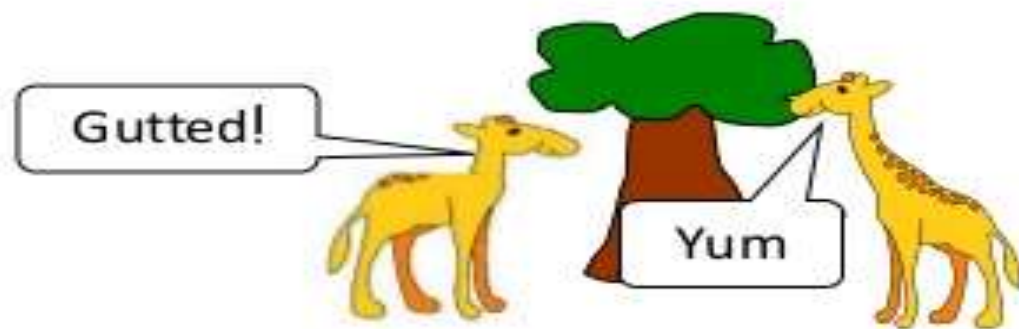
D. Genetic drift

Natural Selection

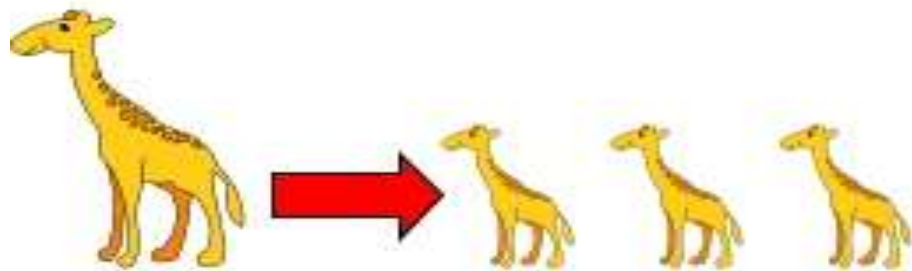
- 1) Each species shows variation:



- 2) There is competition within each species for food, living space, water, mates etc.



- 3) The “better adapted” members of these species are more likely to survive – “Survival of the Fittest”



- 4) These survivors will pass on their better genes to their offspring who will also show this beneficial variation.

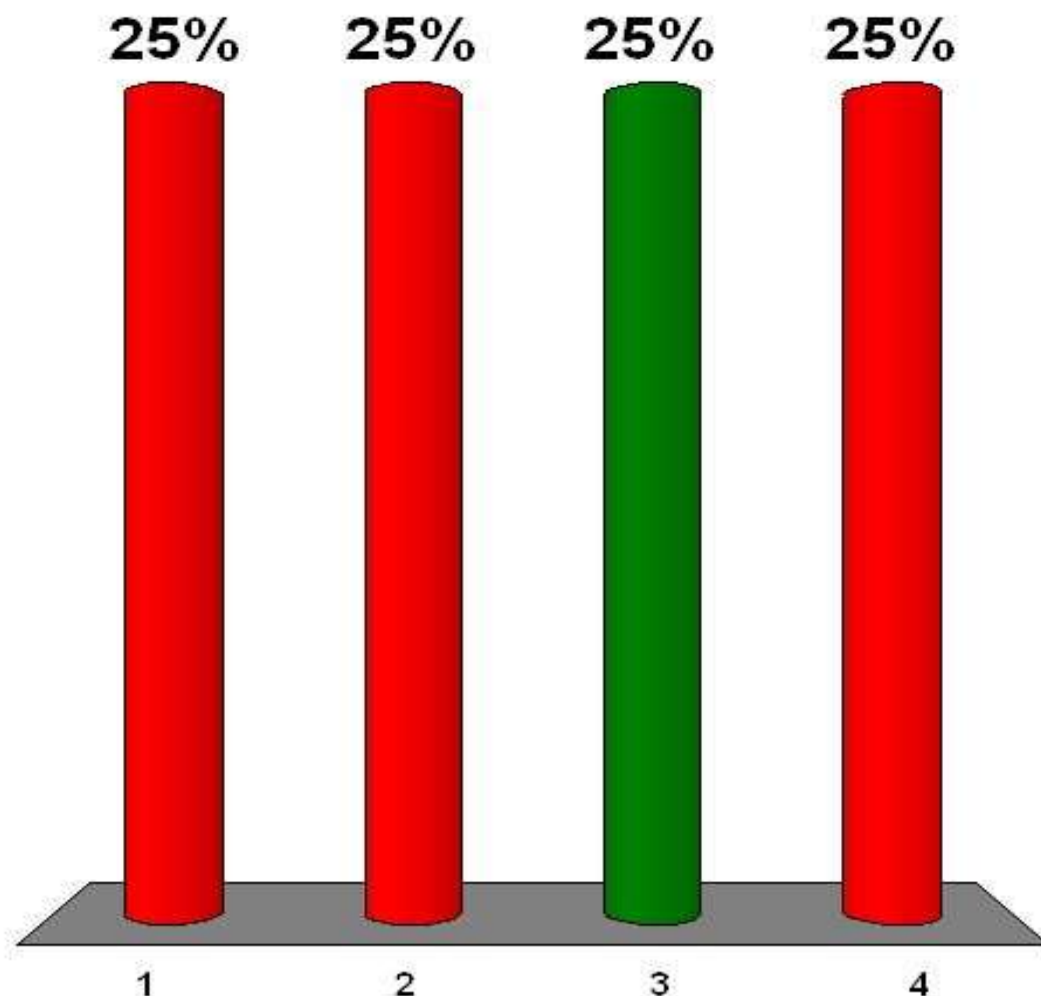
B

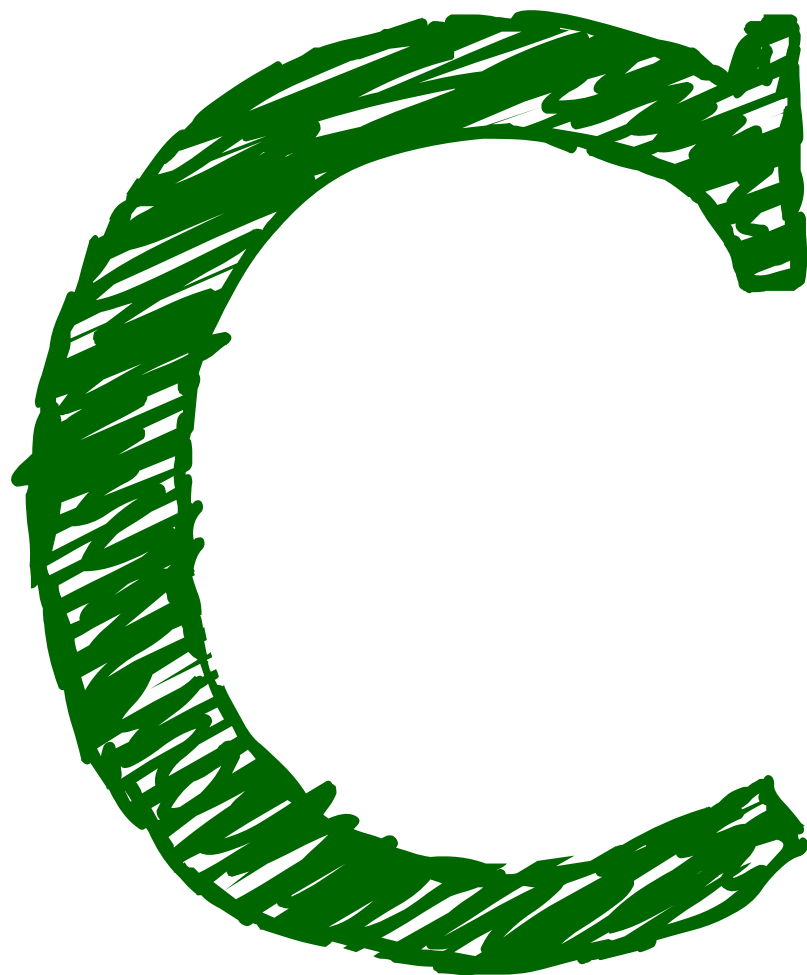
115. Which technology can lead to the formation of a new genetic trait?

- A. Tissue culture
- B. Embryo transplant
- C. Exposure to radiation
- D. Asexual reproduction

In many kinds of organisms, inheritable differences are due mostly to

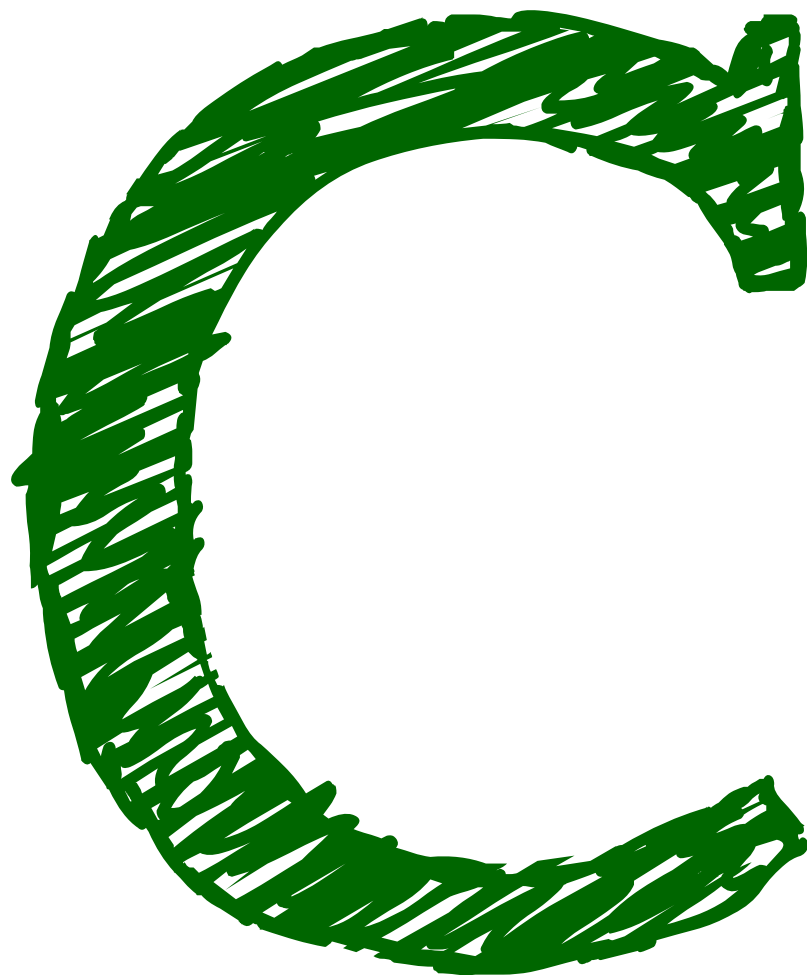
1. mutations during gamete formation.
2. polygenic traits.
3. gene shuffling during gamete formation.
4. the effects of radiation.





116. Of the following, which is an example of natural altered state of awareness?

- A. Sedation
- B. Intoxication
- C. Sleep
- D. Drug “high”



117. Object, person, or situation
which when perceived causes action
or reaction?

A. sound

B. stimulus

C. sight

D. scent

B

118. Which characterizes psychologically healthy people?

- A. concerned with goals beyond themselves
- B. feels superior over others
- C. can't accept others as they are
- D. pre-occupied only with their interest



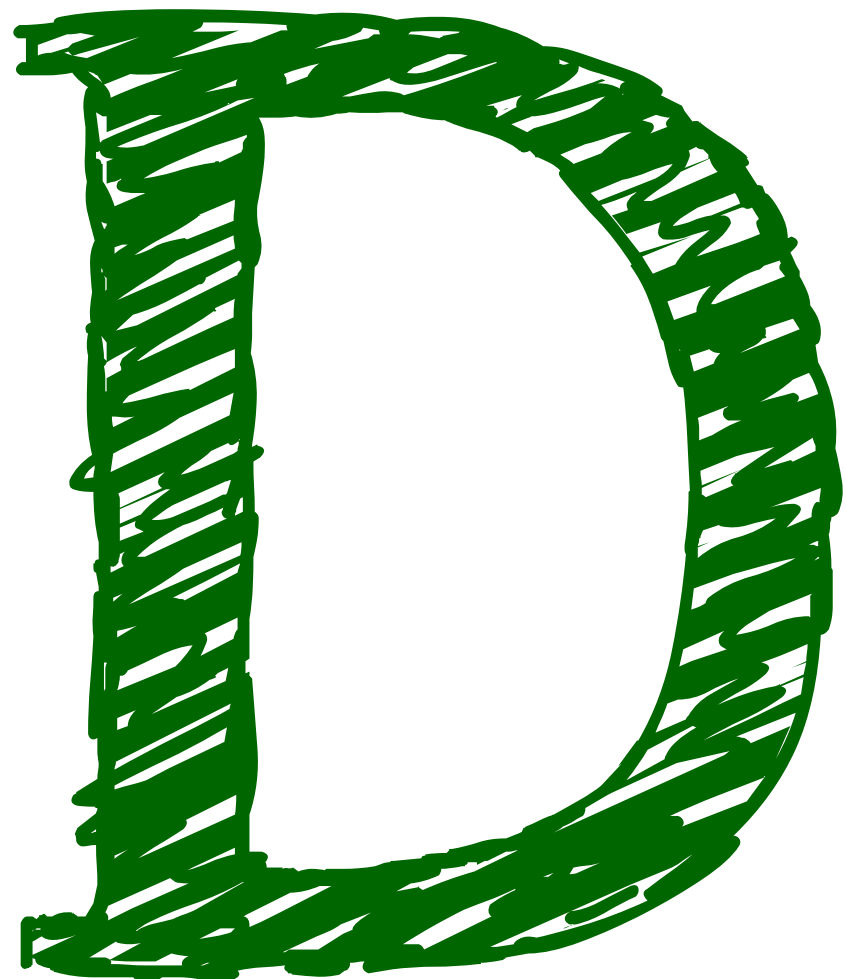
119. Who was the orator of the Reform Movement and first editor of La Solidaridad?

- A. Jose Rizal
- B. Marcelo del Pilar
- C. Juan Luna
- D. Graciano Lopez Jaena

Mga Repormista

- Graciano Lopez Jaena
 - Dakilang Mananalumpati.
 - Nagsulat ng mga kuwentong satiriko laban sa katiwalian ng mga prayle at opisyal na Espanyol.
 - Unang patnugot ng La Solidaridad.





120. Which is the fundamental law of the Philippines?

- A. The Philippine Constitution
- B. The Ten Commandments
- C. The Civil and Crime Codes
- D. The Bill of Rights

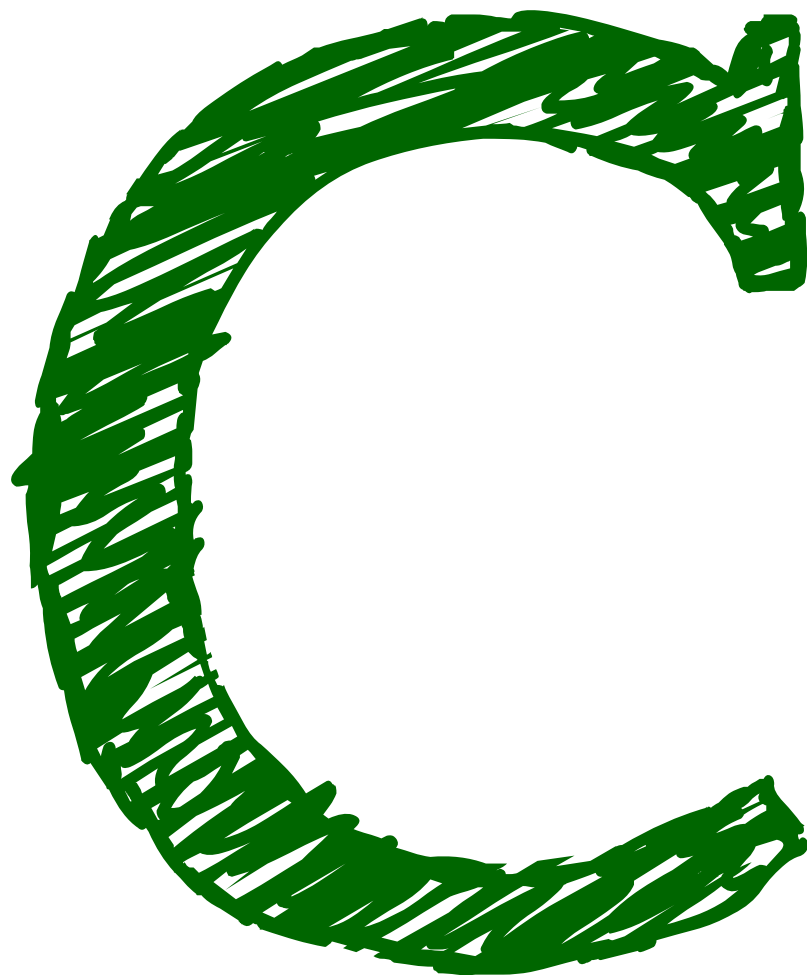


121. What law passed on August 1909 by the U.S. Congress established a partial free foreign trade for the Philippines?

- A. Underwood-Simmons Act
- B. Bell Trade
- C. Payne-Aldrich Act
- D. Treaty of General Relations

Trade, Commerce & Industry

- Free trade relations with the US
 - 1902 : Reduction of 25% was allowed on goods coming from the Philippines
 - 1909 : American Congress passed the Payne-Aldrich Tariff Act
 - All PH exports, except rice, were allowed to enter American markets free of duty w/in certain quota limits
 - 1913 : American Congress passed the Simmons-Underwood Tariff Act
 - Quota limitations were abolished on PH export product like hemp, sugar and tobacco



122. Which state policy provides for preferential attention to the welfare of the less fortunate members of the Philippine society?

- A. Bill of rights
- B. Social justice
- C. Distributive justice
- D. Criminal justice

TYPES OF JUSTICE

➤ DISTRIBUTIVE JUSTICE

- Requires distributing society's benefits and burdens fairly.

➤ RETRIBUTIVE JUSTICE

- Requires fairness when blaming or punishing persons for doing wrong.

➤ COMPENSATORY JUSTICE

- Requires restoring to a person what the person lost when he or she was wronged by someone.



Discussion of Social Justice

- No clear cut definition could encompass the scope of social justice and very little direct discussion is held on its meaning in the field of education – it is assumed there is a universal understanding of its meaning.
- Agreement on some common attributes:
 - Fairness
 - Equity
 - A general respect for basic human rights
- Description of such a broad term cannot exist in isolation. Rather, a fluid understanding and pliable implementation needs to happen in order for a very interpretative concept to work effectively and timelessly in an evolving world.

B

123. Which condition contradicts the social justice mandated in the 1987 Constitution?

- a. Sense of dependency among citizens
- b. Creation of economic opportunities for all
- c. Elimination of cultural inequalities
- d. Equitable distribution of wealth

1987 National Constitution (present). Continued...

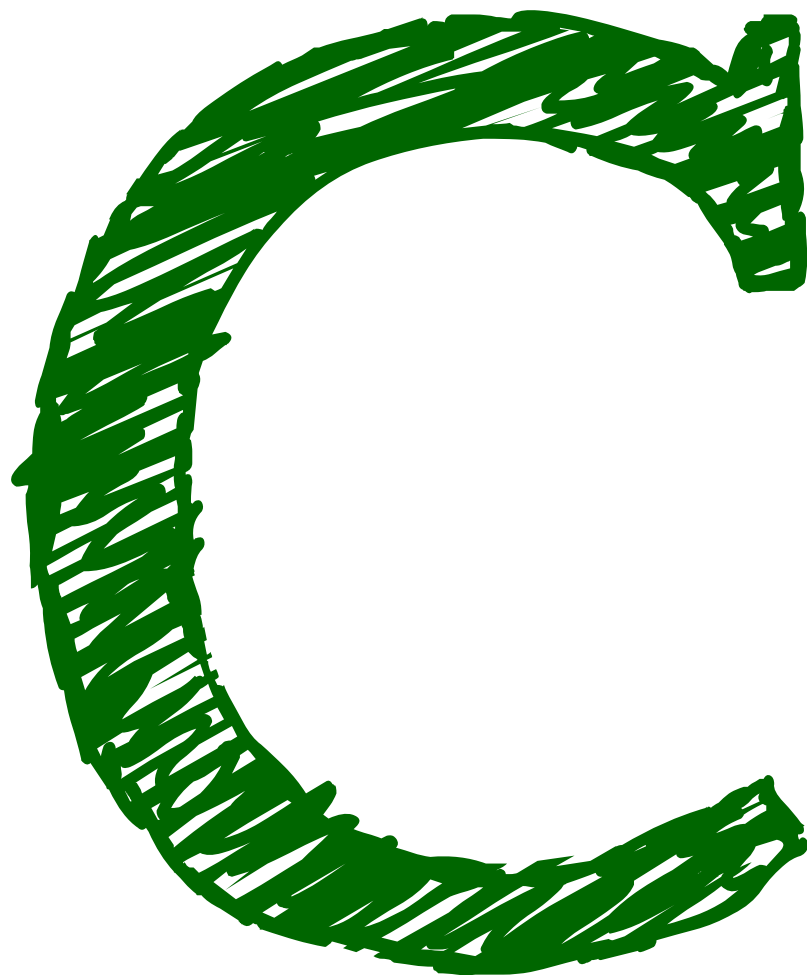
Article 13: Social Justice and Human Rights

- **Section 1.** "The Congress shall give highest priority to the enactment of measures that protect and enhance the right of all the people to human dignity, reduce social, economic, and political inequalities, and remove cultural inequities by equitably diffusing wealth and political power for the common good."
- **Section 13.** "The State shall establish a special agency for disabled person for their rehabilitation, self-development, and self-reliance, and their integration into the mainstream of society."



124. What is the term for the unpleasant sensation or frustration when a traditional-bound Filipino goes to the U.S and observes the open display of love and emotion?

- A. Culture relativism
- B. Culture cosmopolitanism
- C. Culture shock
- D. Culture phobia



125. One of the pledges of the President of the Philippines is to defend its:

A. Constitution

B. Land

C. Citizens

D. Islands

SEC.5. Before they enter on the execution of their office, the President , the Vice-President, or the Acting President shall take the following oath or affirmation:

“ I do solemnly swear (or affirm) that I will faithfully and conscientiously fulfill my duties as President, or Vice-President, or Acting President of the Philippines. Preserve and defend its Constitution, executive it's laws, do justice to everyman, and consecrate myself to the service of the Nation. So help me God. ”(In case of affirmation, last sentence will be omitted.)



126. When teachers brand as “pilosopo” students who ask questions, they communicate the NEGATIVE sociological principle that _____

- A. Asking questions in class is a sign of inattentiveness
- B. Passive and conformity are worth perpetuating
- C. Asking questions in class is a sign of ignorance
- D. teachers are all knowing and cannot be questioned

B

127. Which law set a full free trade policy abolishing the quota limitation on Philippine exports to the United State?

A. Payne-Aldrich Act

B. Underwood-Simmons Act

C. Hare-Hawes Cutting Act

D. Bell Trade

Trade, Commerce & Industry

- Free trade relations with the US
 - 1902 : Reduction of 25% was allowed on goods coming from the Philippines
 - 1909 : American Congress passed the Payne-Aldrich Tariff Act
 - All PH exports, except rice, were allowed to enter American markets free of duty w/in certain quota limits
 - 1913 : American Congress passed the Simmons-Underwood Tariff Act
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Philippine Trade Act

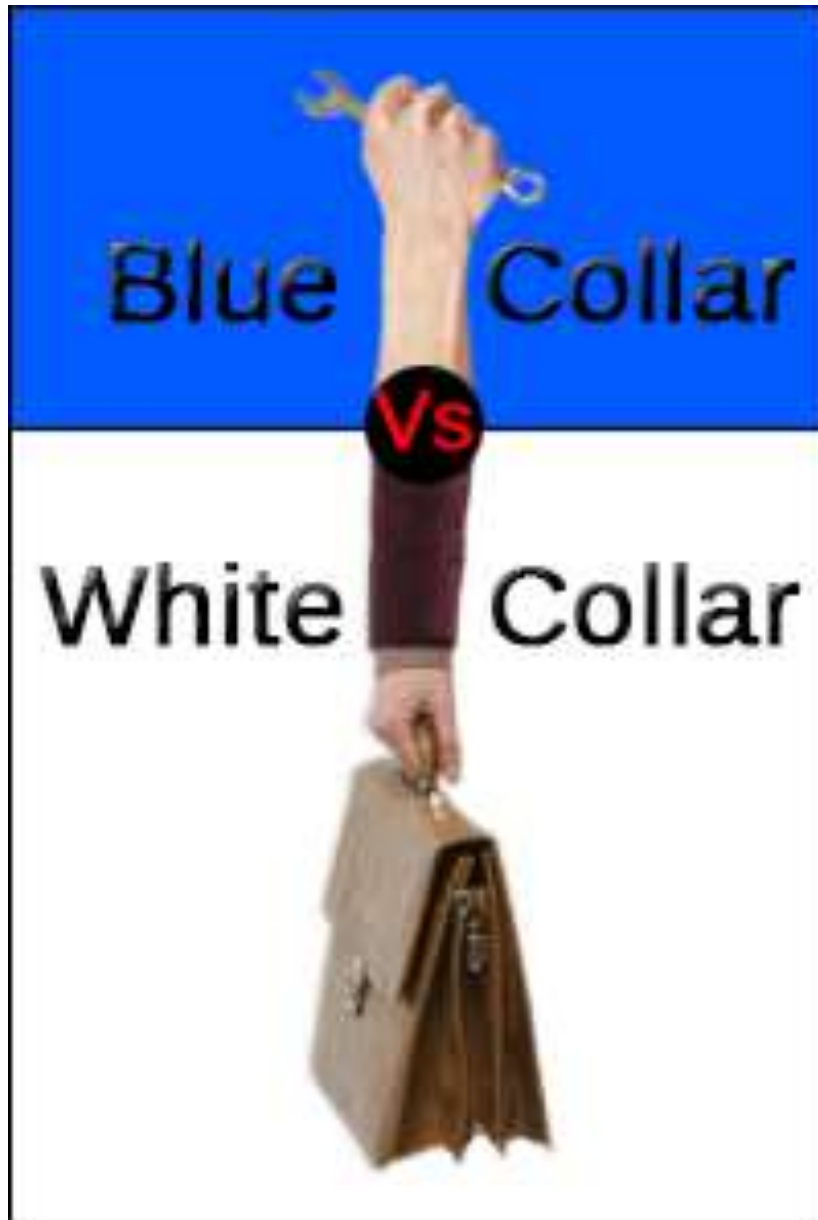
□ 1946

- The Philippine Trade Act, or the Bell Trade Act of 1946 **set rules** for the newly independent country.
- The Philippine **peso** would be **tied to the U.S. dollar**.
- The Philippine **Constitution** would **grant the U.S. equal access** to the country's **natural resources**.
- The act called for **free trade** for the **U.S.** and that the U.S. could **import goods free** of import duties.



128. What problem can result owing to bias among parents and the youth towards manual or technical occupations, e.g. masons, mechanics, plumber, etc.?

- A. Boss mentality
- B. White collar mania
- C. Blue collar mania
- D. Technocrat mind



Blue Collar vs. White Collar



"Dirty job" – work seen as a "noble sacrifice"

- ✧ Work is unpleasant, painful, difficult, and cruel – a dirty job that someone has to do
- ✧ Dirty work is unfit for women

Professionalization of the Academy

- ✧ German research university became popularized
- ✧ "Science is a vocation" – Max Weber
- ✧ Rationality, predictability, and controlling one's environment and outcome would advance society

B

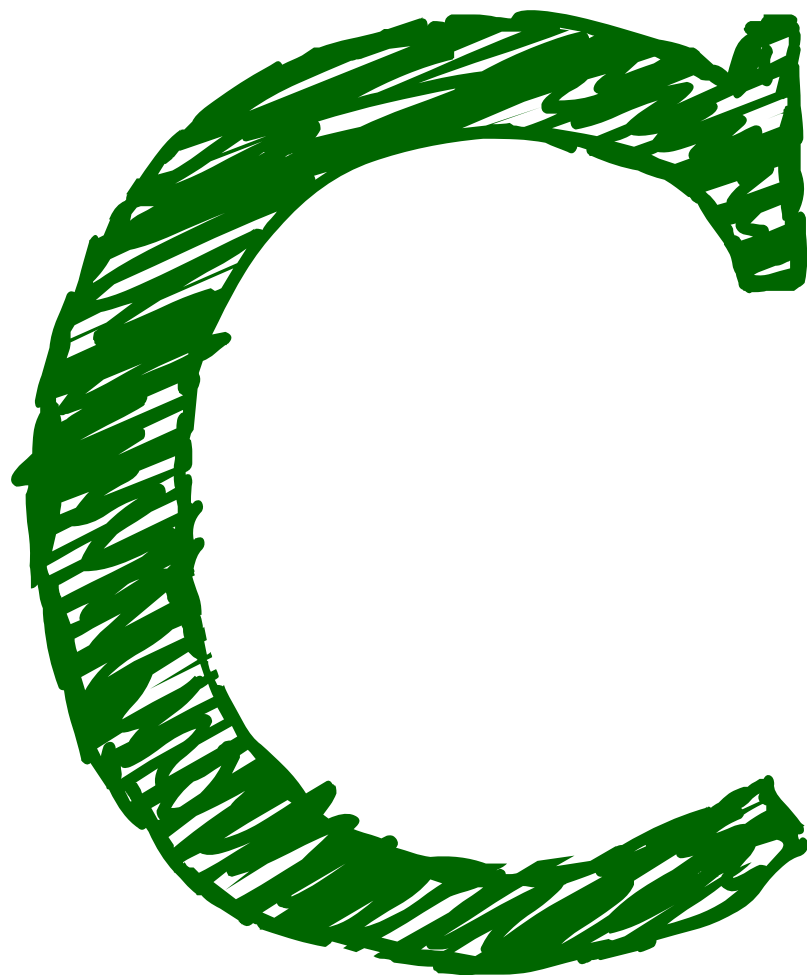
129. When one controls the supply/production of goods, this implies?

A. inflation

B. economic stability

C. monopoly

D. economic equilibrium



130. Which enacted by the U.S. Congress in 1916 established a bicameral legislature in the Philippines with the senator (upper House) and House of Representative (lower House) to be composed of elected Filipinos?

A. Philippine Legislative Law

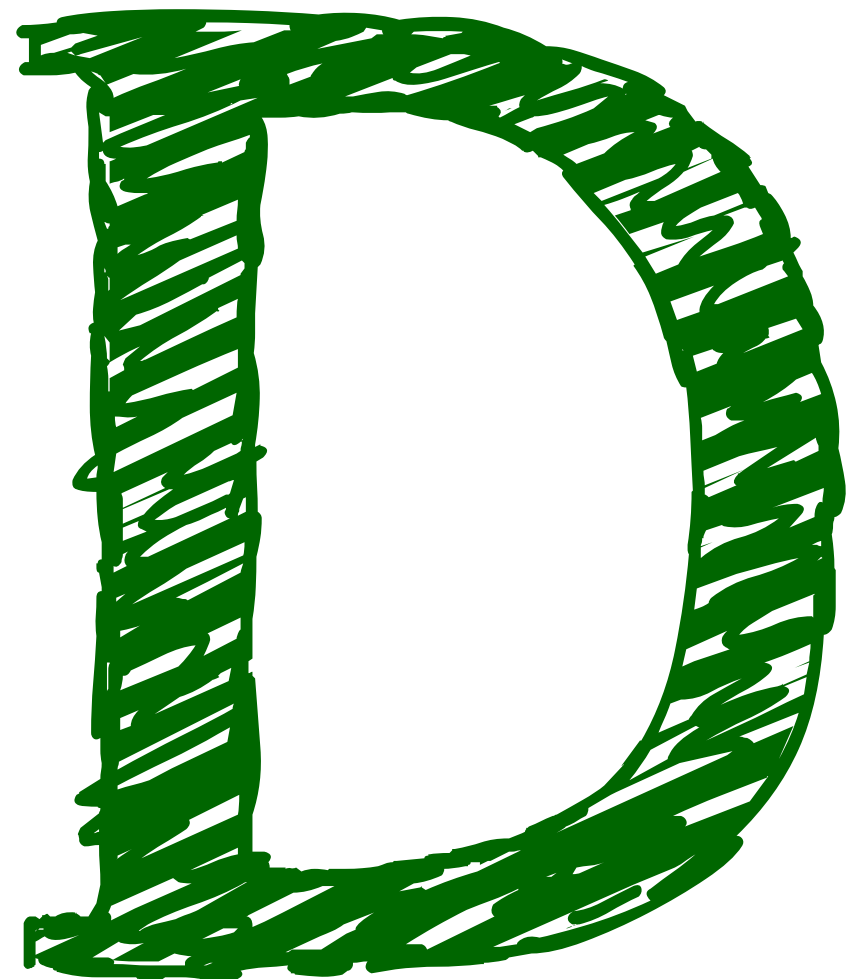
B. McKinley Law

C. Cooper Act

D. Jones Law

The Philippine Legislature, 1915-1935

In 1916, the US Congress enacted the Philippine Autonomy Act also known as Jones Law. The act reorganized the Philippine legislature into a bicameral, Filipino-controlled body. The Senate, which was composed of twenty-four members, replaced the Philippine Commission as the upper chamber. The House of Representatives replaced the Philippine Assembly as the lower chamber. This resulted in the separation of the legislative from the executive branch of the colonial government, which continued to be headed by the American governor-general.



131. What is considered the earliest form of writing in the Philippines?

A. Alibata

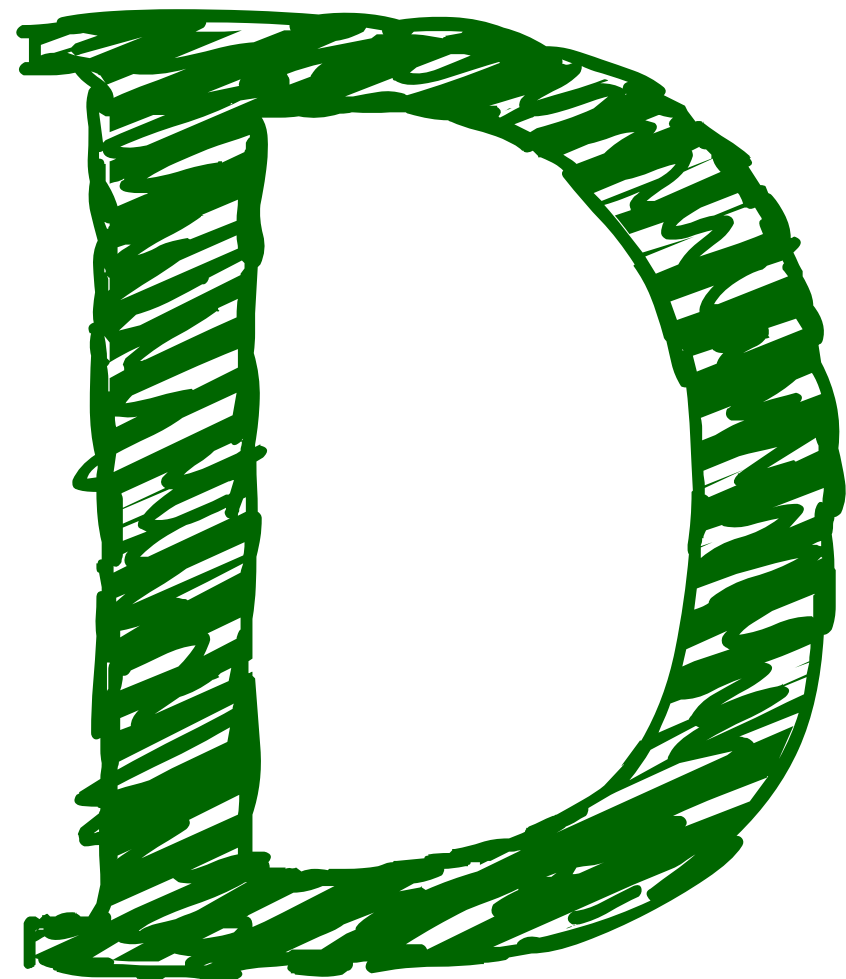
B. Kalantiao

C. Abakada

D. Baybayin

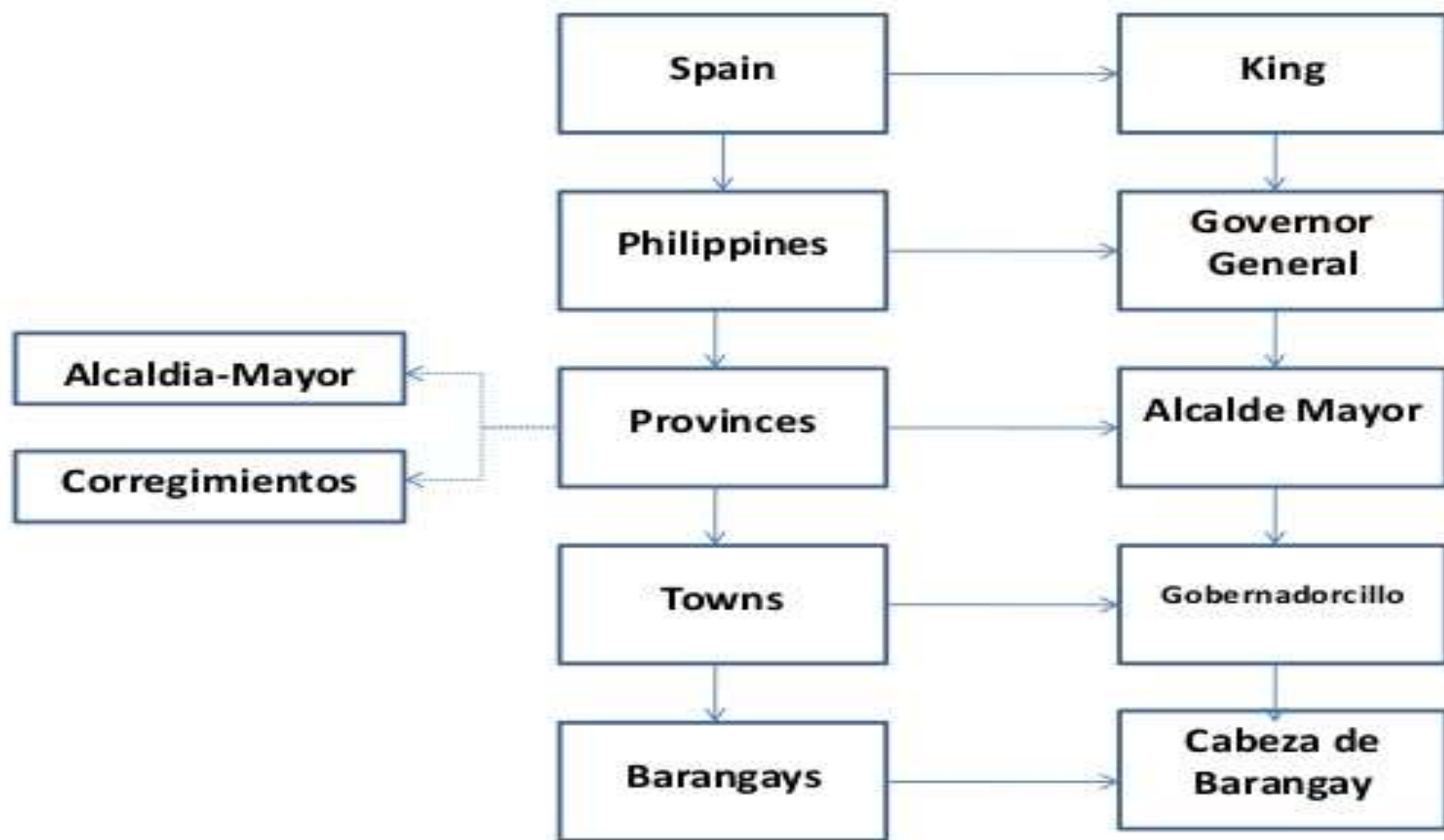
a•li•bá•ta *png*: [*Mal Ara alif + bata*] **1**: alpabetong Arabiko tulad ng pagkakakilala sa Silangan **2**: unang dalawang titik sa alpabetong Arabiko; alif (*a*) at bata (*b*).

bay•bá•yin *png*: [*baybay + in*] **1**: kabuuan ng lahat ng titik ng isang wika **2**: baybay² **3**: pampang **4**: tawag sa sinaunang alpabeto ng mga Filipino



132. During the 17-18th century Hispanic era, what was the highest post a native or Chinese mestizo can attain at the municipal level, equivalent to local bureaucrats namely the mayors today?

- A. Gobernadorcillo
- B. Tenientes de justicia
- C. Cabeza de barangay
- D. Directorcilo



Political Structure in the Philippines during Spanish Colonization



133. During the Hispanic era, native and Chinese male mestizo 16-60 years old is obliged to do forced labor for forty day. How can exemption be made?

- A. Getting a cedula personal
- B. Give sanctorum tax to the church
- C. Paying the bandala
- D. Paying the falla (exemption fee)

Taxation

To support the colony, several forms of taxes and monopolies were imposed.



Direct:

The *tithe* is the payment of the 10% of an individual's annual income to the government. The *sanctorum* is the tax being paid as support to the church. The *tribute(buwis)* is the tax or rent given to the landlord a resident is under. It may be in cash or in kind (tobacco, chickens, produce, gold, blankets, cotton, rice, etc., depending on the region of the country), fixed at 8 reales and later increased to 15 reales.

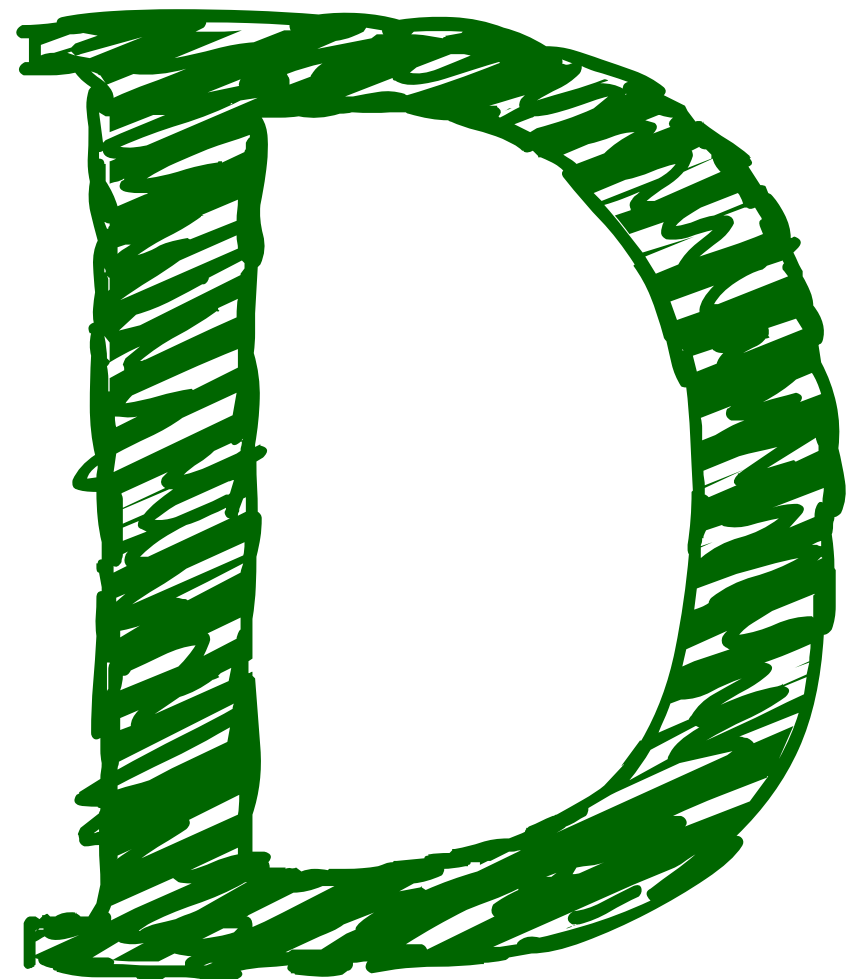
FORCED LABOR (POLO Y SERVICIO)

- 16 to 60 years old must render services for 40 days
- Polistas, name of the laborers
- Falla, exemption fee for the forced labor

GALLEON TRADE

- A trade between Manila and Acapulco of the Chinese and the Spaniards in Manila and in Mexico
- Resulted cultural exchanges but agriculture was neglected and forced labor was more intensified

- Bandala – farmers are forced to sell their produce to the government but would only give them promisory notes.

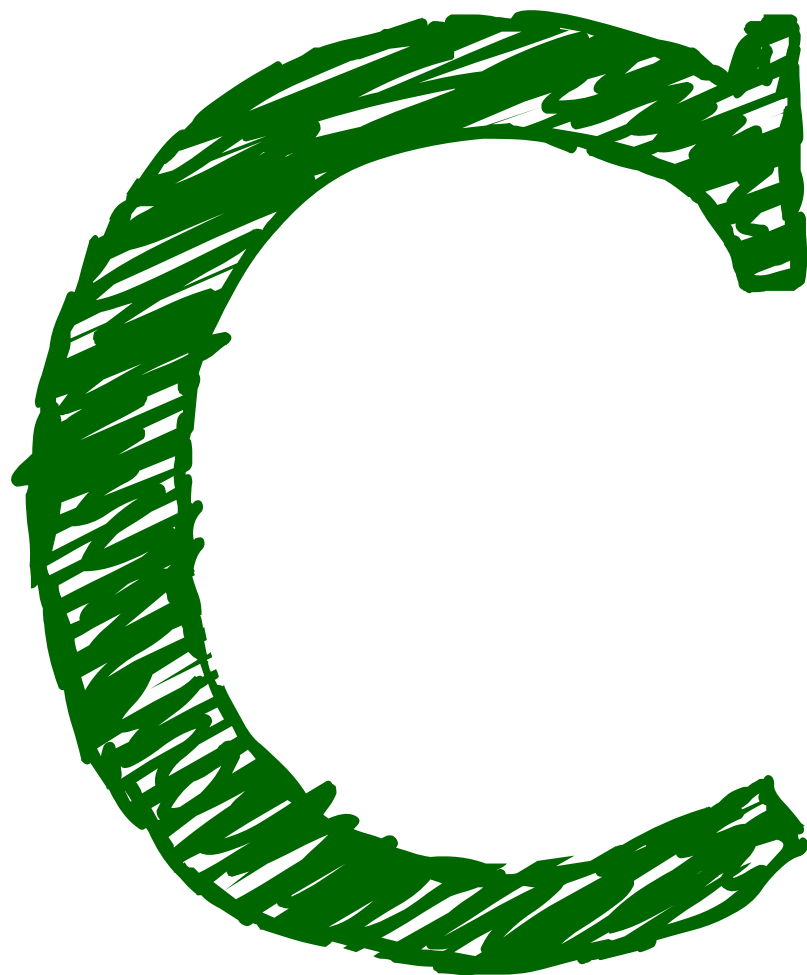


134. Which Spanish policy provided the resettlement of Filipino communities to form town centers of cabeceras?

- A. Assimilation
- B. Encomienda system
- C. Reduccion
- D. Bandala

Reduccion

- Reduccion is the centralization of the Filipino community where churches, convents, *casa real* and plaza complexes can be found. Plazas are where people gather when there are events or celebrations like festivals. By using this system, the Spaniards can easily monitor the movements of Filipinos to prevent protests and to collect the taxes easier. There are also changes in the architectural designs of infrastructures.



135. How can Dr. Jose Rizal's La Liga organization be linked according to present-day social groups/ movements?

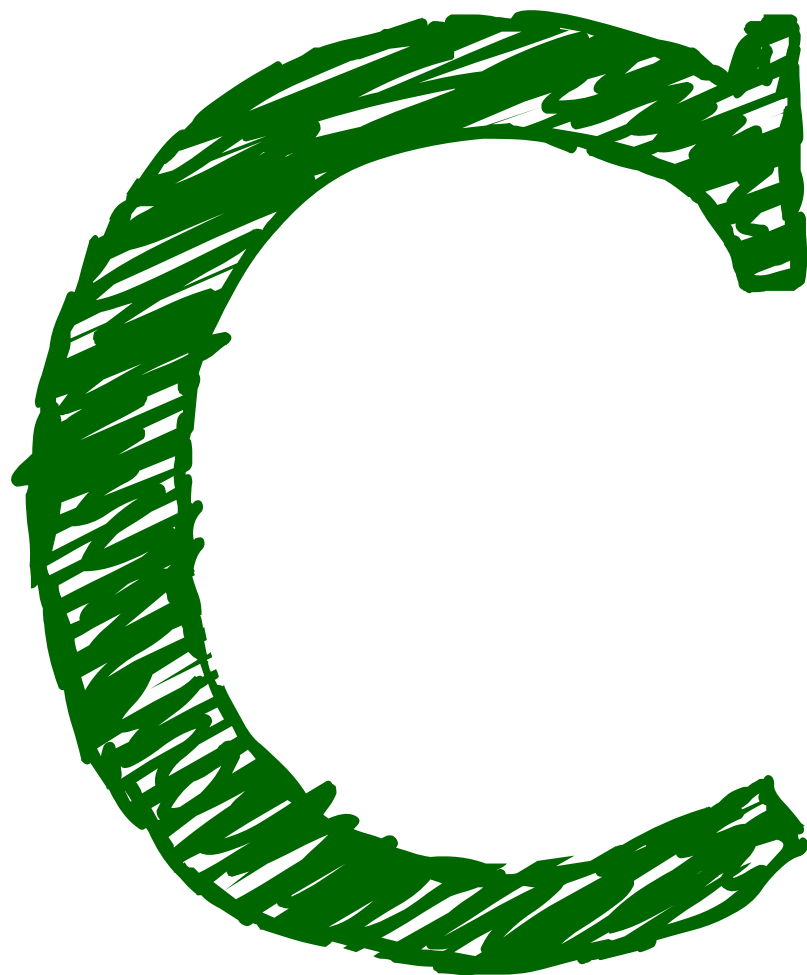
- A. National political party like LP, NP, LAKAS
- B. Shadow government
- C. National socio-civic organization
- D. Street parliamentary movement

ANG KONSTITUSYON NG LA LIGA FILIPINA

Ang layunin ng La Liga Filipina gaya ng isanasaad ng konstitusyon nito ay ang mga sumusuonod:

- ☐ Mapag-isa ang buong kapuluan sa isang katawang buo, malakas, at magkakauri
- ☐ Proteksiyon ng bawat isa para sa pangangailangan ng bawat isa
- ☐ Pagtatanggol laban sa lahat ng karahasan at kawalang katarungan
- ☐ Pagpapaunlad sa Edukasyon, agrikultura, at pangangalakal
- ☐ Pag-aaral at pagpapairal ng mga pagbabago





136. What was the social relevance of Noli and Fili to conditions among Filipino during the late 19th century period?

A. Unfulfilled secularization movement of Gom-Bur-Za

B. Need for violent uprising against Hispanic imperial rule

C. Rule of friars over Hispanic colonial governments

D. Human civil oppression of native Filipinos



137. Who was the literary figure known as “Huseng Sisiw”?

- A. Jose de la Cruz
- B. Jose Rizal
- C. Juan Matapang Cruz
- D. Francisco Baltazar

Si Huseng Sisiw...

- ▶ Ikinapit sa kanya ang taguring ***Huseng Sisiw*** dahil kung may nagpapagawa sa kanya ng patulang liham ng pag-ibig, ang hinihingi niyang kabayaran ay sisiw.
 - ▶ ayon kay Julian Cruz Balmaseda, ang nagturo umano kay Francisco Balagtas kung paano sumulat ng tula.
- 



138. Famous ballerina who brought ballet to children of various economic levels

- A. Ikea Conde
- B. Aiza Seguerra
- C. Nora Aunor
- D. Liza Macuja – Elizalde



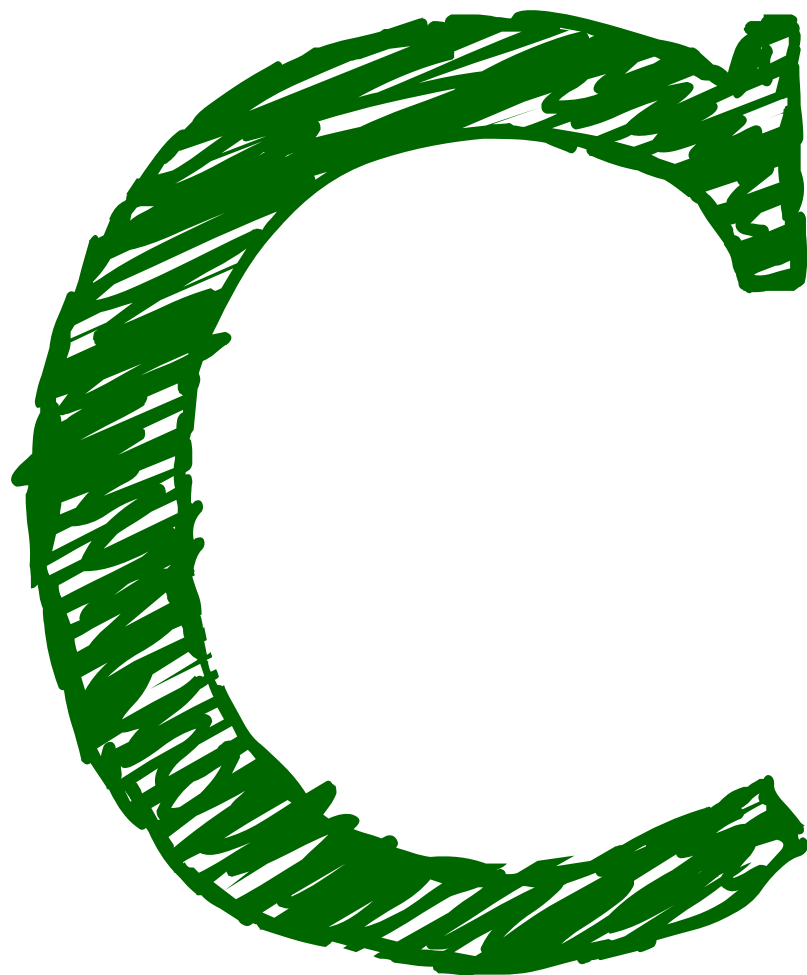
139. Art is product of man's need to express himself with a high degree of sensitivity towards his environment. Who is the painter of fisher folks and farmers of his hometown in Rizal?

- A. Vicente Manansala
- B. Carlos V. Francisco
- C. Anita Magsaysay
- D. Fernando Amorsolo



140. What is the best reason for holism in art education?

- A. Everyone is a potential artist with talents of his/her own
- B. Art means the beautiful in human creativity
- C. Aesthetic development is for total growth
- D. We can be unique artists in our degree of growth



141. Art is the handmaid of religion. Why do Jews and Muslims allow signs and symbols and not human being in religious art?

- a. People might worship images themselves
- b. Signs and symbols display hidden mysteries
- c. Human figures are difficult art forms
- d. they display more traditional themes



142. Taoism is another Chinese philosophy about the same period as Confucianism, but more spiritual rather than ethical interest as it stressed as it stressed that nature contains a divine impulse that directs all life. Who is the philosopher behind Taoism?

- a. Confucius
- b. Lao-Tzu
- c. The Buddha
- d. Siddharta Gautama

B

143. Due to close family ties, Filipino families are generally _____.

A. Nuclear

B. Extended

C. Divided

D. Secular

Types of Families

- ▶ **Nuclear Family**
 - Father, Mother, Children
- ▶ **Extended Family**
 - Grandparents, Aunts, Uncles, Cousins
- ▶ **Single-Parent Family**
 - One parent with children
- ▶ **Blended Family**
 - Stepparents, Stepbrothers, Stepsisters
- ▶ **Adoptive Family**
 - Family with children not biologically linked to parents
- ▶ **Childless Family**
 - Couple who does not have children



B

144. What contained the concrete promise of the Americans to Filipinos Independence “as soon as stable government can be established?”

A. Preamble of the Jones Law

B. Fairfield Bill of 1924

C. Preamble of the Commonwealth
Government

D. First 1919 Parliamentary Mission to the
U.S.



PREAMBLE OF JONES LAW

Whereas it was never the intention of the people of United States in the incipency of the war with Spain to make it a war of conquest or for territorial aggrandizement; and

Whereas it is, as it has always been, the purpose of the people of the United States to withdraw their sovereignty over Philippine Islands and to recognize their independence as soon as a stable government can be established therein; and

Whereas for the speedy accomplishment of such purpose it is desirable to place in the hands of the people of the Philippines as large a control of their domestic affairs as can be given them without, in the meantime, impairing the exercise of the rights of sovereignty by the people of the United States, in order that, by the use and exercise of popular franchise and governmental powers, they may be the better prepared to fully assume the responsibilities and enjoy all the privileges of complete independence: Therefore

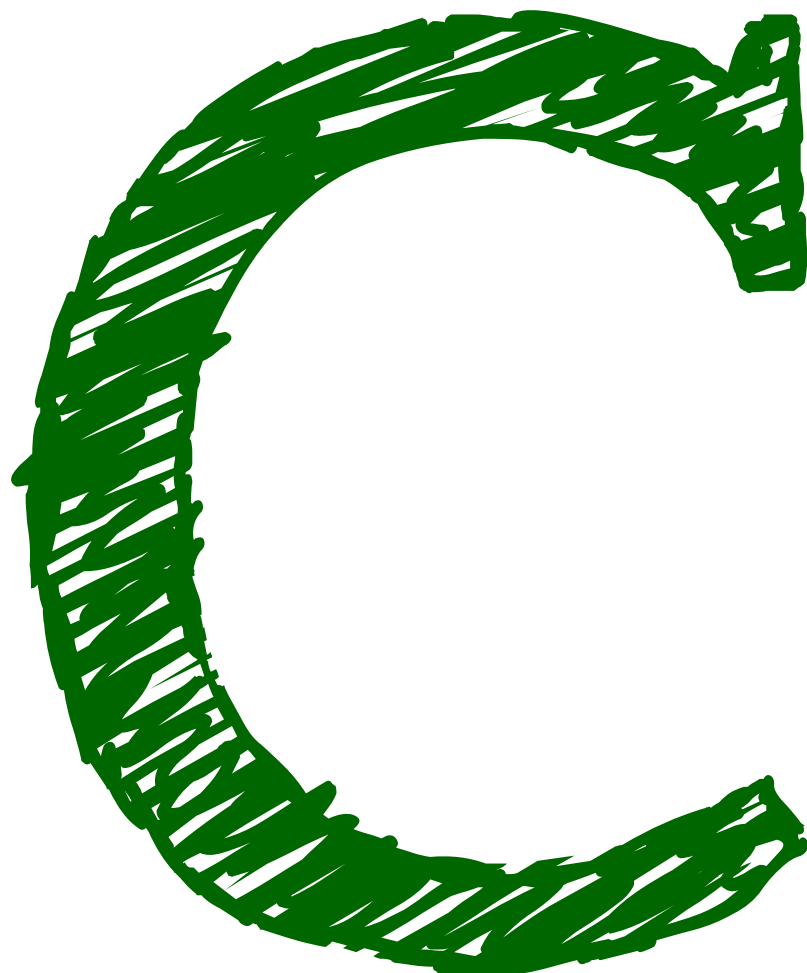
145. What symbol is used to make the text bold?

A. Ctrl+X

B. Ctrl+O

C. Ctrl+B

D. Ctrl+V



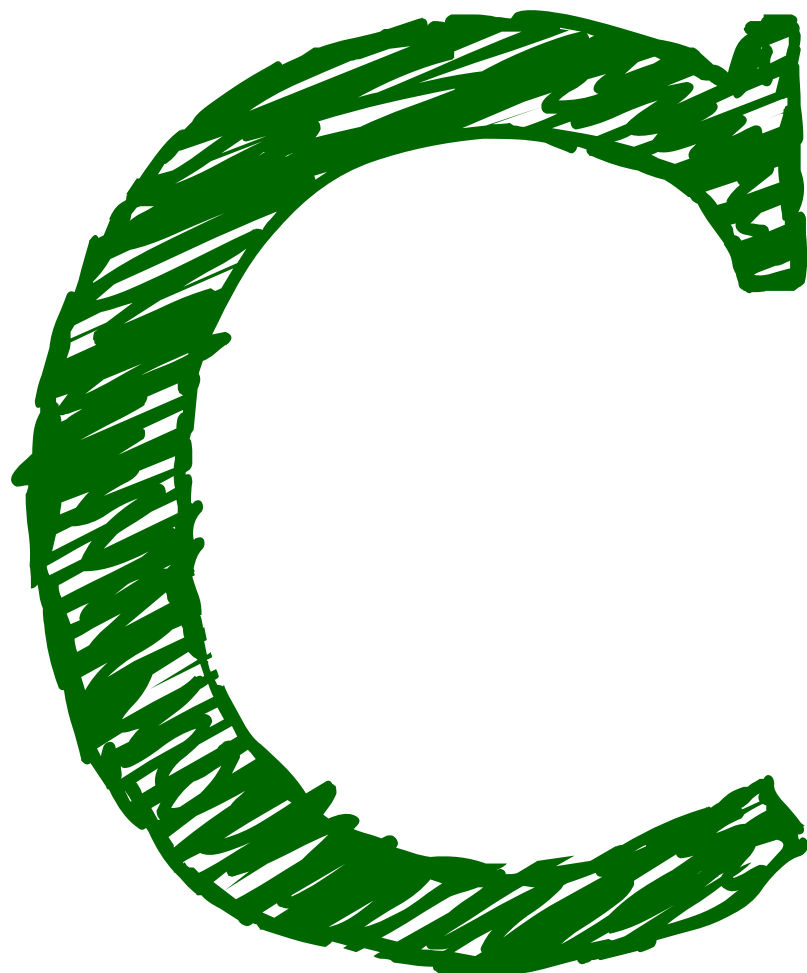
146. _____ means the graphics will be displayed more than once.

A. texting

B. timing

C. tiling

D. tagging



147. Which is the brain of the computer?

- A. RAM
- B. CPU
- C. Program
- D. Password

B

148. Ctrl is a symbol for _____.

- A. control key
- B. enter key
- C. backspace
- D. alternate key



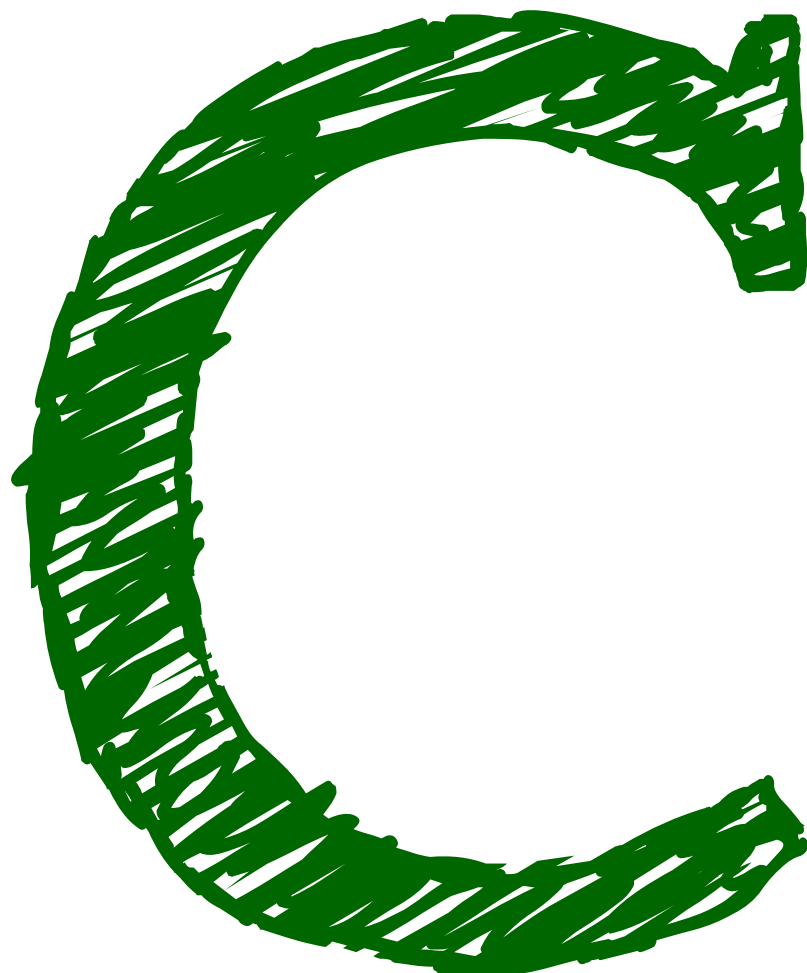
149. The acronym HTML means
Hyper Text Mark-up _____.

A. Line

B. Letter

C. Language

D. Label



150. Three-dimensional image reproduced from a pattern of interference is called _____.

- A. Internet
- B. Hologram
- C. Audio-visual
- D. Network

B